Supplementary Material

From Proxemics to Foundation Models: A Survey of Socially Aware Robot Navigation

This supplement contains the screening records, per-work coding tables and supporting analyses underlying the survey. The main manuscript presents condensed summaries, while the supplementary tables provide the corresponding detailed records. Each table is also provided as a machine-readable CSV file (`Supplementary_Data_tables_CSV.zip`). The discussion is in the main manuscript and is not repeated here.

Literature cut-off: 19 October 2026. The final coverage search was conducted on 26 October 2026 and restricted to works dated up to the literature cut-off.

Corpus component	n
Unique works	98
Direct foundation-model (FM) social-navigation papers	45
Classical and learning-based works	12
Adjacent FM navigation works	13
Datasets, simulators and benchmarks	15
Surveys	13
Representative pre-FM comparison set (not exhaustive; outside the corpus count; includes the 12 classical and learning works)	40

Contents

Item	Content	Relation to the manuscript
Table S1, part A (`S1_screening_record.csv`)	Part A: all 284 candidates identified in the expansion search and the final cut-off check (three arXiv search passes, general web search, backward citation chasing; arXiv, Crossref and proceedings at the cut-off), each with its screening level, decision and reason	Section 2.2
Table S1, part B (`S1b_final_search_record.csv`)	Part B: all 270 unique works of the final coverage search (IEEE Xplore, OpenAlex, forward citation chasing, and a limited manual Google Scholar check of one results page; Google Scholar was not searched systematically), with retrieval source, decision and reason; lists the 3 unresolved works	Section 2.2
Table S2 (`S2_publication_status.csv`)	Publication status of all 98 corpus works: venue, status class, peer review, DOI and relation of the analysed copy to the published version	Section 2.4; Table S11
Table S3 (`S3_adjacent_foundation_models.csv`)	The 13 adjacent FM navigation works: mechanism, task, concept inherited by social-navigation methods, and why each is *not* evidence of social competence	Section 4
Table S4 (`S4_full_final_coding.csv`)	Final coding of the 85 non-survey corpus works on Dimensions A, B, D and E, with evaluated (Yes) and addressed-only (Partial) capability leaves	Sections 5-9
Table S5 (`S5_prefm_comparison_set.csv`)	The 40-work representative pre-FM comparison set: selection, coding source, links and coding (part A); summary by paradigm (part B, `S5b_prefm_by_paradigm.csv`; formerly main-text Table 4)	Sections 2.3, 3 and 7
Table S6 (`S6_robustness.csv`)	Headline counts for all direct papers, the peer-reviewed subset and papers public before the Cai et al. cut-off	Section 10.5
Table S7 (`S7_capability_coverage.csv`)	Capability counts per leaf: direct evaluated and addressed, pre-FM evaluated, datasets evaluated	Section 7; Fig. 7
Table S8 (`S8_code_dictionary.csv`)	Full code dictionary of the five coding dimensions and the cross-cutting attributes	Section 5; Fig. 4
Table S9 (`S9_interface_evidence.csv`)	Counts behind Fig. 6: papers per interface and, within them, systematic robot evaluation, FM-independent safety guard and remote inference; evidence by publication year	Sections 5 and 9; Fig. 6

Table S10 (`S10_metric_families.csv`)	Full metric-family table (formerly a main-text table): metric names found, level, what each family measures and fails to measure, direct papers using it	Section 8; Fig. 8
Table S11 (`S11_direct_architecture.csv`)	Per-paper architecture of every direct paper: publication status, FM models, inputs, Dimensions A, B and D, planner or controller, temporal coupling, FM-independent safety guard, other attributes and main contribution (formerly main-text Table 5)	Sections 5 and 6; Table 4
Table S12 (`S12_direct_evidence.csv`)	Per-paper evaluated capabilities and evidence: EV1-EV6, trials, statistics, key metrics, runtime, uncertainty, hallucination mitigation, code/data release and key limitation (formerly main-text Table 6)	Sections 6, 7 and 9
Table S13 (`S13_datasets_benchmarks.csv`)	Per-resource inventory of datasets, simulators and benchmarks: size, sensors, human trajectories, group/intention, preference and multi-action labels, closed loop, VLM-readiness, FM role and limitations (the main-text Table 5 groups them into families)	Section 8; Table 5
Fig. S1	FM roles (Dimension B) per direct paper, grouped by primary interface	Section 5.2; Table S11
Fig. S2	Evidence profile (Dimension E) per direct paper, grouped by primary interface	Section 9; Table S12
Fig. S3	Capability coding (Dimension C) per direct paper	Section 7
Fig. S4	Dated timeline of the paradigms of social representation (overlapping bands; no causal ordering)	Section 3; Table 3

Notation

Publication status (S2; Status column of S11):

Class	Meaning
A	Published and peer-reviewed, verified against a publisher record
B	Accepted according to the authors (or, for one work, a citing paper); proceedings not yet indexed at the cut-off
C	Preprint only
D	Conference paper; acceptance verified, peer-review status not independently established

Classes B and D are never counted as peer-reviewed.

Screening (Table S1): part A records the expansion search and the final cut-off check (19 October 2026); part B records the final coverage search (26 October 2026). The two records are kept separate and their counts are not merged.

- *First found* and *Found in passes* (part A): P1–P3 are the three arXiv search passes, plus "web", "citation" and "final check" (the final cut-off check).
- *Screening level*: title/abstract or full text.
- *Decision* (part A): INCLUDE; EXCLUDE; BACKGROUND ONLY (cited but not coded); RELATED VERSION (an alternative version of a corpus work); EXCLUDE (WITHDRAWN). For included works, *Corpus entry* gives the citation key used in the manuscript.
- *Decision* (part B): INCLUDE; EXCLUDE; UNRESOLVED (not coded: full text not accessible); ALREADY KNOWN (already in the corpus or the part A record); ALREADY SCREENED (screened in part A under a title variant).
- Exclusion reasons (parts A and B; criteria A–C as defined in manuscript Section 2.1). Many records give a specific reason in place of, or after, the code:

Code	Reason
R1	Not robot/embodied navigation (fails A)
R2	Autonomous driving, UAV or other non-ground-robot domain (fails A)
R3	Robot manipulation / HRI / perception without mobile-navigation behaviour (fails A)
R4	Embodied/robot navigation without an explicit social-navigation capability (fails C)
R5	Social navigation without an FM role in the navigation method (fails B)
R6	Navigation assistance for people: the person navigates, not a robot (fails A)

R7	Survey, review, position paper or thesis outside the screening scope; surveys enter the corpus only through the seed set (part B only)
R8	Central claims or data not verifiable after full-text assessment, irrespective of publication status (exclusion rule in manuscript Section 2.2; one record, part B only)

Taxonomy abbreviations (S4, S11, S12):

Dimension / attribute	Codes
A, social representation (multi-label)	A0 not represented · A1 kinematic agent · A2 predicted motion · A3 social-space field · A4 relational structure · A5 symbolic/linguistic · A6 implicit/latent
B, FM role (multi-label)	R1 scene/context interpretation · R2 human-state/relation inference · R3 social-norm reasoning and behaviour selection · R4 instruction/preference grounding · R5 human-facing communication
C, social capability (15 leaves)	SAF1 collision · SAF2 near-collision/risk margin · SPA1 personal space · SPA2 group/conversation space · SPA3 encounter conventions · SPA4 crowd navigation · HUM1 motion understanding/prediction · HUM2 intention/activity · NOR1 etiquette/politeness · NOR2 explicit social-instruction following · NOR3 context-dependent conduct · NOR4 cultural/norm adaptation · EXP1 perceived safety/comfort · EXP2 legibility/predictability · EXP3 naturalness
C values	**Yes:** evaluated using a dedicated metric or a systematic scenario-specific success criterion that required the capability. **Partial:** addressed without such a measure. An evaluated capability is also addressed, so addressed counts comprise Yes and Partial.
D, navigation interface (categorical, not a maturity order)	D0 non-executed decision · D1 semantic/social state · D2 candidate selection · D3 objective/constraint/parameter shaping · D4 goal/waypoint/trajectory · D5 runtime latent conditioning · D6 direct action · D7 training-time supervision · D8 human-facing channel
E, evidence profile (one dimension represented through six separately coded attributes; not combined into a score)	EV1 offline evaluation · EV2 closed-loop simulation · EV3 physical-robot evidence (none/demo/systematic) · EV4 real people in robot trials (none/simulated-only/staged/unscripted/present-type-unreported/not-reported) · EV5 human evaluation (none/text raters/observers-video/co-present participants) · EV6 open-world or long-duration deployment. EV3, EV4 and EV5 record different forms of evidence and are not interchangeable
Cross-cutting attributes (S11, S12)	temporal coupling · FM-independent safety guard · uncertainty treatment · hallucination mitigation · inference location · memory · privacy constraint

Reporting conventions:

- Details a paper does not report are coded "Not reported". Counts are therefore lower bounds.
- Runtime, trial and metric entries are given as the papers report them, and are not comparable across papers.
- Table S10 counts studies reporting a runtime or latency evaluation metric (21/45). In addition, 2 direct studies report numeric runtime or latency in their system descriptions or discussion but not as an evaluation metric; these are included in the 23/45 overall runtime-reporting count (Table S12, *Runtime reported*) but not in this metric-family count.

Note on the SNEI dataset (main-text Table 5; Tables S4 and S13)

SNEI quantities differ across sources and are not interchangeable:

- the originating paper reports about 2K social scenarios, 1,961 images (summed over interaction types) and more than 40K question–answer pairs;
- the public release contains 1,508 annotated rows in two batches;

- one paper reports 810 conversations, which equals the first public batch;
- several papers use a 325-image subset (265 training / 60 test).

Its source data are reported inconsistently: the originating paper describes it as built from SCAND, while a later paper by a different group states SCAND and MuSoHu. This discrepancy could not be resolved from the available records. The sensor entry "front-facing robot RGB image (from SCAND)" in Table S13 records the originating paper's statement.

Pre-FM comparison set (S5): "Not determinable" marks evidence that the reference-survey descriptions used for coding do not establish; such works are excluded from evidence denominators. Every work in Table S5 (parts A and B) is cited by its key and listed in full in the reference list at the end of this supplement.

Reproducibility

- Every table in this supplement, like every count in the manuscript, is generated from the per-paper records.
- Coding followed the five-dimensional scheme defined in the main manuscript (Section 5). Repeat-coding and metric-based consistency checks (Section 2.5) were used to assess internal consistency; these checks do not constitute inter-rater reliability.

Contents

Table S1, part A. Screening record of the expansion search and the final cut-off check (19 October 2026) 6

Table S1, part B. Screening record of the final coverage search (26 October 2026) 14

Table S2. Publication status of all corpus works 23

Table S3. Adjacent foundation-model navigation works (not social-navigation evidence) 28

Table S4. Full final coding of all non-survey corpus works 29

Table S5, part A. Pre-FM comparison set: per-work selection and coding 31

Table S5, part B. Pre-FM comparison set by paradigm 35

Table S6. Robustness of headline counts 36

Table S7. Social-capability coverage 37

Table S8. Code dictionary 38

Table S9. Navigation interfaces and evidence 40

Table S10. Metric families 41

Table S11. Architecture of the direct papers (per paper) 43

Table S12. Capabilities and evidence of the direct papers (per paper) 45

Table S13. Datasets, simulators and benchmarks (per resource) 47

Fig. S1. FM roles per direct paper (Dimension B), grouped by primary navigation interface 48

Fig. S2. Evidence profile per direct paper (Dimension E), grouped by primary navigation interface 49

Fig. S3. Capability coding per direct paper (Dimension C) 50

Fig. S4. Dated timeline of the paradigms of social representation. Each band runs from the approximate first appearance of a paradigm to the literature cut-off; markers show representative works at their publication year. Overlapping bands indicate coexistence, not replacement or causal progression 51

References cited in Table S5 52

Table S1, part A. Screening record of the expansion search and the final cut-off check (19 October 2026)

284 rows. Machine-readable version: `S1_screening_record.csv`.

Candidate id (arXiv id or key)	First found (search pass P1-P3, web, citation)	Found in passes	Title	First author	Date	Decision	Corpus entry (cite key)	Screening level	Reason
2303.14880	citation	citation	Toward Human-Like Social Robot Navigation: ... (MuSoHu)	D. M. Nguyen et al.	2023	INCLUDE	Nguyen2023MuSoHu	full text	included after full-text screening
2305.14310	P1	P1	Navigating Prompt Complexity for Zero-Shot Classification: A Study of Large Language Models in Compu	Yida Mu et al.	2023-05-23	EXCLUDE		title/abstract	R1: Not robot/embodied navigation (non-robotics use of "navigate", NLP/social-science/LLM-evaluation topic)
2306.01874	citation	citation	SACSoN: Scalable Autonomous Control for Social Navigation	N. Hirose et al.	2024	INCLUDE	Hirose2024SACSoN	full text	included after full-text screening
2306.16740	P1	P1	Principles and Guidelines for Evaluating Social Robot Navigation Algorithms	Anthony Francis et al.	2023-06-29	INCLUDE	Francis2025PrinciplesGuidelines	full text	already in the seed corpus
2308.12537	citation	citation	HuBo-VLM: Unified VLM designed for HUman roBOt interaction tasks	Z. Dong et al.	2023	EXCLUDE		full text	HuBo-VLM: VLM for HRI perception tasks (detection/ grounding); no navigation (fails A)
2309.02561	P1	P1	Physically Grounded Vision-Language Models for Robotic Manipulation	Jensen Gao et al.	2023-09-05	EXCLUDE		title/abstract	R3: Robot manipulation / HRI / perception without mobile-navigation behaviour (fails A)
2309.12568	P1	P1	A Study on Learning Social Robot Navigation with Multimodal Perception	Bhabaranjan Panigrahi et al.	2023-09-22	EXCLUDE		title/abstract	R5: Social navigation without an FM role in the navigation method (fails B)
2310.02457	P1	P1	The Empty Signifier Problem: Towards Clearer Paradigms for Operationalising "Alignment" in Large Lan	Hannah Rose Kirk et al.	2023-10-03	EXCLUDE		title/abstract	R1: Not robot/embodied navigation (non-robotics use of "navigate", NLP/social-science/LLM-evaluation topic)
2311.08244	P1	P1	Language and Sketching: An LLM-driven Interactive Multimodal Multitask Robot Navigation Framework	Weiqin Zu et al.	2023-11-14	EXCLUDE		title/abstract	R4: Embodied/robot navigation without an explicit social-navigation capability (fails C)
2312.16016	P1	P1	V-STRONG: Visual Self-Supervised Traversability Learning for Off-road Navigation	Sanghun Jung et al.	2023-12-26	EXCLUDE		title/abstract	R4: Embodied/robot navigation without an explicit social-navigation capability (fails C)
2401.12275	P1	P1	Multi-Agent Dynamic Relational Reasoning for Social Robot Navigation	Jiachen Li et al.	2024-01-22	EXCLUDE		title/abstract	R5: Social navigation without an FM role in the navigation method (fails B)
2401.13311	P3	P3	ConTextual: Evaluating Context-Sensitive Text-Rich Visual Reasoning in Large Multimodal Models	Rohan Wadhawan et al.	2024-01-24	EXCLUDE		title/abstract	R1: Not robot/embodied navigation (non-robotics use of "navigate", NLP/social-science/LLM-evaluation topic)
2401.17083	P1	P1	Online Robot Navigation and Manipulation with Distilled Vision-Language Models	Kangcheng Liu	2024-01-30	EXCLUDE		title/abstract	R4: Embodied/robot navigation without an explicit social-navigation capability (fails C)
2402.03494	P1	P1	Beyond Text: Utilizing Vocal Cues to Improve Decision Making in LLMs for Robot Navigation Tasks	Xingpeng Sun et al.	2024-02-05	EXCLUDE		full text	Beyond Text: vocal cues for LLM instruction disambiguation; no social-navigation behaviour (fails C)
2402.15420	P1	P1+P3	PREDILECT: Preferences Delineated with Zero-Shot Language-based Reasoning in Reinforcement Learning	Simon Holk et al.	2024-02-23	INCLUDE	Holk2024PREDILECT	full text	included after full-text screening
2403.15648	P1	P1	Unifying Large Language Model and Deep Reinforcement Learning for Human-in-Loop Interactive Socially	Weizheng Wang et al.	2024-03-22	INCLUDE	Wang2024SALM	full text	included after full-text screening
2404.00210	P1	P1+P3	VLM-Social-Nav: Socially Aware Robot Navigation through Scoring using Vision-Language Models	Daeun Song et al.	2024-03-30	INCLUDE	Song2025VLMSocialNav	full text	already in the seed corpus
2406.02425	P1	P1	CoNav: A Benchmark for Human-Centered Collaborative Navigation	Changhao Li et al.	2024-06-04	EXCLUDE		full text	CoNav: human-centered collaborative navigation benchmark; FM used only to animate the simulated humans, not in the navigation method, and th
2406.08824	P1	P1	LLM-Driven Robots Risk Enacting Discrimination, Violence, and Unlawful Actions	Andrew Hundt et al.	2024-06-13	EXCLUDE		title/abstract	R3: Robot manipulation / HRI / perception without mobile-navigation behaviour (fails A)
2406.19236	P1	P1	Human-Aware Vision-and-Language Navigation: Bridging Simulation to Reality with Dynamic Human Intera	Heng Li et al.	2024-06-27	INCLUDE	Li2024HAVLN	full text	already in the seed corpus
2407.09652	P3	P3	How Chinese are Chinese Language Models? The Puzzling Lack of Language Policy in China's LLMs	Andrea W Wen-Yi et al.	2024-07-12	EXCLUDE		title/abstract	R1: Not robot/embodied navigation (non-robotics use of "navigate", NLP/social-science/LLM-evaluation topic)
2407.09890	P1	P1	Speech-Guided Sequential Planning for Autonomous Navigation using Large Language Model Meta AI 3 (Ll	Alkesh K. Srivastava et al.	2024-07-13	EXCLUDE		title/abstract	R4: Embodied/robot navigation without an explicit social-navigation capability (fails C)
2407.12425	P1	P1	Navigating the Noisy Crowd: Finding Key Information for Claim Verification	Haisong Gong et al.	2024-07-17	EXCLUDE		title/abstract	R1: Not robot/embodied navigation (non-robotics use of "navigate", NLP/social-science/LLM-evaluation topic)
2408.07910	P1	P1	DM2RM: Dual-Mode Multimodal Ranking for Target Objects and Receptacles Based on Open-Vocabulary Inst	Ryosuke Korekata et al.	2024-08-15	EXCLUDE		title/abstract	R3: Robot manipulation / HRI / perception without mobile-navigation behaviour (fails A)
2409.04965	P1	P1+P3	Socially-Aware Robot Navigation Enhanced by Bidirectional Natural Language Conversations Using Large	Congcong Wen et al.	2024-09-08	INCLUDE	Wen2025HSACLLM	full text	already in the seed corpus
2409.08027	P1	P1	iLLuMinaTE: An LLM-XAI Framework Leveraging Social Science Explanation Theories Towards Actionable S	Vinitra Swamy et al.	2024-09-12	EXCLUDE		title/abstract	R1: Not robot/embodied navigation (non-robotics use of "navigate", NLP/social-science/LLM-evaluation topic)
2409.12471	P1	P1+P3	Arena 4.0: A Comprehensive ROS2 Development and Benchmarking Platform for Human-centric Navigation U	Volodymyr Shcherbyna1 et al.	2024-09-19	INCLUDE	Shcherbyna2025Arena40	full text	already in the seed corpus
2409.13244	P1	P1	From Cognition to Precognition: A Future-Aware Framework for Social Navigation	Zeying Gong et al.	2024-09-20	BACKGROUND ONLY		full text	Falcon (ICRA 2025): RL social navigation + Social-HM3D/ MP3D benchmark; no FM (fails B); benchmark used by HUMA
2409.13393	web	web	Hey Robot! Personalizing Robot Navigation Through MPC with an LLM	D. Martinez-Baselga et al.	2025	INCLUDE	MartinezBaselga2025HeyRobot	full text	included after full-text screening
2409.13675	P1	P1	OLiVia-Nav: An Online Lifelong Vision Language Approach for Mobile Robot Social Navigation	Siddarth Narasimhan et al.	2024-09-20	INCLUDE	Narasimhan2025OLiViaNav	full text	already in the seed corpus
2409.18084	P1	P1	GSON: A Group-based Social Navigation Framework with Large Multimodal Model	Shangyi Luo et al.	2024-09-26	INCLUDE	Luo2025GSON	full text	already in the seed corpus
2409.20445	citation	citation	Robot Navigation Using Physically Grounded VLMs in Outdoor Environments (VLM-GroNav)	M. Elnoor et al.	2024	EXCLUDE		full text	VLM-GroNav: VLM terrain/traversability grounding outdoors; no social capability (fails C)
2410.04005	P1	P1	Enhancing the Travel Experience for People with Visual Impairments through Multimodal Interaction: N	He Zhang et al.	2024-10-05	EXCLUDE		title/abstract	R6: Navigation assistance for people (the human navigates, not a robot) (fails A)
2410.04302	P1	P1	PANav: Toward Privacy-Aware Robot Navigation via Vision-Language Models	Bangguo Yu et al.	2024-10-05	INCLUDE	Yu2024PANav	full text	included after full-text screening
2410.05243	P1	P1	Navigating the Digital World as Humans Do: Universal Visual Grounding for GUI Agents	Boyu Gou et al.	2024-10-07	EXCLUDE		title/abstract	R1: Not robot/embodied navigation (non-robotics use of "navigate", NLP/social-science/LLM-evaluation topic)
2410.15044	P1	P1	Adanonymizer: Interactively Navigating and Balancing the Duality of Privacy and Output Performance i	Shuning Zhang et al.	2024-10-19	EXCLUDE		title/abstract	R1: Not robot/embodied navigation (non-robotics use of "navigate", NLP/social-science/LLM-evaluation topic)

Candidate id (arXiv id or key)	First found (search pass P1-P3, web, citation)	Found in passes	Title	First author	Date	Decision	Corpus entry (cite key)	Screening level	Reason
2411.01796	P3	P3	Constrained Human-AI Cooperation: An Inclusive Embodied Social Intelligence Challenge	Weihua Du et al.	2024-11-04	EXCLUDE		title/abstract	R4: Embodied/robot navigation without an explicit social-navigation capability (fails C)
2412.01250	P1	P1	Collaborative Instance Object Navigation: Leveraging Uncertainty-Awareness to Minimize Human-Agent D	Francesco Taioli et al.	2024-12-02	EXCLUDE		title/abstract	R4: Embodied/robot navigation without an explicit social-navigation capability (fails C)
2412.02655	P1	P1	LLM-Enhanced Path Planning: Safe and Efficient Autonomous Navigation with Instructional Inputs	Pranav Doma et al.	2024-12-03	EXCLUDE		full text	LLM path planning for industrial pick-and-place; pedestrian distance is one incidental constraint, evaluated only qualitatively (C not met)
2412.05893	P1	P1	doScenes: An Autonomous Driving Dataset with Natural Language Instruction for Human Interaction and	Parthib Roy et al.	2024-12-08	EXCLUDE		title/abstract	R2: Autonomous driving, UAV or other non-ground-robot domain
2412.19595	P1	P1+P3	SocRATES: Towards Automated Scenario-based Testing of Social Navigation Algorithms	Shashank Rao Marpally et al.	2024-12-27	INCLUDE	Marpally2026SocRATES	full text	included after full-text screening
2412.20230	P1	P1	Leveraging Large Language Models for Enhancing Autonomous Vehicle Perception	Athanasios Karagounis	2024-12-28	EXCLUDE		title/abstract	R2: Autonomous driving, UAV or other non-ground-robot domain
2501.01439	P3	P3	Probabilistic Mission Design for Neuro-Symbolic Unmanned Aircraft Systems	Simon Kohaut et al.	2024-12-25	EXCLUDE		title/abstract	R2: Autonomous driving, UAV or other non-ground-robot domain
2501.01544	P2	P2	Many of Your DPOs are Secretly One: Attempting Unification Through Mutual Information	Rasul Tutnov et al.	2025-01-02	EXCLUDE		title/abstract	R1: Not robot/embodied navigation (non-robotics use of "navigate", NLP/social-science/LLM-evaluation topic)
2501.02092	P1	P1	Human and Machine: How Software Engineers Perceive and Engage with AI-Assisted Code Reviews Compared	Adam Alami et al.	2025-01-03	EXCLUDE		title/abstract	R1: Not robot/embodied navigation (non-robotics use of "navigate", NLP/social-science/LLM-evaluation topic)
2501.05925	P3	P3	Navigating Tomorrow: Reliably Assessing Large Language Models Performance on Future Event Prediction	Petraq Nako et al.	2025-01-10	EXCLUDE		title/abstract	R1: Not robot/embodied navigation (non-robotics use of "navigate", NLP/social-science/LLM-evaluation topic)
2501.09024	P1	P1+P3	Social-LLaVA: Enhancing Robot Navigation through Human-Language Reasoning in Social Spaces	Amirreza Payandeh et al.	2024-12-30	INCLUDE	Payandeh2025SocialLLaVA	full text	already in the seed corpus
2501.10105	P1	P1	Universal Actions for Enhanced Embodied Foundation Models	Jinliang Zheng et al.	2025-01-17	EXCLUDE		title/abstract	R3: Robot manipulation / HRI / perception without mobile-navigation behaviour (fails A)
2501.12128	P1	P1+P2	Evaluating Efficiency and Engagement in Scripted and LLM-Enhanced Human-Robot Interactions	Tim Schreiter et al.	2025-01-21	EXCLUDE		title/abstract	R3: Robot manipulation / HRI / perception without mobile-navigation behaviour (fails A)
2502.04756	P3	P3	Concept Navigation and Classification via Open-Source Large Language Model Processing	Maël Kubli	2025-02-07	EXCLUDE		title/abstract	R1: Not robot/embodied navigation (non-robotics use of "navigate", NLP/social-science/LLM-evaluation topic)
2502.16101	P3	P3	Worse than Zero-shot? A Fact-Checking Dataset for Evaluating the Robustness of RAG Against Misleadin	Linda Zeng et al.	2025-02-22	EXCLUDE		title/abstract	R1: Not robot/embodied navigation (non-robotics use of "navigate", NLP/social-science/LLM-evaluation topic)
2502.16671	P2	P2	MimeQA: Towards Socially-Intelligent Nonverbal Foundation Models	Hengzhi Li et al.	2025-02-23	EXCLUDE		title/abstract	R1: Not robot/embodied navigation (non-robotics use of "navigate", NLP/social-science/LLM-evaluation topic)
2502.19024	P1	P1	Ground-level Viewpoint Vision-and-Language Navigation in Continuous Environments	Zerui Li et al.	2025-02-26	EXCLUDE		title/abstract	R4: Embodied/robot navigation without an explicit social-navigation capability (fails C)
2503.02572	P1	P1	RaceVLA: VLA-based Racing Drone Navigation with Human-like Behaviour	Valerii Serpiva et al.	2025-03-04	EXCLUDE		title/abstract	R2: Autonomous driving, UAV or other non-ground-robot domain
2503.07557	P1	P1+P2+P3	AutoSpatial: Visual-Language Reasoning for Social Robot Navigation through Efficient Spatial Reasoni	Yangzhe Kong et al.	2025-03-10	INCLUDE	Kong2025AutoSpatial	full text	already in the seed corpus
2503.08367	P3	P3	Embodied Crowd Counting	Runling Long et al.	2025-03-11	EXCLUDE		title/abstract	R4: Embodied/robot navigation without an explicit social-navigation capability (fails C)
2503.09758	P1	P1+P3	Multi-Agent LLM Actor-Critic Framework for Social Robot Navigation	Weizheng Wang et al.	2025-03-12	INCLUDE	Wang2025SAMALM	full text	included after full-text screening
2503.09820	P1	P1	ViLAM: Distilling Vision-Language Reasoning into Attention Maps for Social Robot Navigation	Mohamed Elnoor et al.	2025-03-12	INCLUDE	Elnoor2026ViLAM	full text	already in the seed corpus
2503.12333	P1	P1	GameChat: Multi-LLM Dialogue for Safe, Agile, and Socially Optimal Multi-Agent Navigation in Constra	Vagul Mahadevan et al.	2025-03-16	EXCLUDE		full text	GameChat: multi-robot LLM dialogue for deadlock resolution among robots; no humans (fails C)
2503.12731	P1	P1	Navigating Heat Exposure: Simulation of Route Planning Based on Visual Language Model Agents	Haoran Ma et al.	2025-03-17	EXCLUDE		title/abstract	R6: Navigation assistance for people (the human navigates, not a robot) (fails A)
2503.12844	P3	P3	GuideDog: A Real-World Egocentric Multimodal Dataset for Blind and Low-Vision Accessibility-Aware Gu	Junhyeok Kim et al.	2025-03-17	EXCLUDE		title/abstract	R6: Navigation assistance for people (the human navigates, not a robot) (fails A)
2503.13975	P1	P1	Navigating Rifts in Human-LLM Grounding: Study and Benchmark	Omar Shaikh et al.	2025-03-18	EXCLUDE		title/abstract	R1: Not robot/embodied navigation (non-robotics use of "navigate", NLP/social-science/LLM-evaluation topic)
2503.14229	P1	P1+P2	HA-VLN 2.0: An Open Benchmark and Leaderboard for Human-Aware Navigation in Discrete and Continuous	Yifei Dong et al.	2025-03-18	INCLUDE	Dong2026HAVLN20	full text	already in the seed corpus
2503.14484	P2	P2	Gricean Norms as a Basis for Effective Collaboration	Fardin Saad et al.	2025-03-18	EXCLUDE		title/abstract	R1: Not robot/embodied navigation (non-robotics use of "navigate", NLP/social-science/LLM-evaluation topic)
2503.16441	P2	P2	Safe and Efficient Social Navigation through Explainable Safety Regions Based on Topological Feature	Victor Toscano-Duran et al.	2025-02-14	EXCLUDE		title/abstract	R5: Social navigation without an FM role in the navigation method (fails B)
2503.16475	P1	P1	LLM-Glasses: GenAI-driven Glasses with Haptic Feedback for Navigation of Visually Impaired People	Issatay Tokmurziyev et al.	2025-03-04	EXCLUDE		title/abstract	R6: Navigation assistance for people (the human navigates, not a robot) (fails A)
2503.21986	P1	P1	Socially Constructed Treatment Plans: Analyzing Online Peer Interactions to Understand How Patients	Madhusudan Basak et al.	2025-03-27	EXCLUDE		title/abstract	R1: Not robot/embodied navigation (non-robotics use of "navigate", NLP/social-science/LLM-evaluation topic)
2503.22040	P1	P1	Navigating the Risks of Using Large Language Models for Text Annotation in Social Science Research	Hao Lin et al.	2025-03-27	EXCLUDE		title/abstract	R1: Not robot/embodied navigation (non-robotics use of "navigate", NLP/social-science/LLM-evaluation topic)
2504.01588	P2	P2	Building Knowledge from Interactions: An LLM-Based Architecture for Adaptive Tutoring and Social Rea	Luca Garello et al.	2025-04-02	EXCLUDE		title/abstract	R1: Not robot/embodied navigation (non-robotics use of "navigate", NLP/social-science/LLM-evaluation topic)
2504.05477	P1	P1+P2	Trust Through Transparency: Explainable Social Navigation for Autonomous Mobile Robots via Vision-La	Oluwadamilola Sotomi et al.	2025-04-07	INCLUDE	Sotomi2025TransparentSocialNavVLM	full text	included after full-text screening
2504.13898	P2	P2	Social Human Robot Embodied Conversation (SHREC) Dataset: Benchmarking Foundational Models' Social R	Dong Won Lee et al.	2025-04-07	EXCLUDE		title/abstract	R3: Robot manipulation / HRI / perception without mobile-navigation behaviour (fails A)
2505.01754	P3	P3	Unraveling Media Perspectives: A Comprehensive Methodology Combining Large Language Models, Topic Mo	Orlando Jähde et al.	2025-05-03	EXCLUDE		title/abstract	R1: Not robot/embodied navigation (non-robotics use of "navigate", NLP/social-science/LLM-evaluation topic)
2505.01881	P1	P1+P2+P3	PhysNav-DG: A Novel Adaptive Framework for Robust VLM-Sensor Fusion in Navigation Applications	Trisanth Srinivasan et al.	2025-05-03	EXCLUDE		title/abstract	R4: Embodied/robot navigation without an explicit social-navigation capability (fails C)
2505.01931	P1	P1	Semantic Intelligence: Integrating GPT-4 with A Planning in Low-Cost Robotics	Jesse Barkley et al.	2025-05-03	EXCLUDE		title/abstract	R4: Embodied/robot navigation without an explicit social-navigation capability (fails C)
2505.03434	P1	P1+P2	Procedural Memory Is Not All You Need: Bridging Cognitive Gaps in LLM-Based Agents	Schaun Wheeler et al.	2025-05-06	EXCLUDE		title/abstract	R1: Not robot/embodied navigation (non-robotics use of "navigate", NLP/social-science/LLM-evaluation topic)
2505.04354	P1	P1	Optimization Problem Solving Can Transition to Evolutionary Agentic Workflows	Wenhao Li et al.	2025-05-07	EXCLUDE		title/abstract	R1: Not robot/embodied navigation (non-robotics use of "navigate", NLP/social-science/LLM-evaluation topic)

Candidate id (arXiv id or key)	First found (search pass P1-P3, web, citation)	Found in passes	Title	First author	Date	Decision	Corpus entry (cite key)	Screening level	Reason
2505.16498	P1	P1	Human-like Semantic Navigation for Autonomous Driving using Knowledge Representation and Large Langu	Augusto Luis Ballardini et al.	2025-05-22	EXCLUDE		title/abstract	R2: Autonomous driving, UAV or other non-ground-robot domain
2505.17461	P2	P2	Diagnosing Vision Language Models' Perception by Leveraging Human Methods for Color Vision Deficienc	Kazuki Hayashi et al.	2025-05-23	EXCLUDE		title/abstract	R1: Not robot/embodied navigation (non-robotics use of "navigate", NLP/social-science/LLM-evaluation topic)
2505.18553	P1	P1	Applying Ontologies and Knowledge Augmented Large Language Models to Industrial Automation: A Decisi	John Oyekan et al.	2025-05-24	EXCLUDE		title/abstract	R1: Not robot/embodied navigation (non-robotics use of "navigate", NLP/social-science/LLM-evaluation topic)
2505.21479	P2	P2	Are Language Models Consequentialist or Deontological Moral Reasoners?	Keenan Samway et al.	2025-05-27	EXCLUDE		title/abstract	R1: Not robot/embodied navigation (non-robotics use of "navigate", NLP/social-science/LLM-evaluation topic)
2505.23179	P3	P3	DIP-R1: Deep Inspection and Perception with RL Looking Through and Understanding Complex Scenes	Sungjune Park et al.	2025-05-29	EXCLUDE		title/abstract	R1: Not robot/embodied navigation (non-robotics use of "navigate", NLP/social-science/LLM-evaluation topic)
2505.24315	P2	P2	InteractAnything: Zero-shot Human Object Interaction Synthesis via LLM Feedback and Object Affordanc	Jinlu Zhang et al.	2025-05-30	EXCLUDE		title/abstract	R1: Not robot/embodied navigation (non-robotics use of "navigate", NLP/social-science/LLM-evaluation topic)
2506.00900	P2	P2	SocialEval: Evaluating Social Intelligence of Large Language Models	Jinfeng Zhou et al.	2025-06-01	EXCLUDE		title/abstract	R1: Not robot/embodied navigation (non-robotics use of "navigate", NLP/social-science/LLM-evaluation topic)
2506.02556	P3	P3	Sign Language: Towards Sign Understanding for Robot Autonomy	Ayush Agrawal et al.	2025-06-03	EXCLUDE		title/abstract	R4: Embodied/robot navigation without an explicit social-navigation capability (fails C)
2506.06606	P2	P2	Stacey: Promoting Stochastic Steepest Descent via Accelerated $\{}ell_p$-Smooth Nonconvex Optimization	Xinyu Luo et al.	2025-06-07	EXCLUDE		title/abstract	R1: Not robot/embodied navigation (non-robotics use of "navigate", NLP/social-science/LLM-evaluation topic)
2506.06737	P2	P2	C-PATH: Conversational Patient Assistance and Triage in Healthcare System	Qi Shi et al.	2025-06-07	EXCLUDE		title/abstract	R1: Not robot/embodied navigation (non-robotics use of "navigate", NLP/social-science/LLM-evaluation topic)
2506.13276	P1	P1	Navigating the Black Box: Leveraging LLMs for Effective Text-Level Graph Injection Attacks	Yuefei Lyu et al.	2025-06-16	EXCLUDE		title/abstract	R1: Not robot/embodied navigation (non-robotics use of "navigate", NLP/social-science/LLM-evaluation topic)
2506.14233	P1	P1+P2	Narrate2Nav: Real-Time Visual Navigation with Implicit Language Reasoning in Human-Centric Environme	Amirreza Payandeh et al.	2025-06-17	INCLUDE	Payandeh2026Narrate2Nav	full text	included after full-text screening
2506.20566	P1	P1+P2+P3	HRIBench: Benchmarking Vision-Language Models for Real-Time Human Perception in Human-Robot Interact	Zhonghao Shi et al.	2025-06-25	EXCLUDE		title/abstract	R3: Robot manipulation / HRI / perception without mobile-navigation behaviour (fails A)
2506.22493	P1	P1+P3	A Detailed Factor Analysis for the Political Compass Test: Navigating Ideologies of Large Language M	Sadia Kamal et al.	2025-06-24	EXCLUDE		title/abstract	R1: Not robot/embodied navigation (non-robotics use of "navigate", NLP/social-science/LLM-evaluation topic)
2506.23046	P2	P2	SoMi-ToM: Evaluating Multi-Perspective Theory of Mind in Embodied Social Interactions	Xianzhe Fan et al.	2025-06-29	EXCLUDE		title/abstract	R1: Not robot/embodied navigation (non-robotics use of "navigate", NLP/social-science/LLM-evaluation topic)
2506.24019	P2	P2	Ella: Embodied Social Agents with Lifelong Memory	Hongxin Zhang et al.	2025-06-30	EXCLUDE		title/abstract	R1: Not robot/embodied navigation (non-robotics use of "navigate", NLP/social-science/LLM-evaluation topic)
2507.02103	P3	P3	What Neuroscience Can Teach AI About Learning in Continuously Changing Environments	Daniel Durstewitz et al.	2025-07-02	EXCLUDE		title/abstract	R1: Not robot/embodied navigation (non-robotics use of "navigate", NLP/social-science/LLM-evaluation topic)
2507.02592	P1	P1	WebSailor: Navigating Super-human Reasoning for Web Agent	Kuan Li et al.	2025-07-03	EXCLUDE		title/abstract	R1: Not robot/embodied navigation (non-robotics use of "navigate", NLP/social-science/LLM-evaluation topic)
2507.04686	P2	P2	MOSU: Autonomous Long-range Robot Navigation with Multi-modal Scene Understanding	Jing Liang et al.	2025-07-07	INCLUDE	Liang2025MOSU	full text	included after full-text screening
2507.11001	P2	P2+P3	Learning to Tune Like an Expert: Interpretable and Scene-Aware Navigation via MLLM Reasoning and CVA	Yanbo Wang et al.	2025-07-15	INCLUDE	Wang2026LENav	full text	included after full-text screening
2507.11334	P2	P2	CogDDN: A Cognitive Demand-Driven Navigation with Decision Optimization and Dual-Process Thinking	Yuehao Huang et al.	2025-07-15	EXCLUDE		title/abstract	R4: Embodied/robot navigation without an explicit social-navigation capability (fails C)
2507.12273	P3	P3	Next-Gen Museum Guides: Autonomous Navigation and Visitor Interaction with an Agentic Robot	Luca Garello et al.	2025-07-16	EXCLUDE		full text	Museum guide robot: LLM used for visitor Q&A; navigation by SLAM; FM is not used in the navigation method (fails B)
2507.19672	P2	P2	Alignment and Safety in Large Language Models: Safety Mechanisms, Training Paradigms, and Emerging C	Haoran Lu et al.	2025-07-25	EXCLUDE		title/abstract	R1: Not robot/embodied navigation (non-robotics use of "navigate", NLP/social-science/LLM-evaluation topic)
2508.01192	P2	P2	Unified Generation-Refinement Planning: Bridging Guided Flow Matching and Sampling-Based MPC for Soc	Kazuki Mizuta et al.	2025-08-02	EXCLUDE		title/abstract	R5: Social navigation without an FM role in the navigation method (fails B)
2508.02505	P3	P3	Would you let a humanoid play storytelling with your child? A usability study on LLM-powered narrati	Maria Lombardi et al.	2025-08-04	EXCLUDE		title/abstract	R3: Robot manipulation / HRI / perception without mobile-navigation behaviour (fails A)
2508.03138	P1	P1+P2	Language as Cost: Proactive Hazard Mapping using VLM for Robot Navigation	Mintaek Oh et al.	2025-08-05	EXCLUDE		full text	Language as Cost: VLM hazard mapping for navigation; hazards not social (fails C); adjacent
2508.05129	P1	P1	Navigating Through Paper Flood: Advancing LLM-based Paper Evaluation through Domain-Aware Retrieval	Wuqiang Zheng et al.	2025-08-07	EXCLUDE		title/abstract	R1: Not robot/embodied navigation (non-robotics use of "navigate", NLP/social-science/LLM-evaluation topic)
2508.05646	P2	P2	A Humanoid Social Robot as a Teaching Assistant in the Classroom	Thomas Sievers	2025-07-25	EXCLUDE		title/abstract	R3: Robot manipulation / HRI / perception without mobile-navigation behaviour (fails A)
2508.09036	P1	P1	Can We Trust AI to Govern AI? Benchmarking LLM Performance on Privacy and AI Governance Exams	Zane Witherspoon et al.	2025-08-12	EXCLUDE		title/abstract	R1: Not robot/embodied navigation (non-robotics use of "navigate", NLP/social-science/LLM-evaluation topic)
2508.10287	P1	P1	JRDB-Reasoning: A Difficulty-Graded Benchmark for Visual Reasoning in Robotics	Simindokht Jahangard et al.	2025-08-14	EXCLUDE		title/abstract	R3: Robot manipulation / HRI / perception without mobile-navigation behaviour (fails A)
2508.11093	P2	P2	Utilizing Vision-Language Models as Action Models for Intent Recognition and Assistance	Cesar Alan Contreras et al.	2025-08-14	EXCLUDE		full text	GUIDER-VLM: human-robot collaboration intent inference; RO-MAN workshop; not social navigation (fails A/C)
2508.12087	P1	P1	MAPF-World: Action World Model for Multi-Agent Path Finding	Zhanjiang Yang et al.	2025-08-16	EXCLUDE		title/abstract	R4: Embodied/robot navigation without an explicit social-navigation capability (fails C)
2508.19004	P2	P2	AI Models Exceed Individual Human Accuracy in Predicting Everyday Social Norms	Pontus Strimling et al.	2025-08-26	EXCLUDE		title/abstract	R1: Not robot/embodied navigation (non-robotics use of "navigate", NLP/social-science/LLM-evaluation topic)
2509.01035	P2	P2+P3	We Politely Insist: Your LLM Must Learn the Persian Art of Taarof	Nikta Gohari Sadr et al.	2025-09-01	EXCLUDE		title/abstract	R1: Not robot/embodied navigation (non-robotics use of "navigate", NLP/social-science/LLM-evaluation topic)
2509.02425	P2	P2	OpenGuide: Assistive Object Retrieval in Indoor Spaces for Individuals with Visual Impairments	Yifan Xu et al.	2025-09-02	EXCLUDE		title/abstract	R6: Navigation assistance for people (the human navigates, not a robot) (fails A)
2509.07819	P2	P2	LLMs in Wikipedia: Investigating How LLMs Impact Participation in Knowledge Communities	Moyan Zhou et al.	2025-09-09	EXCLUDE		title/abstract	R1: Not robot/embodied navigation (non-robotics use of "navigate", NLP/social-science/LLM-evaluation topic)
2509.08757	P1	P1+P2+P3	SocialNav-SUB: Benchmarking VLMs for Scene Understanding in Social Robot Navigation	Michael J. Munje et al.	2025-09-10	INCLUDE	Munje2025SocialNavSUB	full text	already in the seed corpus
2509.10317	P3	P3	Robot guide with multi-agent control and automatic scenario generation with LLM	Elizaveta D. Moskovskaya et al.	2025-09-12	EXCLUDE		full text	Tour-guide robot with LLM-generated storytelling scenarios; no navigation behaviour evaluated (fails A/C)
2509.11648	P2	P2	EthicsMH: A Pilot Benchmark for Ethical Reasoning in Mental Health AI	Sai Kartheek Reddy Kasu	2025-09-15	EXCLUDE		title/abstract	R1: Not robot/embodied navigation (non-robotics use of "navigate", NLP/social-science/LLM-evaluation topic)
2509.14063	P2	P2	Language Conditioning Improves Accuracy of Aircraft Goal Prediction in Non-Towered Airspace	Sundhar Vinodh Sangeetha et al.	2025-09-17	EXCLUDE		title/abstract	R2: Autonomous driving, UAV or other non-ground-robot domain

Candidate id (arXiv id or key)	First found (search pass P1-P3, web, citation)	Found in passes	Title	First author	Date	Decision	Corpus entry (cite key)	Screening level	Reason
2509.17941	P1	P1	ComposableNav: Instruction-Following Navigation in Dynamic Environments via Composable Diffusion	Zichao Hu et al.	2025-09-22	BACKGROUND ONLY		full text	ComposableNav (CoRL 2025): social instruction primitives via diffusion; FM only assumed upstream, not used (fails B); compares with VLM soci
2509.21189	P1	P1	Human-like Navigation in a World Built for Humans	Bhargav Chandaka et al.	2025-09-25	EXCLUDE		full text	ReasonNav (“Human-like navigation in a world built for humans”): VLM reads signs/asks for directions; no social-compliance capability evalua
2509.22537	P2	P2	The Emergence of Altruism in Large-Language-Model Agents Society	Haoyang Li et al.	2025-09-26	EXCLUDE		title/abstract	R1: Not robot/embodied navigation (non-robotics use of “navigate”, NLP/social-science/LLM-evaluation topic)
2510.00339	P3	P3	Navigating the Synchrony-Stability Frontier in Adaptive Chatbots	T. James Brandt	2025-09-30	EXCLUDE		title/abstract	R1: Not robot/embodied navigation (non-robotics use of “navigate”, NLP/social-science/LLM-evaluation topic)
2510.01049	P2	P2	KeySG: Hierarchical Keyframe-Based 3D Scene Graphs	Abdelrhman Werby et al.	2025-10-01	EXCLUDE		title/abstract	R4: Embodied/robot navigation without an explicit social-navigation capability (fails C)
2510.01642	P1	P1	FailSafe: Reasoning and Recovery from Failures in Vision-Language-Action Models	Zijun Lin et al.	2025-10-02	EXCLUDE		title/abstract	R3: Robot manipulation / HRI / perception without mobile-navigation behaviour (fails A)
2510.19687	P2	P2	Are Large Language Models Sensitive to the Motives Behind Communication?	Addison J. Wu et al.	2025-10-22	EXCLUDE		title/abstract	R1: Not robot/embodied navigation (non-robotics use of “navigate”, NLP/social-science/LLM-evaluation topic)
2510.21860	P1	P1	Butter-Bench: Evaluating LLM Controlled Robots for Practical Intelligence	Callum Sharrock et al.	2025-10-23	EXCLUDE		title/abstract	R4: Embodied/robot navigation without an explicit social-navigation capability (fails C)
2510.23509	P1	P1+P2+P3	Logic-Guided Socially-aware Robot Navigation World Model	Weizheng Wang et al.	2025-10-27	EXCLUDE (WITHDRAWN)		full text	NaviWM: withdrawn by authors on arXiv; recorded separately as withdrawn; not counted
2510.24594	P3	P3	Detecting the Use of Generative AI in Crowdsourced Surveys: Implications for Data Integrity	Dapeng Zhang et al.	2025-10-28	EXCLUDE		title/abstract	R1: Not robot/embodied navigation (non-robotics use of “navigate”, NLP/social-science/LLM-evaluation topic)
2510.25224	P2	P2	ProMediate: A Socio-cognitive framework for evaluating proactive agents in multi-party negotiation	Ziyi Liu et al.	2025-10-29	EXCLUDE		title/abstract	R1: Not robot/embodied navigation (non-robotics use of “navigate”, NLP/social-science/LLM-evaluation topic)
2510.27033	P1	P1	A Multi-Modal Neuro-Symbolic Approach for Spatial Reasoning-Based Visual Grounding in Robotics	Simindokht Jahangard et al.	2025-10-30	EXCLUDE		title/abstract	R3: Robot manipulation / HRI / perception without mobile-navigation behaviour (fails A)
2511.03676	P2	P2	Unconscious and Intentional Human Motion Cues for Expressive Robot-Arm Motion Design	Taito Tashiro et al.	2025-11-05	EXCLUDE		title/abstract	R3: Robot manipulation / HRI / perception without mobile-navigation behaviour (fails A)
2511.08639	P2	P2+P3	The Journal of Prompt-Engineered (Moral) Philosophy Or: Why AI-Assisted Ethics Research Requires Pro	Michele Loi	2025-11-10	EXCLUDE		title/abstract	R1: Not robot/embodied navigation (non-robotics use of “navigate”, NLP/social-science/LLM-evaluation topic)
2511.10810	P1	P1	HARNESS: Human-Agent Risk Navigation and Event Safety System for Proactive Hazard Forecasting in Hig	Ran Elgedawy et al.	2025-11-13	EXCLUDE		title/abstract	R1: Not robot/embodied navigation (non-robotics use of “navigate”, NLP/social-science/LLM-evaluation topic)
2511.12232	citation	citation	SocialNav-Map: Dynamic Mapping with Human Trajectory Prediction for Zero-Shot Social Navigation	Not recorded	2025	EXCLUDE		full text	SocialNav-Map: dynamic mapping + trajectory prediction for zero-shot social navigation; no FM in the method (fails B)
2511.13524	P1	P1	FreeAskWorld: An Interactive and Closed-Loop Simulator for Human-Centric Embodied AI	Yuhang Peng et al.	2025-11-17	EXCLUDE		full text	FreeAskWorld: interactive human-centric simulator; no robot social-navigation capability evaluated (fails C)
2511.17581	P2	P2	EgoCogNav: Cognition-aware Human Egocentric Navigation	Zhiwen Qiu et al.	2025-11-15	EXCLUDE		title/abstract	R6: Navigation assistance for people (the human navigates, not a robot) (fails A)
2511.21135	P1	P1+P2	SocialNav: Training Human-Inspired Foundation Model for Socially-Aware Embodied Navigation	Ziyi Chen et al.	2025-11-26	INCLUDE	Chen2026SocialNavCVPR	full text	already in the seed corpus
2512.01078	P2	P2	SimWorld: An Open-ended Realistic Simulator for Autonomous Agents in Physical and Social Worlds	Jiawei Ren et al.	2025-11-30	EXCLUDE		title/abstract	R4: Embodied/robot navigation without an explicit social-navigation capability (fails C)
2512.07177	P1	P1+P2	Using Vision-Language Models as Proxies for Social Intelligence in Human-Robot Interaction	Fanjun Bu et al.	2025-12-08	EXCLUDE		title/abstract	R3: Robot manipulation / HRI / perception without mobile-navigation behaviour (fails A)
2512.09920	P1	P1+P2+P3	LISN: Language-Instructed Social Navigation with VLM-based Controller Modulating	Junting Chen et al.	2025-12-10	INCLUDE	Chen2026LISN	full text	included after full-text screening
2512.10046	P1	P1	SimWorld-Robotics: Synthesizing Photorealistic and Dynamic Urban Environments for Multimodal Robot N	Yan Zhuang et al.	2025-12-10	EXCLUDE		full text	SimWorld-Robotics (NeurIPS 2025): urban simulator with pedestrians; FM-agnostic benchmark without social-navigation metrics (fails B/C)
2512.12302	P2	P2	From Human Intention to Action Prediction: Intention-Driven End-to-End Autonomous Driving	Huan Zheng et al.	2025-12-13	EXCLUDE		title/abstract	R2: Autonomous driving, UAV or other non-ground-robot domain
2512.14757	P1	P1+P2+P3	SocialNav-MoE: A Mixture-of-Experts Vision Language Model for Socially Compliant Navigation with Rei	Tomohito Kawabata et al.	2025-12-15	INCLUDE	Kawabata2025SocialNavMoE	full text	already in the seed corpus
2512.16019	P1	P1+P3	Few-Shot Inference of Human Perceptions of Robot Performance in Social Navigation Scenarios	Qiping Zhang et al.	2025-12-17	INCLUDE	Zhang2025FewShotPerception	full text	included after full-text screening
2512.16755	P1	P1	CitySeeker: How Do VLMS Explore Embodied Urban Navigation With Implicit Human Needs?	Siqi Wang et al.	2025-12-18	EXCLUDE		title/abstract	R4: Embodied/robot navigation without an explicit social-navigation capability (fails C)
2512.18582	P2	P2	Wireless Copilot: An AI-Powered Partner for Navigating Next-Generation Wireless Complexity	Haoxiang Luo et al.	2025-12-21	EXCLUDE		title/abstract	R1: Not robot/embodied navigation (non-robotics use of “navigate”, NLP/social-science/LLM-evaluation topic)
2512.19992	P2	P2	S^{3}IT: A Benchmark for Spatially Situated Social Intelligence Test	Zhe Sun et al.	2025-12-23	EXCLUDE		title/abstract	R1: Not robot/embodied navigation (non-robotics use of “navigate”, NLP/social-science/LLM-evaluation topic)
2512.20206	P2	P2	TongSIM: A General Platform for Simulating Intelligent Machines	Zhe Sun et al.	2025-12-23	EXCLUDE		title/abstract	R4: Embodied/robot navigation without an explicit social-navigation capability (fails C)
2512.21055	P3	P3	Making AI Functional with Workarounds: An Insider’s Account of Invisible Labour in Organisational Po	Shang Chieh Lee et al.	2025-12-24	EXCLUDE		title/abstract	R1: Not robot/embodied navigation (non-robotics use of “navigate”, NLP/social-science/LLM-evaluation topic)
2512.21722	P1	P1+P2	MAction-SocialNav: Multi-Action Socially Compliant Navigation via Reasoning-enhanced Prompt Tuning	Zishuo Wang et al.	2025-12-25	INCLUDE	Wang2026MActionSocialNav	full text	already in the seed corpus
2512.22867	P1	P1+P2+P3	MUSON: A Reasoning-oriented Multimodal Dataset for Socially Compliant Navigation in Urban Environmen	Zhuonan Liu et al.	2025-12-28	INCLUDE	Liu2026MUSON	full text	already in the seed corpus
2601.00978	P2	P2	From Perception to Symbolic Task Planning: Vision-Language Guided Human-Robot Collaborative Structur	Yanyi Chen et al.	2026-01-02	EXCLUDE		title/abstract	R3: Robot manipulation / HRI / perception without mobile-navigation behaviour (fails A)
2601.01546	P1	P1	Improving Behavioral Alignment in LLM Social Simulations via Context Formation and Navigation	Letian Kong et al.	2026-01-04	EXCLUDE		title/abstract	R1: Not robot/embodied navigation (non-robotics use of “navigate”, NLP/social-science/LLM-evaluation topic)
2601.05904	P3	P3	Can AI mediation improve democratic deliberation?	Michael Henry Tessler et al.	2026-01-09	EXCLUDE		title/abstract	R1: Not robot/embodied navigation (non-robotics use of “navigate”, NLP/social-science/LLM-evaluation topic)
2601.13809	P1	P1+P2	DroneVLA: VLA based Aerial Manipulation	Fawad Mehboob et al.	2026-01-20	EXCLUDE		title/abstract	R2: Autonomous driving, UAV or other non-ground-robot domain
2601.14622	P1	P1	Probing Prompt Design for Socially Compliant Robot Navigation with Vision Language Models	Ling Xiao et al.	2026-01-21	RELATED VERSION	Xiao2026PromptingVLMsJSAI	full text	Probing Prompt Design: related, differently titled version of the JSAI 2026 corpus paper (same authors/topic); not a separate work

Candidate id (arXiv id or key)	First found (search pass P1-P3, web, citation)	Found in passes	Title	First author	Date	Decision	Corpus entry (cite key)	Screening level	Reason
2602.01959	P1	P1	Boosting metacognition in entangled human-AI interaction to navigate cognitive-behavioral drift	Ezequiel Lopez-Lopez et al.	2026-02-02	EXCLUDE		title/abstract	R1: Not robot/embodied navigation (non-robotics use of "navigate", NLP/social-science/LLM-evaluation topic)
2602.02220	P1	P1	LangMap: A Human-Verified Benchmark for Hierarchical Open-Vocabulary Goal Navigation	Bo Miao et al.	2026-02-02	EXCLUDE		title/abstract	R4: Embodied/robot navigation without an explicit social-navigation capability (fails C)
2602.02459	P1	P1+P2	TIC-VLA: A Think-in-Control Vision-Language-Action Model for Robot Navigation in Dynamic Environment	Zhiyu Huang et al.	2026-02-02	EXCLUDE		full text	TIC-VLA: navigation VLA in dynamic environments; no explicit social-navigation evaluation (fails C); adjacent
2602.04456	P2	P2	Growth First, Care Second? Tracing the Landscape of LLM Value Preferences in Everyday Dilemmas	Zhiyi Chen et al.	2026-02-04	EXCLUDE		title/abstract	R1: Not robot/embodied navigation (non-robotics use of "navigate", NLP/social-science/LLM-evaluation topic)
2602.09002	P1	P1+P2+P3	From Obstacles to Etiquette: Robot Social Navigation with VLM-Informed Path Selection	Zilin Fang et al.	2026-02-09	INCLUDE	Fang2026ObstaclestoEtiquette	full text	already in the seed corpus
2602.09472	P2	P2	LLM-Grounded Dynamic Task Planning with Hierarchical Temporal Logic for Human-Aware Multi-Robot Hand	Shuyuan Hu et al.	2026-02-10	EXCLUDE		title/abstract	R3: Robot manipulation / HRI / perception without mobile-navigation behaviour (fails A)
2602.10910	P1	P1	Safe mobility support system using crowd mapping and avoidance route planning using VLM	Sena Saito et al.	2026-02-11	INCLUDE	Saito2025VLMCrowdMapping	full text	included after full-text screening
2602.12597	P1	P1+P2+P3	PISHYAR: A Socially Intelligent Smart Cane for Indoor Social Navigation and Multimodal Human-Robot I	Mahdi Haghighat Joo et al.	2026-02-13	EXCLUDE		full text	PISHYAR: smart cane (human-carried device), not a mobile robot (fails A)
2602.15864	P1	P1	ReasonNavi: Human-Inspired Global Map Reasoning for Zero-Shot Embodied Navigation	Yuzhuo Ao et al.	2026-01-26	EXCLUDE		title/abstract	R4: Embodied/robot navigation without an explicit social-navigation capability (fails C)
2602.17734	P1	P1	Five Fatal Assumptions: Why T-Shirt Sizing Systematically Fails for AI Projects	Raja Soundaramourty et al.	2026-02-18	EXCLUDE		title/abstract	R1: Not robot/embodied navigation (non-robotics use of "navigate", NLP/social-science/LLM-evaluation topic)
2602.18606	P1	P1	OVerSeeC: Open-Vocabulary Costmap Generation from Satellite Images and Natural Language	Rwik Rana et al.	2026-02-20	EXCLUDE		title/abstract	R4: Embodied/robot navigation without an explicit social-navigation capability (fails C)
2602.18710	P2	P2	Many AI Analysts, One Dataset: Navigating the Agentic Data Science Multiverse	Martin Bertran et al.	2026-02-21	EXCLUDE		title/abstract	R1: Not robot/embodied navigation (non-robotics use of "navigate", NLP/social-science/LLM-evaluation topic)
2602.19983	P1	P1	Contextual Safety Reasoning and Grounding for Open-World Robots	Zachary Ravichandran et al.	2026-02-23	BACKGROUND ONLY		full text	CORE: VLM contextual safety constraints enforced by CBFs; social restriction is one sub-case among hazards; primary contribution is contextu
2602.23400	P2	P2	U-CAN: Utility-Aware Contrastive Attenuation for Efficient Unlearning in Generative Recommendation	Zezheng Wu et al.	2026-02-26	EXCLUDE		title/abstract	R1: Not robot/embodied navigation (non-robotics use of "navigate", NLP/social-science/LLM-evaluation topic)
2603.03942	P2	P2	Lightweight Visual Reasoning for Socially-Aware Robots	Alessio Galatolo et al.	2026-03-04	EXCLUDE		full text	Lightweight Visual Reasoning for Socially-Aware Robots: generic VLM reasoning module, no navigation evaluation (fails A/C)
2603.04217	P3	P3	When Do Language Models Endorse Limitations on Human Rights Principles?	Keenan Samway et al.	2026-03-04	EXCLUDE		title/abstract	R1: Not robot/embodied navigation (non-robotics use of "navigate", NLP/social-science/LLM-evaluation topic)
2603.10703	P1	P1	WalkGPT: Grounded Vision-Language Conversation with Depth-Aware Segmentation for Pedestrian Navigati	Rafi Ibn Sultan et al.	2026-03-11	EXCLUDE		title/abstract	R6: Navigation assistance for people (the human navigates, not a robot) (fails A)
2603.11290	P2	P2	A Causal Approach to Predicting and Improving Human Perceptions of Social Navigation Robots	Maximilian Diehl et al.	2026-03-11	EXCLUDE		full text	Causal approach to predicting human perceptions of social-navigation robots: no FM (fails B); background for EXP1
2603.11447	P1	P1+P2+P3	Enhancing Lightweight Vision Language Models through Group Competitive Learning for Socially Complia	Xinyu Zhang et al.	2026-03-12	INCLUDE	Zhang2026GCL	full text	already in the seed corpus
2603.12696	P1	P1	HaltNav: Reactive Visual Halting over Lightweight Topological Priors for Robust Vision-Language Navi	Zihui Yu et al.	2026-03-13	EXCLUDE		title/abstract	R4: Embodied/robot navigation without an explicit social-navigation capability (fails C)
2603.13698	P1	P1+P2+P3	SAATT Nav: a Socially Aware Autonomous Transparent Transportation Navigation Framework for Wheelchai	Yutong Zhang et al.	2026-03-14	INCLUDE	Zhang2026SAATTNav	full text	included after full-text screening
2603.14341	P1	P1	Generation of Human Comprehensible Access Control Policies from Audit Logs	Gautam Kumar et al.	2026-03-15	EXCLUDE		title/abstract	R1: Not robot/embodied navigation (non-robotics use of "navigate", NLP/social-science/LLM-evaluation topic)
2603.15359	citation	citation	NavThinker: Action-Conditioned World Models for Coupled Prediction and Planning in Social Navigation	T. Hu, Z. Gong, L. Kong et al.	2026	EXCLUDE		full text	NavThinker: world model + RL social navigation; pretrained vision encoder (Depth Anything) only, no language/ multimodal reasoning (fails B u
2603.15624	P1	P1	Exploring the Use of VLMs for Navigation Assistance for People with Blindness and Low Vision	Yu Li et al.	2026-01-26	EXCLUDE		title/abstract	R6: Navigation assistance for people (the human navigates, not a robot) (fails A)
2603.16013	P1	P1	Safety Case Patterns for VLA-based driving systems: Insights from SimLingo	Gerhard Yu et al.	2026-03-16	EXCLUDE		title/abstract	R2: Autonomous driving, UAV or other non-ground-robot domain
2603.16642	P3	P3	When AI Navigates the Fog of War	Ming Li et al.	2026-03-17	EXCLUDE		title/abstract	R1: Not robot/embodied navigation (non-robotics use of "navigate", NLP/social-science/LLM-evaluation topic)
2603.17510	P1	P1	Interpreting Context-Aware Human Preferences for Multi-Objective Robot Navigation	Tharun Sethuraman et al.	2026-03-18	INCLUDE	Sethuraman2026ContextAwarePreferences	full text	included after full-text screening
2603.19533	P2	P2	Pedestrian Crossing Intent Prediction via Psychological Features and Transformer Fusion	Sima Ashayer et al.	2026-03-20	EXCLUDE		title/abstract	R2: Autonomous driving, UAV or other non-ground-robot domain
2603.20231	P2	P2	Moral Mazes in the Era of LLMs	Dang Nguyen et al.	2026-03-06	EXCLUDE		title/abstract	R1: Not robot/embodied navigation (non-robotics use of "navigate", NLP/social-science/LLM-evaluation topic)
2603.20664	P1	P1	E-SocialNav: Efficient Socially Compliant Navigation with Language Models	Ling Xiao et al.	2026-03-21	INCLUDE	Xiao2026ESocialNav	full text	already in the seed corpus
2603.25422	P1	P1	Navigating the Prompt Space: Improving LLM Classification of Social Science Texts Through Prompt Eng	Erkan Gunes et al.	2026-03-26	EXCLUDE		title/abstract	R1: Not robot/embodied navigation (non-robotics use of "navigate", NLP/social-science/LLM-evaluation topic)
2603.26898	P3	P3	Magic Words or Methodical Work? Challenging Conventional Wisdom in LLM-Based Political Text Annotati	Lorcan McLaren et al.	2026-03-27	EXCLUDE		title/abstract	R1: Not robot/embodied navigation (non-robotics use of "navigate", NLP/social-science/LLM-evaluation topic)
2603.28032	P1	P1	CARLA-Air: Fly Drones Inside a CARLA World – A Unified Infrastructure for Air-Ground Embodied Intel	Tianle Zeng et al.	2026-03-30	EXCLUDE		title/abstract	R2: Autonomous driving, UAV or other non-ground-robot domain
2603.28890	P1	P1	Bootstrap Perception Under Hardware Depth Failure for Indoor Robot Navigation	Nishant Pushparaju et al.	2026-03-30	EXCLUDE		title/abstract	R4: Embodied/robot navigation without an explicit social-navigation capability (fails C)
2604.01742	P3	P3	Dense Point-to-Mask Optimization with Reinforced Point Selection for Crowd Instance Segmentation	Hongru Chen et al.	2026-04-02	EXCLUDE		title/abstract	R1: Not robot/embodied navigation (non-robotics use of "navigate", NLP/social-science/LLM-evaluation topic)
2604.02032	P1	P1	IndoorCrowd: A Multi-Scene Dataset for Human Detection, Segmentation, and Tracking with an Automated	Sebastian-Ion Nae et al.	2026-04-02	EXCLUDE		title/abstract	R3: Robot manipulation / HRI / perception without mobile-navigation behaviour (fails A)
2604.05525	P1	P1+P2	CrowdVLA: Embodied Vision-Language-Action Agents for Context-Aware Crowd Simulation	Juyeong Hwang et al.	2026-04-07	BACKGROUND ONLY		full text	CrowdVLA: VLA agents for crowd simulation (pedestrians, not a robot) (fails A); relevant to simulated-human realism
2604.07502	P1	P1	Beyond Human-Readable: Rethinking Software Engineering Conventions for the Agentic Development Era	Dmytro Ustynov	2026-04-08	EXCLUDE		title/abstract	R1: Not robot/embodied navigation (non-robotics use of "navigate", NLP/social-science/LLM-evaluation topic)
2604.07973	P1	P1	How Far Are Large Multimodal Models from Human-Level Spatial Action? A Benchmark for Goal-Oriented E	Baining Zhao et al.	2026-04-09	EXCLUDE		title/abstract	R4: Embodied/robot navigation without an explicit social-navigation capability (fails C)

Candidate id (arXiv id or key)	First found (search pass P1-P3, web, citation)	Found in passes	Title	First author	Date	Decision	Corpus entry (cite key)	Screening level	Reason
2604.11721	P2	P2	Evaluating Cooperation in LLM Social Groups through Elected Leadership	Ryan Faulkner et al.	2026-04-13	EXCLUDE		title/abstract	R1: Not robot/embodied navigation (non-robotics use of "navigate", NLP/social-science/LLM-evaluation topic)
2604.16022	P2	P2	SocialGrid: A Benchmark for Planning and Social Reasoning in Embodied Multi-Agent Systems	Hikaru Shindo et al.	2026-04-17	EXCLUDE		title/abstract	R1: Not robot/embodied navigation (non-robotics use of "navigate", NLP/social-science/LLM-evaluation topic)
2604.19522	P1	P1+P2	GenerativeMPC: VLM-RAG-guided Whole-Body MPC with Virtual Impedance for Bimanual Mobile Manipulation	Marcelino Julio Fernando et al.	2026-04-21	EXCLUDE		title/abstract	R3: Robot manipulation / HRI / perception without mobile-navigation behaviour (fails A)
2604.23970	P1	P1	LLM-Guided Agentic Floor Plan Parsing for Accessible Indoor Navigation of Blind and Low-Vision Peopl	Aydin Ayanzadeh et al.	2026-04-27	EXCLUDE		title/abstract	R6: Navigation assistance for people (the human navigates, not a robot) (fails A)
2604.26839	P1	P1+P2+P3	Walk With Me: Long-Horizon Social Navigation for Human-Centric Outdoor Assistance	Lingfeng Zhang et al.	2026-04-29	INCLUDE	Zhang2026WalkWithMe	full text	included after full-text screening
2605.00943	P2	P2	ARIS: Agentic and Relationship Intelligence System for Social Robots	Stavya Datta et al.	2026-05-01	EXCLUDE		title/abstract	R3: Robot manipulation / HRI / perception without mobile-navigation behaviour (fails A)
2605.02759	P2	P2	DynoSLAM: Dynamic SLAM with Generative Graph Neural Networks for Real-World Social Navigation	Danil Tokhchukov et al.	2026-05-04	EXCLUDE		title/abstract	R5: Social navigation without an FM role in the navigation method (fails B)
2605.03287	P2	P2	Attention: What Prevents Young Adults from Speaking Up Against Cyberbullying in an LLM-Powered Socia	Qian Yang et al.	2026-05-05	EXCLUDE		title/abstract	R1: Not robot/embodied navigation (non-robotics use of "navigate", NLP/social-science/LLM-evaluation topic)
2605.06863	P2	P2	Bi3: A Biplatform, Bicultural, Biperson Dataset for Social Robot Navigation	Andrew Stratton et al.	2026-05-07	BACKGROUND ONLY		full text	Bi3: bicultural social-navigation dataset without FM (fails B); relevant to NOR4 cultural adaptation
2605.07496	P2	P2	PathPainter: Transferring the Generalization Ability of Image Generation Models to Embodied Navigati	Yijin Wang et al.	2026-05-08	EXCLUDE		title/abstract	R4: Embodied/robot navigation without an explicit social-navigation capability (fails C)
2605.07550	P1	P1	Mind the Gap: Geometrically Accurate Generative Reconstruction from Disjoint Views	Grzegorz Wilczynski et al.	2026-05-08	EXCLUDE		title/abstract	R1: Not robot/embodied navigation (non-robotics use of "navigate", NLP/social-science/LLM-evaluation topic)
2605.11782	P1	P1+P3	Urban Risk-Aware Navigation via VQA-Based Event Maps for People with Low Vision	Antoni Valls et al.	2026-05-12	EXCLUDE		title/abstract	R6: Navigation assistance for people (the human navigates, not a robot) (fails A)
2605.11859	P1	P1	EvoNav: Evolutionary Reward Function Design for Robot Navigation with Large Language Models	Zhikai Zhao et al.	2026-05-12	INCLUDE	Zhao2026EvoNav	full text	included after full-text screening
2605.13321	P1	P1+P2	HCSG: Human-Centric Semantic-Geometric Reasoning for Vision-Language Navigation	Haoxuan Xu et al.	2026-05-13	INCLUDE	Xu2026HCSG	full text	already in the seed corpus
2605.13706	P1	P1	Identifying AI Web Scrapers Using Canary Tokens	Steven Seiden et al.	2026-05-13	EXCLUDE		title/abstract	R1: Not robot/embodied navigation (non-robotics use of "navigate", NLP/social-science/LLM-evaluation topic)
2605.16979	P1	P1+P2	NORM-Nav: Zero-Shot Mobile Robot Navigation with Natural Language Behavioral Constraints	Dongjie Huo et al.	2026-05-16	INCLUDE	Huo2026NORMNav	full text	included after full-text screening
2605.17364	P3	P3	NewsLens: A Multi-Agent Framework for Adversarial News Bias Navigation	Joy Bose	2026-05-17	EXCLUDE		title/abstract	R1: Not robot/embodied navigation (non-robotics use of "navigate", NLP/social-science/LLM-evaluation topic)
2605.23187	P1	P1+P2	IntentionNav: A Benchmark for Intent-Driven Object Navigation from Implicit Human Instruction	Lin Qian et al.	2026-05-22	EXCLUDE		title/abstract	R4: Embodied/robot navigation without an explicit social-navigation capability (fails C)
2605.24319	P3	P3	Omissive Bias in Religious Representation: Benchmarking LLM Answers to Everyday Ethical Decision-mak	David Wingate et al.	2026-05-23	EXCLUDE		title/abstract	R1: Not robot/embodied navigation (non-robotics use of "navigate", NLP/social-science/LLM-evaluation topic)
2605.24562	P2	P2	PEDESTRIANQA: A Benchmark for Vision-Language Models on Pedestrian Intention and Trajectory Predicti	Naman Mishra et al.	2026-05-23	EXCLUDE		full text	PedestrianQA: VLM pedestrian-intention benchmark for autonomous driving (fails A)
2605.31196	P1	P1+P3	Probing Collision Grounding in Vision-Language Models for Safe Human-Robot Collaboration	Jun Wang et al.	2026-05-29	EXCLUDE		title/abstract	R3: Robot manipulation / HRI / perception without mobile-navigation behaviour (fails A)
2605.31574	P1	P1	Can Generative AI help people navigate Radical Moral Disagreements? The CONSIDER prototype	William Hohnen-Ford et al.	2026-05-29	EXCLUDE		title/abstract	R1: Not robot/embodied navigation (non-robotics use of "navigate", NLP/social-science/LLM-evaluation topic)
2606.02313	P2	P2	Towards Precise Intent-Aligned VLA Aerial Navigation via Expert-Guided GRPO	Tianyang Chen et al.	2026-06-01	EXCLUDE		title/abstract	R2: Autonomous driving, UAV or other non-ground-robot domain
2606.02775	P1	P1	AURA: Action-Gated Memory for Robot Policies at Constant VRAM	Josef Chen	2026-06-01	EXCLUDE		title/abstract	R3: Robot manipulation / HRI / perception without mobile-navigation behaviour (fails A)
2606.05896	P2	P2	Resonant Minds: Closed-Loop Social Avatars with Theory of Mind	Jianxu Shangguan et al.	2026-06-04	EXCLUDE		title/abstract	R1: Not robot/embodied navigation (non-robotics use of "navigate", NLP/social-science/LLM-evaluation topic)
2606.08029	P1	P1+P2	IntentNav: Learning Spatial-Visual Object Navigation from Human Demonstrations	Yuxin Cai et al.	2026-06-06	EXCLUDE		title/abstract	R4: Embodied/robot navigation without an explicit social-navigation capability (fails C)
2606.08792	P3	P3	The Amplifying Mirror: Locating and Steering the Partisan Direction inside a Large Language Model	Wendy K. Tam	2026-06-07	EXCLUDE		title/abstract	R1: Not robot/embodied navigation (non-robotics use of "navigate", NLP/social-science/LLM-evaluation topic)
2606.09024	P3	P3	Personal Salience: Highlighting Is Social, but Individuality Lives in Selection	Kazuki Nakayashiki et al.	2026-06-08	EXCLUDE		title/abstract	R1: Not robot/embodied navigation (non-robotics use of "navigate", NLP/social-science/LLM-evaluation topic)
2606.10495	P1	P1	Act on What You See: Unlocking Safe Social Navigation in Vision-Language-Action Models	Qingzi Wang et al.	2026-06-09	INCLUDE	Wang2026SALSA	full text	included after full-text screening
2606.13397	P2	P2	Mod-Guide: An LLM-based Content Moderation Feedback System to Address Insensitive Speech toward Indi	Dipto Das et al.	2026-06-11	EXCLUDE		title/abstract	R1: Not robot/embodied navigation (non-robotics use of "navigate", NLP/social-science/LLM-evaluation topic)
2606.17577	P3	P3	Surrogate Assisted Pedestrian Protection Design via a Foundation Model Orchestrated Workflow	Osamu Ito et al.	2026-06-16	EXCLUDE		title/abstract	R2: Autonomous driving, UAV or other non-ground-robot domain
2606.17639	P2	P2	ERQA-Plus: A Diagnostic Benchmark for Reasoning in Embodied AI	Hong Yang et al.	2026-06-16	EXCLUDE		title/abstract	R4: Embodied/robot navigation without an explicit social-navigation capability (fails C)
2606.20458	P1	P1	Slow Brain, Fast Planner: Latency-Resilient VLM-Augmented Urban Navigation	Zhenghao "Mark" Peng et al.	2026-06-18	EXCLUDE		full text	Slow Brain, Fast Planner: latency-resilient VLM urban navigation; no explicit social evaluation (fails C); adjacent
2606.23339	P1	P1	When Robots Rate Their Own Interactions: Engagement Validity and the Strangeness Failure	Victor Lockwood et al.	2026-06-22	EXCLUDE		title/abstract	R3: Robot manipulation / HRI / perception without mobile-navigation behaviour (fails A)
2606.23633	P3	P3	AI Exposure Scores: what they measure, what they miss, and what comes next	Campbell Lund et al.	2026-06-22	EXCLUDE		title/abstract	R1: Not robot/embodied navigation (non-robotics use of "navigate", NLP/social-science/LLM-evaluation topic)
2606.28760	P1	P1+P2	Vision-Language Models for Deployable Social Robot Navigation: Bridging Semantic Reasoning and Low-L	Runji Cai et al.	2026-06-27	INCLUDE	Cai2026VLMDeployableSRNSurvey	full text	already in the seed corpus
2606.30404	P1	P1	HUMEMBR: Learning Human Routines for Predictive Embodied Navigation	Samira Huber et al.	2026-06-29	EXCLUDE		full text	HUMEMBR: routine memory for person search/EQA; finding people is not socially compliant navigation (fails C)
2607.01044	P1	P1+P2+P3	Robots Ask the Way: Communication-Enabled Social Navigation	Valentino Sacco et al.	2026-07-01	EXCLUDE		full text	CommNav ("Robots Ask the Way"): communication-enabled social navigation; the proposed COMM model uses structured sighting data without an FM
2607.06724	P1	P1+P2+P3	EvoPlan: Evolutionary Neuro-Symbolic Robot Planning with Spatio-Temporal Guarantees	Bhavya Sai Nukapotula et al.	2026-07-07	EXCLUDE		title/abstract	R4: Embodied/robot navigation without an explicit social-navigation capability (fails C)
2607.09261	P1	P1+P2	Validating Virtual Reality for Studying Multimodal Human-Robot Interaction in Socially Aware Robot N	Hariharan Arunachalam et al.	2026-07-10	EXCLUDE		full text	VR validation of multimodal HRI in socially aware navigation: methodological user study, no FM (fails B)

Candidate id (arXiv id or key)	First found (search pass P1-P3, web, citation)	Found in passes	Title	First author	Date	Decision	Corpus entry (cite key)	Screening level	Reason
2607.10991	P1	P1+P2+P3	Think When It Matters: Conditional VLM Reasoning for Social Navigation with RL Policies	Ali Ahmadi et al.	2026-07-13	INCLUDE	Ahmadi2026HUMA	full text	included after full-text screening
2607.12195	P1	P1	Comparing Semantic Navigation in Humans and Large Language Models using Natural Language Processing	Gabriel Paris-Colombo et al.	2026-07-13	EXCLUDE		title/abstract	R1: Not robot/embodied navigation (non-robotics use of “navigate”, NLP/social-science/LLM-evaluation topic)
2607.16806	P1	P1+P2+P3	Token-Wise Latent Streaming from Slow Reasoners to Fast Planners for Dynamic Vision Language Navigat	Tianshuai Hu et al.	2026-07-18	INCLUDE	Hu2026SPARKVLN	full text	included after full-text screening
2607.16938	P1	P1+P3	What Do They See? Interpreting Complex Road Scenarios Through the Eyes of Vision-Language-Action Mod	Kalpana Panda et al.	2026-07-18	EXCLUDE		title/abstract	R2: Autonomous driving, UAV or other non-ground-robot domain
2607.16956	P1	P1+P2+P3	G2-Nav: Grounded and Guarded Vision-Language Costmaps for Robot Social Navigation	Yuwen Liao et al.	2026-07-18	INCLUDE	Liao2026G2Nav	full text	already in the seed corpus
2607.17394	P1	P1	TaskArtisan: Designing Composable Generative Widgets for LLM-Assisted Analysis	Meng Chen et al.	2026-07-19	EXCLUDE		title/abstract	R1: Not robot/embodied navigation (non-robotics use of “navigate”, NLP/social-science/LLM-evaluation topic)
2607.19850	P1	P1+P2+P3	SOPD-SocialNav: Selective On-Policy Distillation for Vision-Language Social Navigation	Xinyu Zhang et al.	2026-07-22	INCLUDE	Zhang2026SOPDSocialNav	full text	already in the seed corpus
2607.23797	P2	P2	Memory for Attention: Language-Conditioned Re-Perception with a Vision--Language--Motion Map	Dibyendu Ghosh	2026-07-26	EXCLUDE		title/abstract	R4: Embodied/robot navigation without an explicit social-navigation capability (fails C)
2607.25219	P1	P1+P2+P3	SONG: A Photorealistic 3D Gaussian Simulation Platform for Benchmarking Social Navigation	Weiqi Huang et al.	2026-07-28	INCLUDE	Huang2026SONG	full text	included after full-text screening
2607.25620	P2	P2	Beyond Epistemia: Epistemic Schizologia and Large Language Models as Techno-Semiotic Machines	Federico Cabitza et al.	2026-07-28	EXCLUDE		title/abstract	R1: Not robot/embodied navigation (non-robotics use of “navigate”, NLP/social-science/LLM-evaluation topic)
2607.28094	P2	P2	An Instrument to Evaluate Governance Proposals: AI Policy Analysis at Scale	Paulo Carvao et al.	2026-07-30	EXCLUDE		title/abstract	R1: Not robot/embodied navigation (non-robotics use of “navigate”, NLP/social-science/LLM-evaluation topic)
2608.00652	P1	P1+P2	Assistant Placement Aria: A Benchmark for Egocentric Placement Assistance	Amir Belder et al.	2026-08-01	EXCLUDE		title/abstract	R3: Robot manipulation / HRI / perception without mobile-navigation behaviour (fails A)
2608.01456	P2	P2	Long-Horizon Embodied Decision-Making via Multimodal Memory Compression	Bingxuan Li et al.	2026-08-02	EXCLUDE		title/abstract	R4: Embodied/robot navigation without an explicit social-navigation capability (fails C)
2608.06898	P2	P2	How Should I Pick a Foundation Model for My Robot? In Favor of a Community Evaluation Framework for	Eric Nichols et al.	2026-08-07	EXCLUDE		title/abstract	R3: Robot manipulation / HRI / perception without mobile-navigation behaviour (fails A)
2608.08320	P2	P2	Diminishing Returns of Intelligence: The Non-Linear Relationship Between LLM Scale and User Percepti	Amanda Rasille Røn Volf et al.	2026-08-08	EXCLUDE		title/abstract	R1: Not robot/embodied navigation (non-robotics use of “navigate”, NLP/social-science/LLM-evaluation topic)
2608.09946	P1	P1	HoosierHelp: Benchmarking LLM Agents for Social Service Navigation	Yiyang Li et al.	2026-07-03	EXCLUDE		title/abstract	R1: Not robot/embodied navigation (non-robotics use of “navigate”, NLP/social-science/LLM-evaluation topic)
2608.11973	P1	P1	Sci-Surf: Navigating Scientific Literature Discovery through Human Feedback and Intelligent Summariz	Fang Guo et al.	2026-08-12	EXCLUDE		title/abstract	R1: Not robot/embodied navigation (non-robotics use of “navigate”, NLP/social-science/LLM-evaluation topic)
2608.12515	P1	P1+P2	Can Vision-Language Models Assess Proxemic Risk from Egocentric Robot Images?	Vladyslava Rudas et al.	2026-08-12	EXCLUDE		full text	Proxemic-risk assessment from egocentric robot images with VLMs: ECCV workshop, non-archival; offline perception only
2608.13239	P2	P2	Reasoning for Social Audio-Visual Question Answering: Where Do We Stand?	Koen P. de Vries et al.	2026-08-13	EXCLUDE		title/abstract	R1: Not robot/embodied navigation (non-robotics use of “navigate”, NLP/social-science/LLM-evaluation topic)
2608.22035	P2	P2	Ludi${}_{\{}scriptscriptstyle 0.1}$: An Agentic System for Socially Intelligent Robots	Wooseong Chung et al.	2026-08-22	EXCLUDE		title/abstract	R3: Robot manipulation / HRI / perception without mobile-navigation behaviour (fails A)
2608.27456	P3	P3	UrbanGround: From Local Perception to Spatial Agency in a Real-Scale City	Tianjie Ju et al.	2026-08-27	EXCLUDE		full text	UrbanGround: MLLM urban navigation sandbox; no social-navigation capability (fails C)
2608.28968	P2	P2	Efficient GPU Retrieval for Semantic Search	Dhritiman Das et al.	2026-08-29	EXCLUDE		title/abstract	R1: Not robot/embodied navigation (non-robotics use of “navigate”, NLP/social-science/LLM-evaluation topic)
2608.30428	P2	P2	Lies We Can See: Joint Verbal and Non-Verbal Deception by VLM Agents in Embodied Social Interactions	Jaewoo Ahn et al.	2026-08-31	EXCLUDE		title/abstract	R1: Not robot/embodied navigation (non-robotics use of “navigate”, NLP/social-science/LLM-evaluation topic)
2609.02421	P3	P3	UTP-Bench: Uncertainty-aware Travel Planning Benchmark	Etcharla Revanth Rao et al.	2026-09-02	EXCLUDE		title/abstract	R1: Not robot/embodied navigation (non-robotics use of “navigate”, NLP/social-science/LLM-evaluation topic)
2609.08637	P1	P1+P3	Navigating the digital spectrum: Assessing political bias, stability, and downstream fairness in Lar	Luka Debevc et al.	2026-09-08	EXCLUDE		title/abstract	R1: Not robot/embodied navigation (non-robotics use of “navigate”, NLP/social-science/LLM-evaluation topic)
2609.15972	P2	P2	Mind2Dialogue: Training Human-Aware Language Models by Simulating User Mental States	Zixuan Wang et al.	2026-09-14	EXCLUDE		title/abstract	R1: Not robot/embodied navigation (non-robotics use of “navigate”, NLP/social-science/LLM-evaluation topic)
2609.17882	P3	P3	Can VLMs Reliably Assess Sidewalk Accessibility Attributes from Pedestrian-Level Imagery?	Seung Jae Lieu et al.	2026-09-15	EXCLUDE		title/abstract	R6: Navigation assistance for people (the human navigates, not a robot) (fails A)
2609.18326	P2	P2	UAVs Meet Embodied Intelligence: Bridging Human Intents and Flying Dynamics Via Harnessing Physical-	Yonglin Tian et al.	2026-09-16	EXCLUDE		title/abstract	R2: Autonomous driving, UAV or other non-ground-robot domain
2609.21316	P1	P1	NaViRrator: Robot Navigation from Human-Readable Maps through a Learned Visual Route	Ayun Lee et al.	2026-09-18	EXCLUDE		title/abstract	R4: Embodied/robot navigation without an explicit social-navigation capability (fails C)
2609.21504	P1	P1+P3	DPed-VLN: A Benchmark for Socially Compliant Vision-and-Language Navigation in Dynamic Pedestrian En	Haojie Dai et al.	2026-09-18	INCLUDE	Dai2026DPedVLN	full text	included after full-text screening
2609.22813	P1	P1	Commonsense-Grounded Path Planning from Abstract Instructions	Masafumi Endo et al.	2026-09-19	EXCLUDE		title/abstract	R4: Embodied/robot navigation without an explicit social-navigation capability (fails C)
2609.26857	P1	P1	FLINT: Fast Lightweight Inference for Traversability	William Bonilla et al.	2026-09-22	EXCLUDE		title/abstract	R4: Embodied/robot navigation without an explicit social-navigation capability (fails C)
2609.28467	P2	P2	Where Should I Join? Robot Group Joining via Language-Guided Goal Prediction	Zilin Fang et al.	2026-09-23	INCLUDE	Fang2026GroupJoining	full text	included after full-text screening
2609.34220	P2	P2	mmHRI: Towards Privacy-Preserving Human-Robot Interaction with Millimeter-Wave Radar	Junqiao Fan et al.	2026-09-28	EXCLUDE		title/abstract	R3: Robot manipulation / HRI / perception without mobile-navigation behaviour (fails A)
2609.40245	P1	P1+P2+P3	STARS: From Spatiotemporal Dynamics to Social Representations in Human-Robot Interaction	Nathan Tsoi et al.	2026-09-30	EXCLUDE		full text	STARS (CoRL 2026): spatiotemporal social representations (GNN); no FM (fails B)
2610.02750	P1	P1	CSIR: Contextually and Socially Informed Robots for Efficient Person Goal Navigation	Tyler Chung et al.	2026-10-02	EXCLUDE		full text	CSIR: person-goal navigation (search for a recipient); not social-compliance behaviour (fails C)
2610.03828	P1	P1+P2+P3	TACET: Context-Appropriate Acoustic-Social Navigation for Quadrupeds	Sungsan Park et al.	2026-10-02	INCLUDE	Park2026TACET	full text	included after full-text screening
2610.05964	P2	P2	From Social Reasoning to Embodied Interaction: An Agentic Framework for Social Robots	Ziyu Cheng et al.	2026-10-05	EXCLUDE		title/abstract	R3: Robot manipulation / HRI / perception without mobile-navigation behaviour (fails A)
2610.08526	P1	P1	WareFly-VLA: A Vision-Language-Action Framework for UAV Navigation and Human Tracking in Smart Wareh	Thinh D. Le et al.	2026-10-06	EXCLUDE		title/abstract	R2: Autonomous driving, UAV or other non-ground-robot domain
2610.18093	P3	P3	LynxScenes: A Large-Scale Real-World Benchmark for Outdoor Wheel-Legged Robot Perception, Navigation	Jun Tian et al.	2026-10-13	BACKGROUND ONLY		full text	LynxScenes: large real-world wheel-legged robot dataset with navigation instructions; no FM method (fails B)

Candidate id (arXiv id or key)	First found (search pass P1-P3, web, citation)	Found in passes	Title	First author	Date	Decision	Corpus entry (cite key)	Screening level	Reason
2610.18373	P1	P1+P2+P3	DR-VLA: Learning Human-Like Quadruped Navigation from Egocentric Web Videos via Domain-Invariant Sce	Jun Tian et al.	2026-10-13	EXCLUDE		full text	DR-VLA: quadruped navigation from egocentric web videos; no explicit social evaluation (fails C); adjacent
2610.19330	P3	P3	NavAble: A Large-Scale Dataset and Synthetic Data Generation Pipeline for Blind Navigation	Hochul Hwang et al.	2026-10-14	EXCLUDE		title/abstract	R6: Navigation assistance for people (the human navigates, not a robot) (fails A)
2610.19361	P1	P1+P3	VLX-Go: Robust Embodied Tracking through Multi-Stage Learning and On-Policy Refinement	Kyusong Lee et al.	2026-10-14	EXCLUDE		title/abstract	R4: Embodied/robot navigation without an explicit social-navigation capability (fails C)
2610.22092	P1	P1+P2	Onboard Lightweight LLM Reasoning for Human-Aware Safe UAV Navigation	Siwei Cai et al.	2026-10-15	EXCLUDE		full text	Onboard LLM reasoning for human-aware UAV navigation: aerial (fails A); workshop paper
doi:10.1109/ICCVW69036.2025.00262	citation	citation	Towards Proactive Social Robots: Distilling Visual Knowledge from Large VLMs	G. De Simone et al.	2025	EXCLUDE		full text	Proactive social robots (ICCV workshop): VLM distillation for engagement perception, not navigation (fails A)
doi:10.1109/ICRA57147.2024.10611462	web	web	LIM2N (ICRA 2024 version of arXiv 2311.08244 “Language and Sketching”)	Not recorded	2024	EXCLUDE		full text	LIM2N: LLM command/sketch interface for multitask navigation; social behaviour not evaluated (fails C)
doi:10.1109/IROS58592.2024.10802269	web	web	SocialNav-FTI: Field-Theory-Inspired Social-aware Navigation Framework based on Human Behavior	Not recorded	2024	BACKGROUND ONLY		full text	SocialNav-FTI: field-theory social navigation; no FM use identified in the record (fails B)
doi:10.1145/3737611.3776944	web	web	Socially-Aware Robot Navigation Using Large Language Models: System and Evaluation	Not recorded	2026	EXCLUDE		full text	2-page companion abstract (ICDCN 2026); insufficient methods/results for coding
doi:10.1145/3748777.3748807	web	web	LLM-Powered Embodied Intelligence for Socially-Aware Robot Navigation in Human-Robot Interaction	Not recorded	2025	EXCLUDE		full text	4-page short paper (SSTD 2025); insufficient methods/results for coding
doi:10.1609/aaai.v40i22.38941	citation	citation	Learning from Human Gaze: Human-like Robot Social Navigation in Dense Crowds (GazeNav)	Z. Yu et al.	2026	BACKGROUND ONLY		full text	GazeNav (AAAI 2026): gaze-guided social navigation and dataset; no FM (fails B); in Cai Table 2
doi:10.24963/ijcai.2026/696	final check	final check	AHead-VLN: A Hierarchical Reasoning Framework Guided by Foresight for Human-Aware Navigation	J. Zhang et al.	2026	INCLUDE	Zhang2026AHeadVLN	full text	venue-only paper found in the final check at the cut-off (publisher index); included after full-text screening

Table S1, part B. Screening record of the final coverage search (26 October 2026)

270 rows. Machine-readable version: `S1b_final_search_record.csv`.

Title	First author	Date	Venue	DOI	Retrieved by	Status in project	Decision	Screening level	Reason
An LLM-based Architecture for Socially Intelligent Robot Navigation based on Social Cues	Andrea Ruo	2025-08-25	RO-MAN 2025	10.1109/ro-man63969.2025.11217843	IEEE Xplore+OpenAlex search	new to project	INCLUDE	full text	INCLUDE (full text, 2026-10-26): local LLM sets navigation mode, speed bound and social-zone distance enforced by CBFs; Gazebo + offline accuracy test. Corpus entry 2025_Ruo_LLM-Social-Cues
RBTLR: A Framework for Safe Social Robot Navigation in Novel Human-Centric Environments	Nayesha Gandotra	2026-03-12	ACM/IEEE International Conference on Human-Robot Interaction (HRI)	10.1145/3776734.3794437	OpenAlex search	new to project	INCLUDE	full text	INCLUDE (full text, 2026-10-26): CLIP/Qwen social context conditions a learned selector of collision-free candidates; LLM goal assignment and clarification. Early-stage. Corpus entry 2026_Gandotra_RBTLR
Vision Language Action for TurtleBot Mobile Robot Path Planning			ICCRE 2026		IEEE Xplore	new to project	INCLUDE	full text	INCLUDE (full text, 2026-10-26): LLM chooses proceed/yield/avoid/stop around a detected human; independent LiDAR safety layer; Gazebo. Corpus entry 2026_Oush_VLA-TurtleBot
Adaptive Social Robot Navigation Using Vision-Language Models	Ketan Anand	2025-10-14	Springer proceedings in mathematics & statistics	10.1007/978-981-95-3841-6_9	OpenAlex search	new to project	UNRESOLVED (not coded)	title/abstract	U: VLM decodes social cues and commands; Instruct-MuSoHu; real datasets + simulation (abstract). Springer Proc. Math. Stat. 2025; full text paywalled
Enhancing Safe Navigation in Industrial Environments via Vision-Language Human-Robot Interaction	Sepideh Valiollahi	2025-01-01	Mechanisms and machine science	10.1007/978-3-032-02106-9_13	OpenAlex search	new to project	UNRESOLVED (not coded)	title/abstract	U: VLM+LLM human detection, 1 m distance, attention signalling, gesture instructions; Isaac Sim (abstract). Mech. Mach. Sci. 2025; full text paywalled
Polite Detouring Strategy for Social Compliance Obstacle Avoidance of Humanoid Service Robots	Yutong Li	2026-01-01	Lecture notes in electrical engineering	10.1007/978-981-95-9350-7_46	OpenAlex search+OpenAlex forward citations	new to project	UNRESOLVED (not coded)	title/abstract	U: VLM used for pedestrian detection + TRPO reward for polite detouring; Pepper trials (abstract). LNEE 2026; full text paywalled
A Call for Deeper Collaboration Between Robotics and Game Development	Iolanda Leite	2025-08-19	IEEE Conference on Games (CoG)	10.1109/cog64752.2025.11114209	OpenAlex forward citations	new to project	EXCLUDE	title/abstract	R1: Not robot/embodied navigation (incl. non-scholarly repository items)
A comprehensive review of recent advancements in vision-and-language navigation	Jahanzeb Tariq Khan	2026-03-21	Discover Computing	10.1007/s10791-026-09977-z	OpenAlex forward citations	new to project	EXCLUDE	title/abstract	R7: Survey, review, position paper or thesis (outside the screening scope)
A Comprehensive Survey and Systematic Real-World Evaluation of Embodied Vision-and-Language Navigation	Liuyi Wang	2026-01-01	IEEE Transactions on Automation Science and Engineering	10.1109/tase.2026.3711284	OpenAlex forward citations	new to project	EXCLUDE	title/abstract	R7: Survey, review, position paper or thesis (outside the screening scope)
A digital twin framework for patrol robots with real-time path planning using optimised A* algorithm with ROS-Unity integration	Maria Nafees	2026-06-01	Results in Engineering	10.1016/j.rineng.2026.111457	OpenAlex forward citations	new to project	EXCLUDE	title/abstract	R4: Embodied navigation without explicit social capability (fails C)
A Review of Methods for Autonomous Movement of Mobile Robots over Rough Terrain	Anton Moscowsky	2025-02-01	Automation and Remote Control	10.1134/s0005117925601113	OpenAlex forward citations	new to project	EXCLUDE	title/abstract	R7: Survey, review, position paper or thesis (outside the screening scope)
A Survey on Efficient Vision-Language Models	Gaurav Shinde	2025-07-13	Wiley Interdisciplinary Reviews Data Mining and Knowledge Discovery	10.1002/widm.70036	OpenAlex forward citations	new to project	EXCLUDE	title/abstract	R7: Survey, review, position paper or thesis (outside the screening scope)
A Systematic Review on Soft Authority in Llm-Powered Robots: How Emotion, Persona, and Repair Shape User Guidance	Anastasia Nikolova	2026-08-01	IEEE International Conference on Robot and Human Interactive Communication (RO-MAN)	10.1109/ro-man72286.2026.11714622	OpenAlex forward citations	new to project	EXCLUDE	title/abstract	R7: Survey, review, position paper or thesis (outside the screening scope)
A Vision-Language Framework for Assistive Home Robotics			2025		IEEE Xplore	new to project	EXCLUDE	title/abstract	R4: Embodied navigation without explicit social capability (fails C)
A Walk to Remember: Mllm Memory-Driven Visual Navigation*	Sandun S. Vitharana	2025-06-30	International Conference on Ubiquitous Robots (UR)	10.1109/ur65550.2025.11078086	OpenAlex forward citations	new to project	EXCLUDE	title/abstract	R4: Embodied navigation without explicit social capability (fails C)
Active reward learning and iterative trajectory improvement from comparative language feedback	Eisuke Hirota	2025-11-12	The International Journal of Robotics Research	10.1177/02783649251394742	OpenAlex forward citations	new to project	EXCLUDE	title/abstract	R3: Manipulation / HRI / perception without mobile-navigation behaviour (fails A)
Adaptive planning for assistive-care robotic missions	Ioannis Stefanakos	2026-03-10	The International Journal of Robotics Research	10.1177/02783649261420234	OpenAlex forward citations	new to project	EXCLUDE	title/abstract	R4: Embodied navigation without explicit social capability (fails C)
Affinity-Guided Group Discovery for Social Activity Understanding in Videos	Hongtao Sun	2026-01-01	SSRN Electronic Journal	10.2139/ssrn.7411563	OpenAlex forward citations	new to project	EXCLUDE	title/abstract	R3: Manipulation / HRI / perception without mobile-navigation behaviour (fails A)
Agentic Llm-Driven Human-Robot Interaction			2026		IEEE Xplore	new to project	EXCLUDE	title/abstract	R3: Manipulation / HRI / perception without mobile-navigation behaviour (fails A)
AI That Moves With You: A Review of Interactive Technologies Powered by Large Foundation Models for Mobility Impairment	Duosi Dai	2026-04-13	CHI Conference on Human Factors in Computing Systems	10.1145/3772318.3791239	OpenAlex forward citations	new to project	EXCLUDE	title/abstract	R7: Survey, review, position paper or thesis (outside the screening scope)
An Innovative Hierarchical Weed Detection Methodology for Intelligent Site-Specific Weed Management Leveraging Large Multimodal Models	Bo-Han Huang	2026-07-01	IEEE/ASME International Conference on Advanced Intelligent Mechatronics	10.1109/aim65483.2026.11658114	OpenAlex forward citations	new to project	EXCLUDE	title/abstract	R1: Not robot/embodied navigation (incl. non-scholarly repository items)
Arena-Bench 2.0: A Comprehensive Benchmark of Social Navigation Approaches in Collaborative Environments			IROS 2025		IEEE Xplore	new to project	EXCLUDE	title/abstract	R5: Abstract (IEEE): benchmark of learning- and model-based planners; no FM component
AutoFPDesigner: Automated Flight Procedure Design Based on Multi-Agent Large Language Model	Longtao Zhu	2025-12-04	IEEE Transactions on Intelligent Transportation Systems	10.1109/tits.2025.3631667	OpenAlex search	new to project	EXCLUDE	title/abstract	R2: Autonomous driving, UAV or other non-ground-robot domain
Automated Context-Aware Navigation Support for Individuals with Visual Impairment Using Multimodal Language Models in Urban Environments	Alton Chao	2025-10-19	IEEE/CVF International Conference on Computer Vision Workshops (ICCVW)	10.1109/iccvw69036.2025.00696	OpenAlex search	new to project	EXCLUDE	title/abstract	R6: Navigation assistance for people (fails A)
Autonomous Drone Navigation with Natural Language Processing for Human-Centric Smart Manufacturing			2025		IEEE Xplore	new to project	EXCLUDE	title/abstract	R2: Autonomous driving, UAV or other non-ground-robot domain
Autonomous Ground Navigation in Highly Constrained Spaces: Lessons Learned From the Fourth BARN Challenge at ICRA 2025	Xuesu Xiao	2026-03-01	IEEE Robotics & Automation Magazine	10.1109/mra.2025.3639795	OpenAlex forward citations	new to project	EXCLUDE	title/abstract	R4: Embodied navigation without explicit social capability (fails C)
Autonomous Mario Kart in the Wild: Lessons Learned From the Earth Rover Challenge at IROS 2024 [Competitions]	Xuesu Xiao	2025-06-01	IEEE Robotics & Automation Magazine	10.1109/mra.2025.3565211	OpenAlex forward citations	new to project	EXCLUDE	title/abstract	R4: Embodied navigation without explicit social capability (fails C)

Title	First author	Date	Venue	DOI	Retrieved by	Status in project	Decision	Screening level	Reason
Autonomous Urban Patrol Robot for Monitoring Improper Micromobility Parking on Pedestrian Pathways ⋆	Hyeon Min Kim	2026-01-01	SSRN Electronic Journal	10.2139/ssrn.6628598	OpenAlex forward citations	new to project	EXCLUDE	title/abstract	R4: Embodied navigation without explicit social capability (fails C)
BehaviorGPT for Visual Art: A Foundation Model for Aesthetics	Rickard Brüel Gabrielsson	2025-07-15	Zenodo (CERN European Organization for Nuclear Research)	10.5281/zenodo.23576072	OpenAlex search	new to project	EXCLUDE	title/abstract	R1: Not robot/embodied navigation (incl. non-scholarly repository items)
Benchmark Evaluations, Applications, and Challenges of Large Vision Language Models: A Survey	Zongxia Li	2025-01-15	Qeios	10.32388/gxr68q	OpenAlex forward citations	new to project	EXCLUDE	title/abstract	R7: Survey, review, position paper or thesis (outside the screening scope)
Beyond Recognition: Evaluating Visual Perspective Taking in Vision Language Models	Gracjan Góral	2026-06-01		10.1109/cvprf71909.2026.00170	OpenAlex forward citations	new to project	EXCLUDE	title/abstract	R3: Manipulation / HRI / perception without mobile-navigation behaviour (fails A)
BRiDGENav: Belief-Driven Risk-Aware Dynamic Guidance and Event-Triggered Navigation	Nimantha Adikaram	2026-10-05	Journal of Sensor and Actuator Networks	10.3390/jsan15050080	OpenAlex forward citations	new to project	EXCLUDE	title/abstract	R5: Social navigation without an FM role in the navigation method (fails B)
BT-ACTION: A Test-Driven Approach for Modular Understanding of User Instruction Leveraging Behaviour Trees and LLMs	Alexander Leszczynski	2025-08-25	IEEE International Conference on Robot and Human Interactive Communication (RO-MAN)	10.1109/ro-man63969.2025.11217657	OpenAlex forward citations	new to project	EXCLUDE	title/abstract	R3: Manipulation / HRI / perception without mobile-navigation behaviour (fails A)
BUMBLE: Unifying Reasoning and Acting with Vision-Language Models for Building-wide Mobile Manipulation	Rutav Shah	2025-05-19	Proceedings - IEEE International Conference on Robotics and Automation/Proceedings	10.1109/icra55743.2025.11128444	OpenAlex forward citations	new to project	EXCLUDE	title/abstract	R4: Embodied navigation without explicit social capability (fails C)
Chat-Driven Reconfiguration of Model Predictive Control			2026		IEEE Xplore	new to project	EXCLUDE	title/abstract	R4: Embodied navigation without explicit social capability (fails C)
CityWalker: Learning Embodied Urban Navigation from Web-Scale Videos	Xinhao Liu	2024-12-07	Qeios	10.32388/yfv2c5	OpenAlex forward citations	new to project	EXCLUDE	title/abstract	R4: Embodied navigation without explicit social capability (fails C)
Clarifying Constraints in Interactive Robot Learning with Language Feedback	Hannah Kuehn	2026-03-10	ACM/IEEE International Conference on Human-Robot Interaction (HRI)	10.1145/3757279.3785583	OpenAlex forward citations	new to project	EXCLUDE	title/abstract	R3: Manipulation / HRI / perception without mobile-navigation behaviour (fails A)
Closed-Loop Perception–Action Framework for VLM-Based Relational Safety Monitoring in Construction	Chaewon Yang	2026-01-01	SSRN Electronic Journal	10.2139/ssrn.7201051	OpenAlex forward citations	new to project	EXCLUDE	title/abstract	R3: Manipulation / HRI / perception without mobile-navigation behaviour (fails A)
CogAAL: A Privacy-Preserving, Full-Stack Cognitive Architecture Integrating Local VLMs, Heterogeneous Robotics, and IoT for Proactive Ambient Assisted Living			2026		IEEE Xplore	new to project	EXCLUDE	title/abstract	R3: Manipulation / HRI / perception without mobile-navigation behaviour (fails A)
Communicating Object Relations through Robot Gestures	Xiang Pan	2026-03-10	ACM/IEEE International Conference on Human-Robot Interaction (HRI)	10.1145/3757279.3785554	OpenAlex forward citations	new to project	EXCLUDE	title/abstract	R3: Manipulation / HRI / perception without mobile-navigation behaviour (fails A)
Context-Aware Autonomous Drone Navigation Using Large Language Models (LLMs)	Abdul-Manan Khan	2025-08-01	Proceedings of the AAAI Symposium Series	10.1609/aaaiss.v6i1.36039	OpenAlex search	new to project	EXCLUDE	title/abstract	R2: Autonomous driving, UAV or other non-ground-robot domain
Context-aware decision making in autonomous vehicles: Integrating social behavior modeling with large language models	Badri Raj Lamichhane	2025-06-11	Array	10.1016/j.array.2025.100420	OpenAlex search	new to project	EXCLUDE	title/abstract	R2: Autonomous driving, UAV or other non-ground-robot domain
Context-Aware Path Planning: A Unified Approach for Navigation in Heterogeneous Environments	Hamid Didari	2026-09-19	Applied Sciences	10.3390/app16189295	OpenAlex forward citations	new to project	EXCLUDE	title/abstract	R4: Embodied navigation without explicit social capability (fails C)
Copeland Ontologic Registry 0a-0020: A Revision-Tracked Meta-Ontology for Cross-Domain Legibility Analysis	Christopher W Copeland	2026-05-17	Zenodo (CERN European Organization for Nuclear Research)	10.5281/zenodo.20261120	OpenAlex search	new to project	EXCLUDE	title/abstract	R1: Not robot/embodied navigation (incl. non-scholarly repository items)
Deep Learning for Multi-Agent Simulation and Control of Crowds	Andreas C. Panagiotou	2026-02-25	Gnosis	10.82357/gnosis/489	OpenAlex search	new to project	EXCLUDE	title/abstract	R7: Survey, review, position paper or thesis (outside the screening scope)
Development of an AI-Based Personal Assistant for Automated Calendar Management via Natural Language	Κωνσταντίνος Ανδρικόπουλος Αναστάσιος	2026-10-14	Harokopio University	10.26258/heal.hua.7877	OpenAlex search	new to project	EXCLUDE	title/abstract	R1: Not robot/embodied navigation (incl. non-scholarly repository items)
Development of DeepSeek-Based Conversational Robot System for Indoor Navigation Assistance in FKTE1 Building, UniMAP	Mohammed Omar Ahmed Aldini	2025-08-18		10.1109/icomms66553.2025.11200366	OpenAlex forward citations	new to project	EXCLUDE	title/abstract	R6: Navigation assistance for people (fails A)
Discrete to continuous VLN adaptation with collision mitigation through depth sensing	Jahanzeb Tariq Khan	2026-09-17	International Journal of Intelligent Robotics and Applications	10.1007/s41315-026-00592-1	OpenAlex forward citations	new to project	EXCLUDE	title/abstract	R4: Embodied navigation without explicit social capability (fails C)
Dynamic Context Adapters: Efficiently Infusing History into Vision-and-Language Models	Yuhang Song	2026-08-02	Lecture notes in computer science	10.1007/978-3-032-31663-9_8	OpenAlex forward citations	new to project	EXCLUDE	title/abstract	R1: Not robot/embodied navigation (incl. non-scholarly repository items)
Edge-Aided Collaborative Inference for Visual Language Navigation-Based Intelligent Vehicles	Jinkai Zheng	2026-01-01	IEEE Transactions on Network Science and Engineering	10.1109/tnse.2026.3679193	OpenAlex forward citations	new to project	EXCLUDE	title/abstract	R2: Autonomous driving, UAV or other non-ground-robot domain
Embodied AI with Foundation Models for Mobile Service Robots: A Systematic Review	Matthew Lisondra	2026-03-04	Robotics	10.3390/robotics15030055	OpenAlex forward citations	new to project	EXCLUDE	title/abstract	R7: Survey, review, position paper or thesis (outside the screening scope)
Embodied Intelligence Security with Vision-language Models: A Survey	Junxian Duan	2026-07-13	Machine Intelligence Research	10.1007/s11633-025-1626-x	OpenAlex forward citations	new to project	EXCLUDE	title/abstract	R7: Survey, review, position paper or thesis (outside the screening scope)
Enhanced Dynamic Window Approach for socially compliant robot navigation	S. Ashwath	2026-05-06	Frontiers in Robotics and AI	10.3389/frobt.2026.1769678	OpenAlex forward citations	new to project	EXCLUDE	title/abstract	R5: Social navigation without an FM role in the navigation method (fails B)
Enhancing reliability in LLM-integrated robotic systems: A unified approach to security and safety	W. J. Zhang	2025-09-08	Journal of Systems and Software	10.1016/j.jss.2025.112614	OpenAlex forward citations	new to project	EXCLUDE	title/abstract	R3: Manipulation / HRI / perception without mobile-navigation behaviour (fails A)
Enhancing Trust in Autonomous Agents: An Architecture for Accountability and Explainability Through Blockchain and Large Language Models	Laura Fernández-Becerra	2024-01-01	SSRN Electronic Journal	10.2139/ssrn.4773675	OpenAlex search	new to project	EXCLUDE	title/abstract	R1: Not robot/embodied navigation (incl. non-scholarly repository items)
ESearch-R1: Learning Cost-Aware MLLM Agents for Interactive Embodied Search via Reinforcement Learning	Weijie Zhou	2025-12-21	arXiv (Cornell University)	10.48550/arxiv.2512.18571	OpenAlex search	new to project	EXCLUDE	title/abstract	R4: Embodied navigation without explicit social capability (fails C)
Exophora Resolution of Linguistic Instructions with a Demonstrative based on Real-World Multimodal Information			2023		IEEE Xplore	new to project	EXCLUDE	title/abstract	R3: Manipulation / HRI / perception without mobile-navigation behaviour (fails A)
Exploring the Cognitive Capabilities of Large Language Models in Autonomous and Swarm Navigation Systems	Dawid Ewald	2025-12-22	Electronics	10.3390/electronics15010035	OpenAlex forward citations	new to project	EXCLUDE	title/abstract	R2: Autonomous driving, UAV or other non-ground-robot domain
FASTNav: Fine-Tuned Adaptive Small-Language-Models Trained for Multi-Point Robot Navigation	Yuxuan Chen	2024-11-25	IEEE Robotics and Automation Letters	10.1109/lra.2024.3506280	OpenAlex forward citations	new to project	EXCLUDE	title/abstract	R4: Embodied navigation without explicit social capability (fails C)
Flora: Sample-Efficient Preference-Based Rl Via Low-Rank Style Adaptation of Reward Functions	Daniel Marta	2025-05-19	Proceedings - IEEE International Conference on Robotics and Automation/Proceedings	10.1109/icra55743.2025.11127633	OpenAlex forward citations	new to project	EXCLUDE	title/abstract	R3: Manipulation / HRI / perception without mobile-navigation behaviour (fails A)
Follow Me: A Study on the Dynamics of Alignment Between Humans and LLM-Based Social Robots	Jeffrey Sherer	2025-01-01	Lecture notes in computer science	10.1007/978-981-96-3519-1_44	OpenAlex forward citations	new to project	EXCLUDE	title/abstract	R3: Manipulation / HRI / perception without mobile-navigation behaviour (fails A)

Title	First author	Date	Venue	DOI	Retrieved by	Status in project	Decision	Screening level	Reason
Foundation Models for Robotic Tasks: Survey, Challenges and Future Directions	Muhammad Alam	2025-07-22		10.36227/techrxiv.175321822.20121795/v1	OpenAlex forward citations	new to project	EXCLUDE	title/abstract	R7: Survey, review, position paper or thesis (outside the screening scope)
From insight to autonomous execution: VLM-enhanced embodied agents towards digital twin-assisted human-robot collaborative assembly	Changchun Liu	2025-11-05	Robotics and Computer-Integrated Manufacturing	10.1016/j.rcim.2025.103176	OpenAlex forward citations	new to project	EXCLUDE	title/abstract	R3: Manipulation / HRI / perception without mobile-navigation behaviour (fails A)
From Language to Logic: A Theoretical Architecture for VLM-Grounded Safe Navigation	Kristy Sakano	2026-06-15	International Conference on Unmanned Aircraft Systems (ICUAS)	10.1109/icuas69441.2026.11598671	OpenAlex forward citations	new to project	EXCLUDE	title/abstract	R2: Autonomous driving, UAV or other non-ground-robot domain
From Links to Dialogue; Hypertext Challenges and Opportunities in Conversational Navigation	Behnam Rahdari	2025-09-15		10.1145/3720533.3750064	OpenAlex search	new to project	EXCLUDE	title/abstract	R1: Not robot/embodied navigation (incl. non-scholarly repository items)
GazeCoT: Unleashing Social Intelligence in Multimodal LLMs With Gaze-Informed Chain-of-Thought Reasoning	Zhoutong Ye	2026-04-13	CHI Conference on Human Factors in Computing Systems	10.1145/3772318.3790922	OpenAlex forward citations	new to project	EXCLUDE	title/abstract	R3: Manipulation / HRI / perception without mobile-navigation behaviour (fails A)
GRACE: Generating Socially Appropriate Robot Actions Leveraging LLMs and Human Explanations	Fethiye Irmak Doğan	2025-05-19	Proceedings - IEEE International Conference on Robotics and Automation/Proceedings	10.1109/icra55743.2025.11127826	OpenAlex forward citations	new to project	EXCLUDE	title/abstract	R3: Manipulation / HRI / perception without mobile-navigation behaviour (fails A)
Harnessing Driver Metacognition to Advance In-Context Learning for Autonomous Vehicle Social Intelligence	Yin-Chun Lu	2026-09-30	IISE Annual Conference & Expo	10.21872/annual2025_8944	OpenAlex search	new to project	EXCLUDE	title/abstract	R2: Autonomous driving, UAV or other non-ground-robot domain
How do Foundation Models Compare to Skeleton-Based Approaches for Gesture Recognition in Human-Robot Interaction?			2025		IEEE Xplore	new to project	EXCLUDE	title/abstract	R3: Manipulation / HRI / perception without mobile-navigation behaviour (fails A)
Human-aware Robot Navigation: Problem Formulation, Approaches, Challenges	Omar Kashmar	2026-01-01	SSRN Electronic Journal	10.2139/ssrn.6486782	OpenAlex search+OpenAlex forward citations	new to project	EXCLUDE	title/abstract	R7: Survey, review, position paper or thesis (outside the screening scope)
Human-machine collaboration for safe and socially compliant trajectory planning of quadrotor	Zezhong Zhang	2025-01-01		10.32657/10356/182953	OpenAlex search	new to project	EXCLUDE	title/abstract	R7: Survey, review, position paper or thesis (outside the screening scope)
Human-robot Interactive Task Parsing Method Based on Large Language Model			2024		IEEE Xplore	new to project	EXCLUDE	title/abstract	R3: Manipulation / HRI / perception without mobile-navigation behaviour (fails A)
Humanlike representation of social networks in large language models	Jae-Young Son	2026-08-31		10.31234/osf.io/73fpj_v1	OpenAlex search	new to project	EXCLUDE	title/abstract	R1: Not robot/embodied navigation (incl. non-scholarly repository items)
ImpedanceDiffusion: Diffusion-Based Global Path Planning for UAV Swarm Navigation with Generative Impedance Control	Faryal Batool	2026-06-15	International Conference on Unmanned Aircraft Systems (ICUAS)	10.1109/icuas69441.2026.11598661	OpenAlex forward citations	new to project	EXCLUDE	title/abstract	R2: Autonomous driving, UAV or other non-ground-robot domain
ImpedanceGPT: VLM-driven Impedance Control of Swarm of Mini-drones for Intelligent Navigation in Dynamic Environment	Faryal Batool	2025-10-19	IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)	10.1109/iros60139.2025.11247031	OpenAlex forward citations	new to project	EXCLUDE	title/abstract	R2: Autonomous driving, UAV or other non-ground-robot domain
IndoNav: A Benchmark Dataset of Indonesian Pedestrian Scenes for Assistive Navigation of Vision-Impaired People			2026		IEEE Xplore	new to project	EXCLUDE	title/abstract	R6: Navigation assistance for people (fails A)
InsightSee: Advancing Multi-agent Vision-Language Models for Enhanced Visual Understanding	Huaxiang Zhang	2024-08-04	IEEE International Conference on Mechatronics and Automation (ICMA)	10.1109/icma61710.2024.10633164	OpenAlex forward citations	new to project	EXCLUDE	title/abstract	R1: Not robot/embodied navigation (incl. non-scholarly repository items)
Intelligent Indoor Navigation for Home Robots based on Large Language Models			2025		IEEE Xplore	new to project	EXCLUDE	title/abstract	R4: Embodied navigation without explicit social capability (fails C)
Intelligent Spatial Perception by Building Hierarchical 3D Scene Graphs for Indoor Scenarios with the Help of LLMs*	Yao Cheng	2024-08-23	WRC Symposium on Advanced Robotics and Automation (WRC SARA)	10.1109/wrcsara64167.2024.10685765	OpenAlex search	new to project	EXCLUDE	title/abstract	R4: Embodied navigation without explicit social capability (fails C)
Investigating Inverse Reinforcement Learning during Rapid Aiming Movement in Extended Reality and Human-Robot Interaction	Mukund Mitra	2025-05-20	ACM Transactions on Human-Robot Interaction	10.1145/3736423	OpenAlex forward citations	new to project	EXCLUDE	title/abstract	R3: Manipulation / HRI / perception without mobile-navigation behaviour (fails A)
Investigating out-of-the-box Indoor Localization Using Large Language Models and Evacuation Maps	Gabriella Trasciatti	2026-03-16	IEEE International Conference on Pervasive Computing and Communications Workshops and other Affiliated Events (PerCom Workshops)	10.1109/percomworkshops68308.2026.11585232	OpenAlex forward citations	new to project	EXCLUDE	title/abstract	R4: Embodied navigation without explicit social capability (fails C)
It's Time to Get It Right: Improving Analog Clock Reading and Clock-Hand Spatial Reasoning in Vision-Language Models	Jaeha Choi	2026-06-01		10.1109/cvprf71909.2026.00910	OpenAlex forward citations	new to project	EXCLUDE	title/abstract	R1: Not robot/embodied navigation (incl. non-scholarly repository items)
JRDB-Social: A Multifaceted Robotic Dataset for Understanding of Context and Dynamics of Human Interactions Within Social Groups	Simindokht Jahangard	2024-06-16	IEEE Computer Society Conference on Computer Vision and Pattern Recognition/Proceedings - IEEE Computer Society Conference on Computer Vision and Pattern Recognition/Proceedings	10.1109/cvpr52733.2024.02085	OpenAlex search	new to project	EXCLUDE	title/abstract	R3: Manipulation / HRI / perception without mobile-navigation behaviour (fails A)
KiteRunner: Language-Driven Cooperative Local-Global Navigation Policy with UAV Mapping in Outdoor Environments	Shibo Huang	2025-10-19	IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)	10.1109/iros60139.2025.11247564	OpenAlex forward citations	new to project	EXCLUDE	title/abstract	R2: Autonomous driving, UAV or other non-ground-robot domain
LAD-VF: LLM-Automatic Differentiation Enables Fine-Tuning-Free Robot Planning from Formal Methods Feedback			2026		IEEE Xplore	new to project	EXCLUDE	title/abstract	R4: Embodied navigation without explicit social capability (fails C)
LANCAR: Leveraging Language for Context-Aware Robot Locomotion in Unstructured Environments	Chak Lam Shek	2024-10-14	IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)	10.1109/iros58592.2024.10802075	OpenAlex search	new to project	EXCLUDE	title/abstract	R4: Embodied navigation without explicit social capability (fails C)
Language Model Alignment for Conversational Shopping at Amazon	Chen Luo	2025-07-13	International ACM SIGIR Conference on Research and Development in Information Retrieval (SIGIR)	10.1145/3726302.3731955	OpenAlex search	new to project	EXCLUDE	title/abstract	R1: Not robot/embodied navigation (incl. non-scholarly repository items)
Large language models in human-robot collaboration: A systematic review, trends, and challenges	Gang Yuan	2026-01-22	Journal of Manufacturing Systems	10.1016/j.jmsy.2026.01.011	OpenAlex forward citations	new to project	EXCLUDE	title/abstract	R7: Survey, review, position paper or thesis (outside the screening scope)
Learning Semantic Traversability With Egocentric Video and Automated Annotation Strategy			2024		IEEE Xplore	new to project	EXCLUDE	title/abstract	R4: Embodied navigation without explicit social capability (fails C)
Learning When to Yield: Adapting Pre-Trained Action Models for Human-Aware Mobile Robot Navigation	Lina Moretti	2025-12-18	Figshare	10.6084/m9.figshare.31329874	OpenAlex search	new to project	EXCLUDE	full text	R8: Full text read (Figshare PDF supplied by the authors): claimed dataset not locatable, several reference errors, and a frozen manipulation VLA said to output holonomic velocities for a differential-drive robot
Learning-enabled collaborative intelligence in search and rescue robotics: a survey of paradigms, models, and challenges	M. I. R. Shuvo	2026-10-14	Frontiers in Robotics and AI	10.3389/frobt.2026.1832982	OpenAlex forward citations	new to project	EXCLUDE	title/abstract	R7: Survey, review, position paper or thesis (outside the screening scope)
Legibility on Social Robot Navigation: A Survey	Leonardo Maia Nogueira	2025-04-28		10.1109/cros66186.2025.11066134	OpenAlex forward citations	new to project	EXCLUDE	title/abstract	R7: Survey, review, position paper or thesis (outside the screening scope)

Title	First author	Date	Venue	DOI	Retrieved by	Status in project	Decision	Screening level	Reason
Let's Reward Step-by-Step: Step-Aware Contrastive Alignment for Vision-Language Navigation in Continuous Environments	Haoyuan Li	2026-01-01	Lecture notes in computer science	10.1007/978-3-032-37211-6_27	OpenAlex search	new to project	EXCLUDE	title/abstract	R4: Embodied navigation without explicit social capability (fails C)
Leveraging Large Language Models for Modular Robot Navigation	Anlong Zhang	2025-06-06	2025	10.1109/raiie65740.2025.11140161	IEEE Xplore+OpenAlex search	new to project	EXCLUDE	title/abstract	R4: Embodied navigation without explicit social capability (fails C)
Leviathan: A Bio-Inspired, CorTexManus-Driven Marine AGI Architecture for Resilient, Long-Term Deep-Sea Exploration	Khan Tahsin Abrar	2025-08-01		10.22541/au.175407961.15590722/v1	OpenAlex search	new to project	EXCLUDE	title/abstract	R1: Not robot/embodied navigation (incl. non-scholarly repository items)
Llama 3.2 Vision Instruct for Indian Traffic Scene Understanding and Object Detection in Dashcam Footage	Santosh Mohan	2025-10-30	Lecture notes in networks and systems	10.1007/978-3-032-06691-6_14	OpenAlex forward citations	new to project	EXCLUDE	title/abstract	R2: Autonomous driving, UAV or other non-ground-robot domain
LLM - Driven Adaptive Autonomous Robot Navigation via Multimodal Fusion for Diverse Environments			2025		IEEE Xplore	new to project	EXCLUDE	title/abstract	R4: Embodied navigation without explicit social capability (fails C)
LLM Hallucination Incident Report — LLM-2025-1222-HALLU ├─ DOI: 10.5281/zenodo.18368527 — Crimson Hexagon Archive	Jack E. Feist	2026-01-25	Zenodo (CERN European Organization for Nuclear Research)	10.5281/zenodo.18368526	OpenAlex search	new to project	EXCLUDE	title/abstract	R1: Not robot/embodied navigation (incl. non-scholarly repository items)
LLM-based Agentic Workflow on Verbal and Non-verbal Audiovisual Perceptions and Actions for Proactive Situated Human-Robot Interactions	Virgile Sucal	2025-11-10	International Conference on Human-Agent Interaction (HAI)	10.1145/3765766.3765797	OpenAlex forward citations	new to project	EXCLUDE	title/abstract	R3: Manipulation / HRI / perception without mobile-navigation behaviour (fails A)
LLM-Based Task Planning for Mobile Robots Using a Novel SLAM Method for Materials Distribution			2025		IEEE Xplore	new to project	EXCLUDE	title/abstract	R4: Embodied navigation without explicit social capability (fails C)
LLM-Driven Augmented Reality Puppeteer: Controller-Free Voice-Commanded Robot Teleoperation	Yuchong Zhang	2025-01-01	Lecture notes in computer science	10.1007/978-3-031-93539-8_7	OpenAlex forward citations	new to project	EXCLUDE	title/abstract	R3: Manipulation / HRI / perception without mobile-navigation behaviour (fails A)
LLM-Driven Context-Aware Trash Recognition and Disposal with a Mobile Manipulator	Janak Panthi	2026-08-01	IEEE International Conference on Robot and Human Interactive Communication (RO-MAN)	10.1109/ro-man72286.2026.11714991	OpenAlex forward citations	new to project	EXCLUDE	title/abstract	R3: Manipulation / HRI / perception without mobile-navigation behaviour (fails A)
LLM-Driven Safe Robotic Navigation in Construction Sites: A Prompt Engineering Framework with Terrain-Adaptive Path Planning	Xiayu Zhao	2026-01-28	ASCE International Conference on Computing in Civil Engineering (i3CE)	10.1061/9780784486436.007	OpenAlex forward citations	new to project	EXCLUDE	title/abstract	R4: Embodied navigation without explicit social capability (fails C)
LLM-driven symbolic planning and hierarchical imitation learning for long-horizon deformable object assembly	Jiaming Qi	2025-07-28	Robotics and Computer-Integrated Manufacturing	10.1016/j.rcim.2025.103096	OpenAlex forward citations	new to project	EXCLUDE	title/abstract	R3: Manipulation / HRI / perception without mobile-navigation behaviour (fails A)
LLM-Enhanced Mobile Robot Navigation and Scene Description for Indoor Environments			2025		IEEE Xplore	new to project	EXCLUDE	title/abstract	R4: Embodied navigation without explicit social capability (fails C)
LLM-Guided Indoor Navigation with Multimodal Map Understanding	Alberto Coffrini	2025-09-15	International Conference on Indoor Positioning and Indoor Navigation	10.1109/ipin66788.2025.11212934	OpenAlex search	new to project	EXCLUDE	title/abstract	R6: Navigation assistance for people (fails A)
LLM-Guided Multi-Agent System for Natural Language-Based Robot Navigation			2025		IEEE Xplore	new to project	EXCLUDE	title/abstract	R4: Embodied navigation without explicit social capability (fails C)
LLM-in-the-Loop Variable Impedance Control: Towards Safe Generalized and Personalized Robotic Interactions	Junyuan Xue	2026-08-01	IEEE International Conference on Cybernetics and Intelligent Systems (CIS) and IEEE International Conference on Robotics, Automation and Mechatronics (RAM)	10.1109/cis-ram71614.2026.11668493	OpenAlex forward citations	new to project	EXCLUDE	title/abstract	R3: Manipulation / HRI / perception without mobile-navigation behaviour (fails A)
LLM-Powered Social Robots for an Indoor Navigation Assistant			2025		IEEE Xplore	new to project	EXCLUDE	title/abstract	R6: Navigation assistance for people (fails A)
LLM-RLNav: Grounding Natural Language Instructions for Robots via LLM Decomposition and Reinforcement Learning			2025		IEEE Xplore	new to project	EXCLUDE	title/abstract	R4: Embodied navigation without explicit social capability (fails C)
LOG-Nav: Efficient Layout-Aware Object-Goal Navigation with Hierarchical Planning	Jiawei Hou	2026-03-14	Proceedings of the AAAI Conference on Artificial Intelligence	10.1609/aaai.v40i22.38893	OpenAlex search	new to project	EXCLUDE	title/abstract	R4: Embodied navigation without explicit social capability (fails C)
Loop Preset System for Concurrent Multi-Task Execution in Quadruped Locomotion	Dongyeon Kim	2026-07-31	The Journal of Korean Institute of Information Technology	10.14801/jkiit.2026.24.7.35	OpenAlex forward citations	new to project	EXCLUDE	title/abstract	R4: Embodied navigation without explicit social capability (fails C)
Lyapunov-Regularized Meta-Learning Adaptive Control for a Vision-Language Model-Guided Wheeled Humanoid Robot in Power Station Maintenance	Lujia Ran	2026-01-01	IEEE Transactions on Automation Science and Engineering	10.1109/tase.2026.3670845	OpenAlex forward citations	new to project	EXCLUDE	title/abstract	R4: Embodied navigation without explicit social capability (fails C)
Master Roadmap and Index for TPT & LFP-0: A Unified Governance Framework for Semantic Accountability in AGI (Root Index v1.0)	Hon Bor So	2025-12-19	Zenodo (CERN European Organization for Nuclear Research)	10.5281/zenodo.17988589	OpenAlex search	new to project	EXCLUDE	title/abstract	R1: Not robot/embodied navigation (incl. non-scholarly repository items)
Mixed Crowd Navigation: Perception, Interaction, Planning, and Control	Jia Pan	2025-09-16	Annual Review of Control Robotics and Autonomous Systems	10.1146/annurev-control-032024-023929	OpenAlex search	new to project	EXCLUDE	title/abstract	R7: Survey, review, position paper or thesis (outside the screening scope)
Modelling Robot Navigation Recovery Policies from Non-Expert Users' Demonstrations In-The-Wild	Ahmad Faris	2025-12-08		10.1109/roboticcc68732.2025.00014	OpenAlex forward citations	new to project	EXCLUDE	title/abstract	R4: Embodied navigation without explicit social capability (fails C)
Motion and Behavior Planning for Socially Assistive Robots	Tribhi Kathuria	2025-01-01	Deep Blue (University of Michigan)	10.7302/25667	OpenAlex search	new to project	EXCLUDE	title/abstract	R7: Survey, review, position paper or thesis (outside the screening scope)
Multi-Functional AI Robotics: Integrating SLAM, LLM, and Expressive Motion for Human Companionship			2025		IEEE Xplore	new to project	EXCLUDE	title/abstract	R7: Survey, review, position paper or thesis (outside the screening scope)
Multi-modal Knowledge Distillation-based Human Trajectory Forecasting			2025		IEEE Xplore	new to project	EXCLUDE	title/abstract	R3: Manipulation / HRI / perception without mobile-navigation behaviour (fails A)
Multi-sensor fusion and semantic-geometric navigation for robotic monitoring of improper micromobility parking on pedestrian pathways	Hyeon Min Kim	2026-07-17	Expert Systems with Applications	10.1016/j.eswa.2026.133688	OpenAlex forward citations	new to project	EXCLUDE	title/abstract	R4: Embodied navigation without explicit social capability (fails C)
Multimodal Knowledge Retrieval-Augmented Iterative Alignment for Satellite Commonsense Conversation	Qianli Ma	2025-09-01	International Joint Conference on Artificial Intelligence	10.24963/ijcai.2025/908	OpenAlex search	new to project	EXCLUDE	title/abstract	R1: Not robot/embodied navigation (incl. non-scholarly repository items)
Multimodal perception-driven decision-making for human-robot interaction: a survey	Wenzheng Zhao	2025-08-22	Frontiers in Robotics and AI	10.3389/frobt.2025.1604472	OpenAlex forward citations	new to project	EXCLUDE	title/abstract	R7: Survey, review, position paper or thesis (outside the screening scope)
NavDreamer: Video Models as Zero-Shot 3D Navigators	Xijie Huang	2026-07-16	IEEE Robotics and Automation Letters	10.1109/lra.2026.3713727	OpenAlex forward citations	new to project	EXCLUDE	title/abstract	R4: Embodied navigation without explicit social capability (fails C)
Navigation Explainability for Object Avoidance with LLMs for Assistive Robots	Courtney Young	2026-08-01	2026	10.1109/ro-man72286.2026.11715000	IEEE Xplore+OpenAlex search+OpenAlex forward citations	new to project	EXCLUDE	full text	R4: Borderline: LLM explains guide-dog obstacle avoidance to a blind user; no social-navigation capability among people (arXiv 2610.28920 read)

Title	First author	Date	Venue	DOI	Retrieved by	Status in project	Decision	Screening level	Reason
Object detection with multimodal large vision-language models: An in-depth review	Ranjan Sapkota	2025-07-31	Information Fusion	10.1016/j.inffus.2025.103575	OpenAlex forward citations	new to project	EXCLUDE	title/abstract	R7: Survey, review, position paper or thesis (outside the screening scope)
On the Vulnerability of LLM/VLM-Controlled Robotics	Xiyang Wu	2025-10-19	IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)	10.1109/iros60139.2025.11246863	OpenAlex forward citations	new to project	EXCLUDE	title/abstract	R3: Manipulation / HRI / perception without mobile-navigation behaviour (fails A)
On-Edge Task Planning with Large Language Models for Service Robotics	Alejandro González-Cantón	2025-10-13	Lecture notes in computer science	10.1007/978-3-032-08462-0_28	OpenAlex search	new to project	EXCLUDE	title/abstract	R3: Manipulation / HRI / perception without mobile-navigation behaviour (fails A)
Open-Vocabulary, Context-Aware Robot Navigation on Construction Sites: Integrating LLM-Driven Value Map Composition with Hierarchical Scene Graphs	Xiayu Zhao	2026-08-27	ASCE Construction Research Congress (CRC)	10.1061/9780784486979.022	OpenAlex search	new to project	EXCLUDE	title/abstract	R4: Embodied navigation without explicit social capability (fails C)
Paleolithic Maritime Navigation, Co-Evolution of Complex Phonetic Syntax, and the Linguistic-Institutional Lineage of Ancient Japanese Defensive Guilds: From Upper Paleolithic Obsidian Seafaring to the Proto-Historic 'Be' Systems of Mononobe and Hasebe	Yoko Hasebe	2026-10-04	Zenodo (CERN European Organization for Nuclear Research)	10.5281/zenodo.23131981	OpenAlex search	new to project	EXCLUDE	title/abstract	R1: Not robot/embodied navigation (incl. non-scholarly repository items)
Parallel Memetic-Computational Loops: A Thurmondian Analysis of Valence in Biological and Silicon Substrates	Thurmond, Justin	2025-12-19	Zenodo (CERN European Organization for Nuclear Research)	10.5281/zenodo.17982856	OpenAlex search	new to project	EXCLUDE	title/abstract	R1: Not robot/embodied navigation (incl. non-scholarly repository items)
Perception to Action with Vision-Language-Action Models for Fast and Reliable Decision Making in Dynamic Environments	Yeryeong Cho	2025-10-14	International Conference on Information and Communication Technology Convergence (ICTC)	10.1109/ictc66702.2025.11388300	OpenAlex forward citations	new to project	EXCLUDE	title/abstract	R4: Embodied navigation without explicit social capability (fails C)
PIVOT: A Framework for High-Fidelity Facade PV Layouts via Generative Seeding and Differentiable Optimization	Dongxu Zhuang	2026-01-01	IEEE Transactions on Automation Science and Engineering	10.1109/tase.2026.3701997	OpenAlex forward citations	new to project	EXCLUDE	title/abstract	R1: Not robot/embodied navigation (incl. non-scholarly repository items)
POLITE: Preferences Combined with Highlights in Reinforcement Learning	Simon Holk	2024-05-13	Proceedings - IEEE International Conference on Robotics and Automation/Proceedings	10.1109/icra57147.2024.10610505	OpenAlex forward citations	new to project	EXCLUDE	title/abstract	R3: Manipulation / HRI / perception without mobile-navigation behaviour (fails A)
Position Paper: A New Paradigm for Robot Multimodal Understanding and Decision-Making with Large Language Models as the Cognitive Core	Yulai Zhang	2026-06-01		10.1109/ijcnn69854.2026.11728577	OpenAlex forward citations	new to project	EXCLUDE	title/abstract	R7: Survey, review, position paper or thesis (outside the screening scope)
Real-time direction-aware time-to-collision reasoning with pose-guided light detection and ranging grounding for pedestrian safety on autonomous ground vehicles	Trung Thị Hoa Trang Nguyen	2026-07-14	Engineering Applications of Artificial Intelligence	10.1016/j.engappai.2026.115660	OpenAlex forward citations	new to project	EXCLUDE	full text	R5: Full text read (EAAI 2026): YOLOv8-Pose and Kalman-filter time-to-collision reasoning; no FM
Real-World Deployment of an LLM-Enabled Voice-Commanded UGV for Logistics in SAR Missions	Alejandro Jarabo-Peñas	2025-10-29		10.1109/ssrr68451.2025.11391314	OpenAlex search	new to project	EXCLUDE	title/abstract	R4: Embodied navigation without explicit social capability (fails C)
Reducing Latency in LLM-Based Natural Language Commands Processing for Robot Navigation			2025		IEEE Xplore	new to project	EXCLUDE	title/abstract	R4: Embodied navigation without explicit social capability (fails C)
Research on Navigation Methods Based on LLMs	Anlong Zhang	2025-04-22	arXiv (Cornell University)	10.48550/arxiv.2504.15600	OpenAlex search	new to project	EXCLUDE	title/abstract	R4: Embodied navigation without explicit social capability (fails C)
Response Memo — Reclassification Request — LLM-2025-1222-HALLU ├─ DOI: 10.5281/zenodo.18368614 — Crimson Hexagon Archive	Jack E. Feist	2026-01-25	Zenodo (CERN European Organization for Nuclear Research)	10.5281/zenodo.18368613	OpenAlex search	new to project	EXCLUDE	title/abstract	R1: Not robot/embodied navigation (incl. non-scholarly repository items)
Review of automatic row alignment technology for intelligent agricultural machinery in the field	Shibin Cui	2026-03-28	Smart Agricultural Technology	10.1016/j.atech.2026.102059	OpenAlex forward citations	new to project	EXCLUDE	title/abstract	R7: Survey, review, position paper or thesis (outside the screening scope)
RO-VLMap: Real-Time Occupancy-Aware Visual Language Mapping for Robust Robot Navigation			2026		IEEE Xplore	new to project	EXCLUDE	title/abstract	R4: Embodied navigation without explicit social capability (fails C)
Roadmap AI	guru g	2026-04-13	Zenodo (CERN European Organization for Nuclear Research)	10.5281/zenodo.19557176	OpenAlex search	new to project	EXCLUDE	title/abstract	R1: Not robot/embodied navigation (incl. non-scholarly repository items)
Robot Navigation in Complex Scenarios	Jing Liang	2025-01-01	University Libraries (University of Maryland)	10.13016/qwpv-oium	OpenAlex search	new to project	EXCLUDE	title/abstract	R7: Survey, review, position paper or thesis (outside the screening scope)
Robot Navigation in Dynamic and Crowded Environments	Xiuye Tao	2025-06-11	IEEE Transactions on Control Systems Technology	10.1109/tcst.2025.3575687	OpenAlex forward citations	new to project	EXCLUDE	title/abstract	R5: Social navigation without an FM role in the navigation method (fails B)
Robotic-CLIP: Fine-Tuning CLIP on Action Data for Robotic Applications			2025		IEEE Xplore	new to project	EXCLUDE	title/abstract	R3: Manipulation / HRI / perception without mobile-navigation behaviour (fails A)
RSG-VLN: Relevance Semantic Map Guided Vision-Language Navigation			2025		IEEE Xplore	new to project	EXCLUDE	title/abstract	R4: Embodied navigation without explicit social capability (fails C)
Rule-VLN: Bridging Perception and Compliance via Semantic Reasoning and Geometric Rectification	Jiawen Wen	2026-01-01	Lecture notes in computer science	10.1007/978-3-032-37252-9_26	OpenAlex forward citations	new to project	EXCLUDE	full text	R4: Full text read (arXiv 2604.16993): traffic-signage compliance in street-view VLN; no people (fails C)
Safe and Efficient Navigation Considering Texting Pedestrians via Attention-Aware Cost-Map Tuning	Shunya Tadano	2026-08-01	IEEE International Conference on Robot and Human Interactive Communication (RO-MAN)	10.1109/ro-man72286.2026.11714772	OpenAlex forward citations	new to project	EXCLUDE	title/abstract	R5: Abstract (IEEE): attention-aware costmap uses a Random Forest on skeletal features; no FM
SAHABAT: An Interactive LLM-Powered Tour Guide Robot for IIUM Gallery			2026		IEEE Xplore	new to project	EXCLUDE	title/abstract	R4: Embodied navigation without explicit social capability (fails C)
SCAN: Scene-context-aware Navigation for Social VR Exploration	Xuehuai Shi	2025-01-01	SSRN Electronic Journal	10.2139/ssrn.5802728	OpenAlex search	new to project	EXCLUDE	title/abstract	R1: Not robot/embodied navigation (incl. non-scholarly repository items)
SCU-T-MPC: Risk-Aware and Socially Compliant Topological Motion Planning Under Uncertainty	Bo Ye	2026-01-01	IET Cyber-Systems and Robotics	10.1049/csy2.70051	OpenAlex forward citations	new to project	EXCLUDE	title/abstract	R5: Social navigation without an FM role in the navigation method (fails B)
Search-on-Graph: Iterative Informed Navigation for Large Language Model Reasoning on Knowledge Graphs	Jia Ao Sun	2026-08-06	ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD)	10.1145/3770855.3817964	OpenAlex search	new to project	EXCLUDE	title/abstract	R1: Not robot/embodied navigation (incl. non-scholarly repository items)
Select2Plan: Training-Free ICL-Based Planning Through VQA and Memory Retrieval	Davide Buoso	2025-09-04	IEEE Robotics and Automation Letters	10.1109/lra.2025.3606790	OpenAlex forward citations	new to project	EXCLUDE	title/abstract	R4: Embodied navigation without explicit social capability (fails C)
Selected Motion Control Problems for a Robotic Guide Dog	Artur Babiarz	2026-10-01	Applied Sciences	10.3390/app16199744	OpenAlex forward citations	new to project	EXCLUDE	title/abstract	R5: Record without abstract; the title describes motion control for a robotic guide dog and names no FM (fails B)
Semantic orientation for indoor navigation system using large language models	Marzena Halama	2025-11-24	Scientific Reports	10.1038/s41598-025-25281-0	OpenAlex search	new to project	EXCLUDE	title/abstract	R6: Navigation assistance for people (fails A)
SEMANTIC PHYSICS: THE INWARD TURN Competing Ontologies and the Convergence Horizon — Crimson Hexagon Archive	Lee Sharks	2026-02-24	Zenodo (CERN European Organization for Nuclear Research)	10.5281/zenodo.18759452	OpenAlex search	new to project	EXCLUDE	title/abstract	R1: Not robot/embodied navigation (incl. non-scholarly repository items)
Semantics-Aware Path Planning for Mobile Robots in PV Installations	Dimitrios Alexiou	2025-10-15	IEEE International Conference on Imaging Systems and Techniques (IST)	10.1109/ist66504.2025.11268432	OpenAlex forward citations	new to project	EXCLUDE	title/abstract	R4: Embodied navigation without explicit social capability (fails C)

Title	First author	Date	Venue	DOI	Retrieved by	Status in project	Decision	Screening level	Reason
SEQUEL: Semi-Supervised Preference-based RL with Query Synthesis via Latent Interpolation	Daniel Marta	2024-05-13	Proceedings - IEEE International Conference on Robotics and Automation/Proceedings	10.1109/icra57147.2024.10610534	OpenAlex forward citations	new to project	EXCLUDE	title/abstract	R3: Manipulation / HRI / perception without mobile-navigation behaviour (fails A)
Shared Trajectory-Based Multi-Policy Decision-Making for Socially Compliant Robot Navigation in Dense Crowds	T H Wang	2026-01-01	IEEE Transactions on Automation Science and Engineering	10.1109/tase.2026.3711907	OpenAlex forward citations	new to project	EXCLUDE	title/abstract	R5: Abstract (IEEE): sampling-based multi-policy planner; no FM
Shielding for Socially Appropriate Robot Listening Behaviors	Sarah Gillet	2024-08-26	IEEE International Conference on Robot and Human Interactive Communication (RO-MAN)	10.1109/ro-man60168.2024.10731356	OpenAlex forward citations	new to project	EXCLUDE	title/abstract	R3: Manipulation / HRI / perception without mobile-navigation behaviour (fails A)
Simulation of the operation of an autonomous wheelchair for patients with amyotrophic lateral sclerosis with deep reinforcement learning algorithms, large language model, and tracking the patient's iris	Armando de Jesús Plasencia Salgueiro	2026-01-01	Elsevier eBooks	10.1016/b978-0-443-30106-3.00020-8	OpenAlex search	new to project	EXCLUDE	title/abstract	R4: Record without abstract; the title names deep RL, an LLM and iris tracking for the autonomous wheelchair of a patient with ALS but indicates no social-navigation capability among people (fails C)
Situated Intent Alignment for Grounded Goal Evaluation in Assistive Robots	Alicia Mora	2026-08-01	IEEE International Conference on Robot and Human Interactive Communication (RO-MAN)	10.1109/ro-man72286.2026.11714736	OpenAlex forward citations	new to project	EXCLUDE	title/abstract	R3: Manipulation / HRI / perception without mobile-navigation behaviour (fails A)
Smarter following behavior: an adaptive parameter tuning navigation method for unmanned agricultural vehicles	Yihao Wu	2026-06-07	Computers & Industrial Engineering	10.1016/j.cie.2026.112163	OpenAlex forward citations	new to project	EXCLUDE	title/abstract	R4: Embodied navigation without explicit social capability (fails C)
SmartSight: A Multilingual, Context-Aware Navigation and Safety Assistant for the Visually Impaired	Marisetti Harshini	2026-02-06		10.1109/ccic68129.2026.11485917	OpenAlex forward citations	new to project	EXCLUDE	title/abstract	R6: Navigation assistance for people (fails A)
Social cognitive architecture for NPC groups: integration of transformer theory of mind and hierarchical reinforcement learning	Zhenyu Qiu	2026-07-07	Artificial Intelligence Review	10.1007/s10462-026-11630-6	OpenAlex forward citations	new to project	EXCLUDE	title/abstract	R1: Not robot/embodied navigation (incl. non-scholarly repository items)
Social Navigation as Gameplay: A Hybrid Social Physics Engine for Deduction Games	Jingrong Wang	2026-07-30	International Conference on the Foundations of Digital Games (FDG)	10.1145/3815598.3815706	OpenAlex search	new to project	EXCLUDE	title/abstract	R1: Not robot/embodied navigation (incl. non-scholarly repository items)
Social Theory Should Be a Structural Prior for Agentic AI: A Formal Framework for Multi-Agent Social Systems	Lynnette Hui Xian Ng	2026-05-08	arXiv (Cornell University)	10.48550/arxiv.2605.07069	OpenAlex search	new to project	EXCLUDE	title/abstract	R1: Not robot/embodied navigation (incl. non-scholarly repository items)
SociaRI: Enhancing Social Robot Acceptance through Ontology-Based Context Modeling and LLM-Driven Dialogue	Luca Greco	2026-09-27	Lecture notes in computer science	10.1007/978-3-032-39518-4_32	OpenAlex forward citations	new to project	EXCLUDE	title/abstract	R3: Manipulation / HRI / perception without mobile-navigation behaviour (fails A)
Spatial Context-Aware VLM-Assisted Goal Reasoning for Navigation in Guide Service Robots			2026		IEEE Xplore	new to project	EXCLUDE	title/abstract	R4: Embodied navigation without explicit social capability (fails C)
Spiritual Responsibility AI: Aligning Emerging Tech with Natural Intelligence (Polymathic) - 2026 Trending Report created by Xpirit AI in collaboration with other LLM models	Marcelo Marshall De Siqueira	2026-01-03	Zenodo (CERN European Organization for Nuclear Research)	10.5281/zenodo.18140122	OpenAlex search	new to project	EXCLUDE	title/abstract	R1: Not robot/embodied navigation (incl. non-scholarly repository items)
SSR-ZSON: Zero-Shot Object Navigation via Spatial-Semantic Relations within a Hierarchical Exploration Framework			2026		IEEE Xplore	new to project	EXCLUDE	title/abstract	R4: Embodied navigation without explicit social capability (fails C)
STAMP: Spatial-Temporal Anchored Motion Planning for Zero-Shot Continuous Vision-and-Language Navigation	T Liu	2026-06-10	Sensors	10.3390/s26123698	OpenAlex forward citations	new to project	EXCLUDE	title/abstract	R4: Embodied navigation without explicit social capability (fails C)
Standards-Aligned Ethical Gating and Decision Telemetry for LLM-Assisted Mobile Robot Navigation: A Simulation Study			2026		IEEE Xplore	new to project	EXCLUDE	full text	R4: Full text read (BigDataService 2026): no people in any scenario; privacy means stopping at a signed restricted zone; the follower is a robot
TARIC: Memory-Augmented Traversability-Aware Outdoor VLN Under Interrupted Semantic Cues	Tianle Zeng	2026-06-25	IEEE Robotics and Automation Letters	10.1109/lra.2026.3707347	OpenAlex forward citations	new to project	EXCLUDE	title/abstract	R4: Embodied navigation without explicit social capability (fails C)
Temporal Semantic Grounding for Vision-Language Navigation: A Multi-Agent Approach to Persistent Object Tracking in Indoor Environments			2026		IEEE Xplore	new to project	EXCLUDE	title/abstract	R4: Embodied navigation without explicit social capability (fails C)
Texting-While-Walking Detection in Real-World Environments Using Vision-Language Models with Prompt Engineering			2026		IEEE Xplore	new to project	EXCLUDE	title/abstract	R3: Manipulation / HRI / perception without mobile-navigation behaviour (fails A)
The Paradigm Shift from Graphical User Interfaces to Agentic API Ecosystems	Akash Narayan	2026-07-09	Zenodo (CERN European Organization for Nuclear Research)	10.5281/zenodo.21272861	OpenAlex search	new to project	EXCLUDE	title/abstract	R1: Not robot/embodied navigation (incl. non-scholarly repository items)
Towards Generalized Context-Aware Navigation for Mobile Robots in Human-Centered Environments	Narasimhan, Siddarth	2025-10-01	TSpace (University of Toronto)		OpenAlex search	new to project	EXCLUDE	title/abstract	R7: Survey, review, position paper or thesis (outside the screening scope)
Towards Safer Mobile Agents: Scalable Generation and Evaluation of Diverse Scenarios for VLMs	T. Taniguchi	2026-08-03	Lecture notes in computer science	10.1007/978-3-032-31335-5_38	OpenAlex forward citations	new to project	EXCLUDE	title/abstract	R4: Embodied navigation without explicit social capability (fails C)
TraveLLM: Could you plan my new public transit route in face of a network disruption?	Fang, Bowen	2024-07-20	arXiv (Cornell University)	10.48550/arxiv.2407.14926	OpenAlex search	new to project	EXCLUDE	title/abstract	R1: Not robot/embodied navigation (incl. non-scholarly repository items)
TrustNavGPT: Modeling Uncertainty to Improve Trustworthiness of Audio-Guided LLM-Based Robot Navigation			2024		IEEE Xplore	new to project	EXCLUDE	title/abstract	R4: Embodied navigation without explicit social capability (fails C)
Uncertainty-Calibrated Hamilton-Jacobi Value Learning for Safe Crowd Navigation			RA-L 2026		IEEE Xplore	new to project	EXCLUDE	title/abstract	R5: Abstract (IEEE): HJ reachability + U-Net; no FM
UniPR-3D: Towards Universal Visual Place Recognition with Visual Geometry Grounded Transformer	Tianchen Deng	2026-01-01	Lecture notes in computer science	10.1007/978-3-032-37624-4_15	OpenAlex forward citations	new to project	EXCLUDE	title/abstract	R1: Not robot/embodied navigation (incl. non-scholarly repository items)
Unsupervised Interpretable Human Intent Discovery in Teleoperation via Bayesian Nonparametric Learning and Multi-Modal LLMs	Zhaokun Chen	2026-09-02	IEEE Robotics and Automation Letters	10.1109/lra.2026.3730341	OpenAlex forward citations	new to project	EXCLUDE	title/abstract	R3: Manipulation / HRI / perception without mobile-navigation behaviour (fails A)
USER-VLM 360: Personalized Vision Language Models with User-aware Tuning for Social Human-Robot Interactions	Hamed Rahimi	2025-10-11	INTERNATIONAL CONFERENCE ON MULTIMODAL INTERACTION	10.1145/3716553.3750767	OpenAlex forward citations	new to project	EXCLUDE	title/abstract	R3: Manipulation / HRI / perception without mobile-navigation behaviour (fails A)
Ventura: Adapting Image Diffusion Models for Unified Task Conditioned Navigation	Arthur Zhang	2026-06-01	Proceedings - IEEE International Conference on Robotics and Automation/Proceedings	10.1109/icra57385.2026.11696635	OpenAlex search	new to project	EXCLUDE	title/abstract	R4: Embodied navigation without explicit social capability (fails C)
Vision Language Model-based Testing of Industrial Autonomous Mobile Robots	Jiahui Wu	2026-04-12	IEEE/ACM International Conference on Software Engineering: Software Engineering in Practice (ICSE-SEIP)	10.1145/3786583.3786920	OpenAlex search	new to project	EXCLUDE	full text	R4: Borderline, full text read (arXiv 2508.02338): VLM-generated human behaviours to test generic AMR safety requirements (collision, stability, efficiency); not a social-navigation capability

Title	First author	Date	Venue	DOI	Retrieved by	Status in project	Decision	Screening level	Reason
Vision language model-enhanced embodied intelligence for digital twin-assisted human-robot collaborative assembly	Changchun Liu	2025-09-02	Journal of Industrial Information Integration	10.1016/j.jii.2025.100943	OpenAlex forward citations	new to project	EXCLUDE	title/abstract	R3: Manipulation / HRI / perception without mobile-navigation behaviour (fails A)
Vision-and-Language Navigation: A Component-Centric Survey of Interactions, Coupling, and Deployment	Xiangxun Wu	2026-08-12	Applied Sciences	10.3390/app16168050	OpenAlex forward citations	new to project	EXCLUDE	title/abstract	R7: Survey, review, position paper or thesis (outside the screening scope)
Vision-Language Based Real-Time Assistive System for Outdoor Navigation of the Visually Impaired in Indian Urban Environments	Charan Sai Ponnada	2025-12-18	International Symposium on Advanced Electrical and Communication Technologies (ISAECT)	10.1109/isaect68904.2025.11318802	OpenAlex search	new to project	EXCLUDE	title/abstract	R6: Navigation assistance for people (fails A)
Vision-language model-based human-robot collaboration for smart manufacturing: A state-of-the-art survey	Junming Fan	2025-01-22	Frontiers of Engineering Management	10.1007/s42524-025-4136-9	OpenAlex forward citations	new to project	EXCLUDE	title/abstract	R7: Survey, review, position paper or thesis (outside the screening scope)
VISOR: A Vision-Language Model-Based Test Oracle for Testing Robots	Prasun Saurabh	2026-07-05		10.1145/3805760.3814910	OpenAlex forward citations	new to project	EXCLUDE	title/abstract	R4: Embodied navigation without explicit social capability (fails C)
Visual-language-guided task planning for horticultural robots	Jose Cuaran	2026-06-27	Computers and Electronics in Agriculture	10.1016/j.compag.2026.112058	OpenAlex forward citations	new to project	EXCLUDE	title/abstract	R4: Embodied navigation without explicit social capability (fails C)
VLM See, Robot Do: Human Demo Video to Robot Action Plan via Vision Language Model	Beichen Wang	2025-10-19	IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)	10.1109/iros60139.2025.11246682	OpenAlex forward citations	new to project	EXCLUDE	title/abstract	R3: Manipulation / HRI / perception without mobile-navigation behaviour (fails A)
VLM-RLPGS: A Cognitive Framework Using Vision–Language Model and Reinforcement Learning for Push–Grasp Synergy	Thanh Phương Nguyễn	2026-02-22	Arabian Journal for Science and Engineering	10.1007/s13369-026-11170-9	OpenAlex forward citations	new to project	EXCLUDE	title/abstract	R3: Manipulation / HRI / perception without mobile-navigation behaviour (fails A)
Weird AI Experiment: The Compliment Battle	Daniel Rosehill	2026-03-26	Zenodo (CERN European Organization for Nuclear Research)	10.5281/zenodo.19240928	OpenAlex search	new to project	EXCLUDE	title/abstract	R1: Not robot/embodied navigation (incl. non-scholarly repository items)
Where to Fuse in the VLM Era: A Survey on Integrating Knowledge into Object Goal Navigation	Bokeon Suh	2026-01-01	Transactions of Korean Society of Automotive Engineers	10.7467/ksae.2026.34.1.119	OpenAlex forward citations	new to project	EXCLUDE	title/abstract	R7: Survey, review, position paper or thesis (outside the screening scope)
Who Should I Approach? Visitor Selection for Museum Guide Robots Using Video-Language Models	Chanill Park	2026-08-01	IEEE International Conference on Robot and Human Interactive Communication (RO-MAN)	10.1109/ro-man72286.2026.11714912	OpenAlex forward citations	new to project	EXCLUDE	title/abstract	R3: Borderline: VLM ranks museum visitors for approach receptiveness from video clips; no navigation behaviour executed
Your Calendar Is Now a Negotiation	Daniel Rosehill	2026-04-01	Zenodo (CERN European Organization for Nuclear Research)	10.5281/zenodo.19361635	OpenAlex search	new to project	EXCLUDE	title/abstract	R1: Not robot/embodied navigation (incl. non-scholarly repository items)
Çok Modlu Yapay Zekâda Görsel Dil Modelleri: Mimari Temeller ve Sektörel Uygulamalar	İrem Cakcak	2026-03-18	Özgür Yayınları eBooks	10.58830/ozgur.pub1230.c4967	OpenAlex forward citations	new to project	EXCLUDE	title/abstract	R7: Survey, review, position paper or thesis (outside the screening scope)
VLM-GroNav: Robot Navigation Using Physically Grounded Vision-Language Models in Outdoor Environments	Mohamed Elnoor	2025-05-19	Proceedings - IEEE International Conference on Robotics and Automation/Proceedings	10.1109/icra55743.2025.11128264	OpenAlex forward citations	screened in Stage 2	ALREADY SCREENED (Stage 2)	title/abstract	Already screened in Stage 2 under the title variant “Robot Navigation Using Physically Grounded VLMs in Outdoor Environments (VLM-GroNav)”: EXCLUDE (fails C)
Act on What You See: Unlocking Safe Social Navigation in Vision-Language-Action Models	Qingzi Wang	2026-06-09	arXiv (Cornell University)	10.48550/arxiv.2606.10495	OpenAlex search	in corpus	ALREADY KNOWN	-	corpus: 2026_Wang_SALSA
AHead-VLN: A Hierarchical Reasoning Framework Guided by Foresight for Human-Aware Navigation	Junyi Zhang	2026-09-01	International Joint Conference on Artificial Intelligence	10.24963/ijcai.2026/696	OpenAlex search	in corpus	ALREADY KNOWN	-	corpus: 2026_Zhang_AHead-VLN
Arena 4.0: A Comprehensive ROS2 Development and Benchmarking Platform for Human-centric Navigation Using Generative-Model-based Environment Generation	Volodymyr Shcherbyna	2024-09-19	arXiv (Cornell University)	10.48550/arxiv.2409.12471	OpenAlex search	in corpus	ALREADY KNOWN	-	corpus: 2024_Shcherbyna_Arena-4.0
AutoSpatial: Visual-Language Reasoning for Social Robot Navigation through Efficient Spatial Reasoning Learning	Yangzhe Kong	2025-10-19	2025	10.1109/iros60139.2025.11245987	IEEE Xplore+Google Scholar+OpenAlex search+OpenAlex forward citations	in corpus	ALREADY KNOWN	-	corpus: 2025_Kong_AutoSpatial
Behav: Behavioral Rule Guided Autonomy Using VLMs for Robot Navigation in Outdoor Scenes	Kasun Weerakoon	2025-05-19	2025	10.1109/icra55743.2025.11127890	IEEE Xplore+OpenAlex forward citations	in corpus	ALREADY KNOWN	-	corpus: 2025_Weerakoon_BehAV
Beyond Text: Utilizing Vocal Cues to Improve Decision Making in LLMs for Robot Navigation Tasks	Xingpeng Sun	2024-02-05	arXiv (Cornell University)	10.48550/arxiv.2402.03494	OpenAlex search	screened in Stage 2	ALREADY KNOWN	-	screened_stage2: EXCLUDE
CoNVOI: Context-aware Navigation using Vision Language Models in Outdoor and Indoor Environments	Adarsh Jagan Sathyamoorthy	2024-10-14	2024	10.1109/iros58592.2024.10802716	IEEE Xplore+OpenAlex search	in corpus	ALREADY KNOWN	-	corpus: 2024_Sathyamoorthy_CoNVOI
CrowdVLA: Embodied Vision-Language-Action Agents for Context-Aware Crowd Simulation	Juyeong Hwang	2026-04-07	arXiv (Cornell University)	10.48550/arxiv.2604.05525	OpenAlex search	screened in Stage 2	ALREADY KNOWN	-	screened_stage2: BACKGROUND ONLY
DPed-VLN: A Benchmark for Socially Compliant Vision-and-Language Navigation in Dynamic Pedestrian Environments	Haojie Dai	2026-09-18	arXiv (Cornell University)	10.48550/arxiv.2609.21504	OpenAlex search	in corpus	ALREADY KNOWN	-	corpus: 2026_Dai_DPed-VLN
E-Socialnav: Efficient Socially Compliant Navigation with Language Models	Ling Xiao	2026-04-21	2026	10.1109/icassp55912.2026.11462076	IEEE Xplore+OpenAlex forward citations	in corpus	ALREADY KNOWN	-	corpus: 2026_Xiao_E-SocialNav
Enhancing Lightweight Vision Language Models through Group Competitive Learning for Socially Compliant Navigation	Xinyu Zhang	2026-03-12	arXiv (Cornell University)	10.48550/arxiv.2603.11447	OpenAlex search	in corpus	ALREADY KNOWN	-	corpus: 2026_Zhang_GCL
Evaluating Efficiency and Engagement in Scripted and LLM-Enhanced Human-Robot Interactions	Tim Schreiter	2025-03-04	HRI 2025	10.1109/hri61500.2025.10974124	IEEE Xplore+OpenAlex search	screened in Stage 2	ALREADY KNOWN	-	screened_stage2: EXCLUDE
EvoPlan: Evolutionary Neuro-Symbolic Robot Planning with Spatio-Temporal Guarantees	Bhavya Sai Nukapotula	2026-07-07	arXiv (Cornell University)	10.48550/arxiv.2607.06724	OpenAlex search	screened in Stage 2	ALREADY KNOWN	-	screened_stage2: EXCLUDE
Few-Shot Inference of Human Perceptions of Robot Performance in Social Navigation Scenarios	Qiping Zhang	2025-12-17	arXiv (Cornell University)	10.48550/arxiv.2512.16019	OpenAlex search	in corpus	ALREADY KNOWN	-	corpus: 2025_Zhang_FewShot-Perception
From Obstacles to Etiquette: Robot Social Navigation With VLM-Informed Path Selection	Zilin Fang	2026-02-09	RA-L 2026	10.1109/lra.2026.3662586	IEEE Xplore+Google Scholar+OpenAlex search+OpenAlex forward citations	in corpus	ALREADY KNOWN	-	corpus: 2026_Fang_Obstacles-to-Etiquette
G2-Nav: Grounded and Guarded Vision-Language Costmaps for Robot Social Navigation	Yuwen Liao	2026-07-18	arXiv (Cornell University)	10.48550/arxiv.2607.16956	OpenAlex search	in corpus	ALREADY KNOWN	-	corpus: 2026_Liao_G2-Nav
GameChat: Multi-LLM Dialogue for Safe, Agile, and Socially Optimal Multi-Agent Navigation in Constrained Environments			2025		IEEE Xplore	screened in Stage 2	ALREADY KNOWN	-	screened_stage2: EXCLUDE
GenerativeMPC: VLM-RAG-guided Whole-Body MPC with Virtual Impedance for Bimanual Mobile Manipulation	Marcelino Julio Fernando	2026-04-21	arXiv (Cornell University)	10.48550/arxiv.2604.19522	OpenAlex search	screened in Stage 2	ALREADY KNOWN	-	screened_stage2: EXCLUDE
GSON: A Group-Based Social Navigation Framework With Large Multimodal Model	Shangyi Luo	2025-08-01	2025	10.1109/lra.2025.3595038	IEEE Xplore+OpenAlex search+OpenAlex forward citations	in corpus	ALREADY KNOWN	-	corpus: 2025_Luo_GSON

Title	First author	Date	Venue	DOI	Retrieved by	Status in project	Decision	Screening level	Reason
HCSG: Human-Centric Semantic-Geometric Reasoning for Vision-Language Navigation	Haoxuan Xu	2026-05-13	arXiv (Cornell University)	10.48550/arxiv.2605.13321	OpenAlex search	in corpus	ALREADY KNOWN	-	corpus: 2026_Xu_HCSG
Hey Robot! Personalizing Robot Navigation Through Model Predictive Control with a Large Language Model			2025		IEEE Xplore	in corpus	ALREADY KNOWN	-	corpus: 2025_Martinez-Baselga_Hey-Robot
HRIBench: Benchmarking Vision-Language Models for Real-Time Human Perception in Human-Robot Interaction	Zhonghao Shi	2025-06-25	arXiv (Cornell University)	10.48550/arxiv.2506.20566	OpenAlex search	screened in Stage 2	ALREADY KNOWN	-	screened_stage2: EXCLUDE
Human-Aware Social Navigation With Comfort Space for Mobile Robots in Human–Robot Coexistence Environments	Ying Zhang	2025-11-26	IEEE Transactions on Computational Social Systems	10.1109/tcss.2025.3627386	OpenAlex forward citations	screened in Stage 7	ALREADY KNOWN	-	screened_stage7: freshness
Language and Sketching: An LLM-driven Interactive Multimodal Multitask Robot Navigation Framework	Weiqin Zu	2024-05-13	ICRA 2024	10.1109/icra57147.2024.10611462	IEEE Xplore+OpenAlex search	screened in Stage 2	ALREADY KNOWN	-	screened_stage2: EXCLUDE
Language as Cost: Proactive Hazard Mapping using VLM for Robot Navigation			2025		IEEE Xplore	screened in Stage 2	ALREADY KNOWN	-	screened_stage2: EXCLUDE
Learning to Tune Like an Expert: Interpretable and Scene-Aware Navigation via MLLM Reasoning and CVAE-Based Adaptation	Yanbo Wang	2026-01-01	IEEE Transactions on Automation Science and Engineering	10.1109/tase.2026.3673004	OpenAlex forward citations	in corpus	ALREADY KNOWN	-	corpus: 2026_Wang_LE-Nav
Lightweight Visual Reasoning for Socially-Aware Robots			2026		IEEE Xplore	screened in Stage 2	ALREADY KNOWN	-	screened_stage2: EXCLUDE
LISN: Language-Instructed Social Navigation with VLM-based Controller Modulating	Junting Chen	2026-06-01	ICRA 2026	10.1109/icra57385.2026.11695757	IEEE Xplore+Google Scholar+OpenAlex search	in corpus	ALREADY KNOWN	-	corpus: 2026_Chen_LISN
LiSSEN: Generating Language-Grounded Safe Social Navigation in Crowded Environments	Jordan Cosio	2026-08-01	IEEE International Conference on Robot and Human Interactive Communication (RO-MAN)	10.1109/ro-man72286.2026.11714560	OpenAlex search	screened in Stage 7	ALREADY KNOWN	-	screened_stage7: freshness
LLM-Enhanced Path Planning: Safe and Efficient Autonomous Navigation with Instructional Inputs	Pranav Doma	2024-12-03	arXiv (Cornell University)	10.48550/arxiv.2412.02655	OpenAlex search	screened in Stage 2	ALREADY KNOWN	-	screened_stage2: EXCLUDE
MAction-SocialNav: Multi-Action Socially Compliant Navigation via Reasoning-Enhanced Prompt Tuning	Zishuo Wang	2025-12-25	RA-L 2026	10.48550/arxiv.2512.21722	IEEE Xplore+Google Scholar+OpenAlex search+OpenAlex forward citations	in corpus	ALREADY KNOWN	-	corpus: 2026_Wang_MAction-SocialNav
MAPF-World: Action World Model for Multi-Agent Path Finding	Zhanjiang Yang	2026-08-21	RA-L 2026	10.1109/lra.2026.3726342	IEEE Xplore+OpenAlex search	screened in Stage 2	ALREADY KNOWN	-	screened_stage2: EXCLUDE
MOSU: Autonomous Long-range Robot Navigation with Multi-modal Scene Understanding	Jing Liang	2025-07-07	arXiv (Cornell University)		OpenAlex search	in corpus	ALREADY KNOWN	-	corpus: 2025_Liang_MOSU
Multi-Agent LLM Actor-Critic Framework for Social Robot Navigation	Weizheng Wang	2025-03-12	arXiv (Cornell University)	10.48550/arxiv.2503.09758	OpenAlex search	in corpus	ALREADY KNOWN	-	corpus: 2025_Wang_SAMALM
MUSON: A Reasoning-oriented Multimodal Dataset for Socially Compliant Navigation in Urban Environments	Zhuonan Liu	2025-12-28	arXiv (Cornell University)	10.48550/arxiv.2512.22867	OpenAlex search	in corpus	ALREADY KNOWN	-	corpus: 2026_Liu_MUSON
Narrate2Nav: Real-Time Visual Navigation with Implicit Language Reasoning in Human-Centric Environments			2026		IEEE Xplore	in corpus	ALREADY KNOWN	-	corpus: 2026_Payandeh_Narrate2Nav
OLiVia-Nav: An Online Lifelong Vision Language Approach for Mobile Robot Social Navigation	Siddarth Narasimhan	2025-05-19	ICRA 2025	10.1109/icra55743.2025.11128004	IEEE Xplore+OpenAlex search	in corpus	ALREADY KNOWN	-	corpus: 2025_Narasimhan_OLiVia-Nav
Perception–Awareness–Decision: Socially-Aware Robot Navigation and Interaction	Edison Jair Bejarano Sepúlveda	2026-03-12	ACM/IEEE International Conference on Human-Robot Interaction (HRI)	10.1145/3776734.3794394	OpenAlex forward citations	in corpus	ALREADY KNOWN	-	corpus: 2026_Bejarano_PAD
PhysNav-DG: A Novel Adaptive Framework for Robust VLM-Sensor Fusion in Navigation Applications	Trisanth Srinivasan	2025-06-11	CVPRW 2025	10.1109/cvprw67362.2025.00653	IEEE Xplore+OpenAlex search	screened in Stage 2	ALREADY KNOWN	-	screened_stage2: EXCLUDE
PISHYAR: A Socially Intelligent Smart Cane for Indoor Social Navigation and Multimodal Human-Robot Interaction for Visually Impaired People	Mahdi Haghighat Joo	2026-02-13	arXiv (Cornell University)	10.48550/arxiv.2602.12597	OpenAlex search	screened in Stage 2	ALREADY KNOWN	-	screened_stage2: EXCLUDE
PREDILECT: Preferences Delineated with Zero-Shot Language-based Reasoning in Reinforcement Learning	Simon Holk	2024-03-10	HRI 2024	10.1145/3610977.3634970	IEEE Xplore+OpenAlex search	in corpus	ALREADY KNOWN	-	corpus: 2024_Holk_PREDILECT
Probing Collision Grounding in Vision-Language Models for Safe Human-Robot Collaboration	Jun Wang	2026-05-29	arXiv (Cornell University)	10.48550/arxiv.2605.31196	OpenAlex search	screened in Stage 2	ALREADY KNOWN	-	screened_stage2: EXCLUDE
Probing Prompt Design for Socially Compliant Robot Navigation with Vision Language Models	Ling Xiao	2026-01-21	arXiv (Cornell University)	10.48550/arxiv.2601.14622	OpenAlex search	screened in Stage 2	ALREADY KNOWN	-	screened_stage2: RELATED VERSION
Robots Ask the Way: Communication-Enabled Social Navigation	Valentino Sacco	2026-07-01	arXiv (Cornell University)	10.48550/arxiv.2607.01044	OpenAlex search	screened in Stage 2	ALREADY KNOWN	-	screened_stage2: EXCLUDE
SAATT Nav: a Socially Aware Autonomous Transparent Transportation Navigation Framework for Wheelchairs	Yutong Zhang	2026-03-14	arXiv (Cornell University)	10.48550/arxiv.2603.13698	OpenAlex search	in corpus	ALREADY KNOWN	-	corpus: 2026_Zhang_SAATT-Nav
Social robot navigation: a review and benchmarking of learning-based methods	Rashid Alyassi	2025-12-11	Frontiers in Robotics and AI	10.3389/frobt.2025.1658643	OpenAlex forward citations	in corpus	ALREADY KNOWN	-	corpus: 2025_Alyassi_Learning-SocialNav-Review-Benchmark
Social-LLaVA: Enhancing Robot Navigation through Human-Language Reasoning in Social Spaces	Amirreza Payandeh	2024-12-30	arXiv (Cornell University)	10.48550/arxiv.2501.09024	OpenAlex search	screened in Stage 2	ALREADY KNOWN	-	screened_stage2: INCLUDE
Social-LLaVA: Enhancing Social Robot Navigation through Human-Language Reasoning	Amirreza Payandeh	2025-10-19	IROS 2025	10.1109/iros60139.2025.11247618	IEEE Xplore+Google Scholar+OpenAlex search+OpenAlex forward citations	in corpus	ALREADY KNOWN	-	corpus: 2025_Payandeh_Social-LLaVA
SocialGrid: A Benchmark for Planning and Social Reasoning in Embodied Multi-Agent Systems	Hikaru Shindo	2026-04-17	arXiv (Cornell University)	10.48550/arxiv.2604.16022	OpenAlex search	screened in Stage 2	ALREADY KNOWN	-	screened_stage2: EXCLUDE
Socially aware navigation for mobile robots: a survey on deep reinforcement learning approaches	Ibrahim Khalil Kabir	2026-01-01	Applied Intelligence	10.1007/s10489-025-06982-5	OpenAlex forward citations	in corpus	ALREADY KNOWN	-	corpus: 2026_Kabir_DRL-SAN-Survey
Socially-Aware Robot Navigation Enhanced by Bidirectional Natural Language Conversations Using Large Language Models	Congcong Wen	2025-10-19	IROS 2025	10.1109/iros60139.2025.11246738	IEEE Xplore+OpenAlex search	in corpus	ALREADY KNOWN	-	corpus: 2025_Wen_HSAC-LLM
Socialnav-moe: A mixture-of-experts vision language model for socially compliant navigation with reinforcement fine-tuning	Kawabata, Tomohito	2025-12-15	T Kawabata	10.48550/arxiv.2512.14757	Google Scholar+OpenAlex search	in corpus	ALREADY KNOWN	-	corpus: 2025_Kawabata_SocialNav-MoE
Socialnav-sub: Benchmarking vlms for scene understanding in social robot navigation	Michael J. Munje	2025-09-10	MJ Munje	10.48550/arxiv.2509.08757	Google Scholar+OpenAlex search	in corpus	ALREADY KNOWN	-	corpus: 2025_Munje_SocialNav-SUB
SocialNav: Training human-inspired foundation model for socially-aware embodied navigation	Ziyi Chen	2025-11-26	Z Chen	10.48550/arxiv.2511.21135	Google Scholar+OpenAlex search	in corpus	ALREADY KNOWN	-	corpus: 2026_Chen_SocialNav-CVPR

Title	First author	Date	Venue	DOI	Retrieved by	Status in project	Decision	Screening level	Reason
SocRATES: Towards Automated Scenario-based Testing of Social Navigation Algorithms	Shashank Rao Marpally	2024-12-27	arXiv (Cornell University)	10.48550/arxiv.2412.19595	OpenAlex search	in corpus	ALREADY KNOWN	-	corpus: 2024_Marpally_SocRATES
SONG: A Photorealistic 3D Gaussian Simulation Platform for Benchmarking Social Navigation	Weiqi Huang	2026-07-28	arXiv (Cornell University)	10.48550/arxiv.2607.25219	OpenAlex search	in corpus	ALREADY KNOWN	-	corpus: 2026_Huang_SONG
SOPD-SocialNav: Selective On-Policy Distillation for Vision-Language Social Navigation	Xinyu Zhang	2026-08-01	WRC SARA 2026	10.1109/wrcsara71458.2026.11732522	IEEE Xplore+OpenAlex search	in corpus	ALREADY KNOWN	-	corpus: 2026_Zhang_SOPD-SocialNav
STARS: From Spatiotemporal Dynamics to Social Representations in Human-Robot Interaction	Nathan Tsoi	2026-09-30	arXiv (Cornell University)	10.48550/arxiv.2609.40245	OpenAlex search	screened in Stage 2	ALREADY KNOWN	-	screened_stage2: EXCLUDE
Think When It Matters: Conditional VLM Reasoning for Social Navigation with RL Policies	Ali Ahmadi	2026-07-13	arXiv (Cornell University)	10.48550/arxiv.2607.10991	OpenAlex search	in corpus	ALREADY KNOWN	-	corpus: 2026_Ahmadi_HUMA
TongSIM: A General Platform for Simulating Intelligent Machines	Zhe Sun	2025-12-23	arXiv (Cornell University)	10.48550/arxiv.2512.20206	OpenAlex search	screened in Stage 2	ALREADY KNOWN	-	screened_stage2: EXCLUDE
Towards Socially Intelligent Robots: Research on Human-Aware Vision Language Navigation	Jiajun Chen	2026-09-13	Exploring Science Academic Conference Series	10.70267/aitia.2026216222	OpenAlex search	screened in Stage 7	ALREADY KNOWN	-	screened_stage7: freshness
Transparent Social Navigation for Autonomous Mobile Robots Via Vision-Language Models	Oluwadamilola Sotomi	2025-08-25	RO-MAN 2025	10.1109/ro-man63969.2025.11217693	IEEE Xplore+OpenAlex search+OpenAlex forward citations	in corpus	ALREADY KNOWN	-	corpus: 2025_Sotomi_Transparent-Social-Nav-VLM
Unifying Large Language Model and Deep Reinforcement Learning for Human-in-Loop Interactive Socially-aware Navigation	Weizheng Wang	2024-03-22	arXiv (Cornell University)	10.48550/arxiv.2403.15648	OpenAlex search	in corpus	ALREADY KNOWN	-	corpus: 2024_Wang_SALM
UrbanGround: From Local Perception to Spatial Agency in a Real-Scale City	Tianjie Ju	2026-08-27	arXiv (Cornell University)	10.48550/arxiv.2608.27456	OpenAlex search	screened in Stage 2	ALREADY KNOWN	-	screened_stage2: EXCLUDE
ViLAM: Distilling Vision-Language Reasoning into Attention Maps for Social Robot Navigation	Mohamed Elnoor	2025-03-12	arXiv (Cornell University)	10.48550/arxiv.2503.09820	OpenAlex search	in corpus	ALREADY KNOWN	-	corpus: 2026_Elnoor_ViLAM
Vision-Language Models for Deployable Social Robot Navigation: Bridging Semantic Reasoning and Low-Level Control	Runji Cai	2026-06-27	arXiv (Cornell University)	10.48550/arxiv.2606.28760	OpenAlex search	in corpus	ALREADY KNOWN	-	corpus: 2026_Cai_VLM-Deployable-SRN-Survey
VL-TGS: Trajectory Generation and Selection Using Vision Language Models in Mapless Outdoor Environments	Daeun Song	2025-04-10	IEEE Robotics and Automation Letters	10.1109/lra.2025.3559822	OpenAlex forward citations	in corpus	ALREADY KNOWN	-	corpus: 2025_Song_VL-TGS
VLM-Social-Nav: Socially Aware Robot Navigation Through Scoring Using Vision-Language Models	Daeun Song	2024-12-04	RA-L 2025	10.1109/lra.2024.3511409	IEEE Xplore+Google Scholar+OpenAlex search+OpenAlex forward citations	in corpus	ALREADY KNOWN	-	corpus: 2025_Song_VLM-Social-Nav
Walk with me: Long-horizon social navigation for human-centric outdoor assistance	Lingfeng Zhang	2026-04-29	L Zhang	10.48550/arxiv.2604.26839	Google Scholar+OpenAlex search	in corpus	ALREADY KNOWN	-	corpus: 2026_Zhang_Walk-With-Me

Table S2. Publication status of all corpus works

98 rows. Machine-readable version: `S2_publication_status.csv`.

Cite key	Category	Year	Venue	Status class	Peer reviewed	DOI	arXiv relation
Helbing1995SocialForce	S1	1995	Physical Review E	A: published, peer-reviewed	Yes	10.1103/PhysRevE.51.4282	arXiv copy posted 1998 after journal publication; journal version authoritative
Trautman2010UnfreezingIGP	S1	2010	2010 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)	A: published, peer-reviewed	Yes	10.1109/IROS.2010.5654369	Corpus copy is the published version
Kuderer2012FeatureBasedPrediction	S2	2012	Robotics: Science and Systems VIII (RSS 2012)	A: published, peer-reviewed	Yes	10.15607/RSS.2012.VIII.025	Corpus copy is the published version
Kruse2013HumanAwareNavigationSurvey	S0	2013	Robotics and Autonomous Systems	A: published, peer-reviewed	Yes	10.1016/j.robot.2013.05.007	Corpus copy is the published version
Kretzschmar2016SociallyCompliantIRL	S2	2016	The International Journal of Robotics Research	A: published, peer-reviewed	Yes	10.1177/0278364915619772	Corpus copy is an author manuscript; journal version authoritative
Okal2016BayesianIRL	S2	2016	2016 IEEE International Conference on Robotics and Automation (ICRA)	A: published, peer-reviewed	Yes	10.1109/ICRA.2016.7487452	Corpus copy is the published version
Charalampous2017SocialAwareNavigationSurvey	S0	2017	Robotics and Autonomous Systems	A: published, peer-reviewed	Yes	10.1016/j.robot.2017.03.002	Corpus copy is the published version
Chen2017CADRL	S2	2017	2017 IEEE International Conference on Robotics and Automation (ICRA)	A: published, peer-reviewed	Yes	10.1109/ICRA.2017.7989037	Corpus copy is the arXiv author version; published version authoritative
Chen2017SACADRL	S2	2017	2017 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)	A: published, peer-reviewed	Yes	10.1109/IROS.2017.8202312	Corpus copy is the arXiv author version; published version authoritative
Tai2018GAILDepth	S2	2018	2018 IEEE International Conference on Robotics and Automation (ICRA)	A: published, peer-reviewed	Yes	10.1109/ICRA.2018.8460968	Corpus copy is the arXiv author version; published version authoritative
Chen2019SARLCrowdNav	S2	2019	2019 International Conference on Robotics and Automation (ICRA)	A: published, peer-reviewed	Yes	10.1109/ICRA.2019.8794134	Corpus copy is the arXiv author version; published version authoritative
Tsoi2020SEAN	S6	2020	Proceedings of the 8th International Conference on Human-Agent Interaction (HAI	A: published, peer-reviewed	Yes	10.1145/3406499.3418760	Corpus copy is the arXiv author version; published version authoritative
Kastner2021ArenaRosnav	S6	2021	2021 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)	A: published, peer-reviewed	Yes	10.1109/IROS51168.2021.9636226	Corpus copy is the published IEEE version
Moller2021HumanAwareNavigationSurvey	S0	2021	Robotics and Autonomous Systems	A: published, peer-reviewed	Yes	10.1016/j.robot.2021.103837	Corpus copy is the arXiv author version; published version authoritative
Ichter2022SayCan	S3	2022	Proceedings of The 6th Conference on Robot Learning (CoRL 2022), PMLR 205	A: published, peer-reviewed	Yes	none	Corpus copy is the PMLR published version; arXiv 2204.01691v2 alternate
Biswas2022SocNavBench	S6	2022	ACM Transactions on Human-Robot Interaction	A: published, peer-reviewed	Yes	10.1145/3476413	Corpus copy is the arXiv author version; published version authoritative
Gao2022EvaluationSANReview	S0	2022	Frontiers in Robotics and AI	A: published, peer-reviewed	Yes	10.3389/frobt.2021.721317	Corpus copy is the published version
Karnan2022SCAND	S6	2022	IEEE Robotics and Automation Letters	A: published, peer-reviewed	Yes	10.1109/LRA.2022.3184025	Corpus copy is the arXiv author version; published version authoritative
Kim2022GroupEstimation	S2	2022	2022 22nd International Conference on Control, Automation and Systems (ICCAS)	A: published, peer-reviewed	Yes	10.23919/ICCAS55662.2022.10003761	Corpus copy is the published version
Shah2022LMNav	S3	2022	Proceedings of The 6th Conference on Robot Learning (CoRL 2022), PMLR 205	A: published, peer-reviewed	Yes	none	Corpus copy is the PMLR published version; arXiv 2207.04429v2 alternate

Cite key	Category	Year	Venue	Status class	Peer reviewed	DOI	arXiv relation
Zitkovich2023RT2	S3	2023	Proceedings of The 7th Conference on Robot Learning (CoRL 2023), PMLR 229	A: published, peer-reviewed	Yes	none	Corpus copy is the PMLR published version; arXiv 2307.15818v1 alternate
Driess2023PaLME	S3	2023	Proceedings of the 40th International Conference on Machine Learning (ICML 2023)	A: published, peer-reviewed	Yes	none	Corpus copy is the PMLR published version; arXiv 2303.03378v1 alternate
Huang2023VLMaps	S3	2023	2023 IEEE International Conference on Robotics and Automation (ICRA)	A: published, peer-reviewed	Yes	10.1109/ICRA48891.2023.10160969	Corpus copy is the arXiv author version; published version authoritative
Huang2023VoxPoser	S3	2023	Proceedings of The 7th Conference on Robot Learning (CoRL 2023), PMLR 229	A: published, peer-reviewed	Yes	none	Corpus copy is the PMLR published version; arXiv 2307.05973v2 alternate
Liang2023CodeasPolicies	S3	2023	2023 IEEE International Conference on Robotics and Automation (ICRA)	A: published, peer-reviewed	Yes	10.1109/ICRA48891.2023.10160591	Corpus copy is the arXiv author version; published version authoritative
Mavrogiannis2023CoreChallengesSurvey	S0	2023	ACM Transactions on Human-Robot Interaction	A: published, peer-reviewed	Yes	10.1145/3583741	Corpus copy is the arXiv author version; published version authoritative
Nguyen2023MuSoHu	S6	2023	2023 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)	A: published, peer-reviewed	Yes	10.1109/IROS55552.2023.10342447	Not reported
Shah2023GNM	S3	2023	2023 IEEE International Conference on Robotics and Automation (ICRA)	A: published, peer-reviewed	Yes	10.1109/ICRA48891.2023.10161227	Corpus copy is the arXiv author version; published version authoritative
Shah2023ViNT	S3	2023	Proceedings of The 7th Conference on Robot Learning (CoRL 2023), PMLR 229	A: published, peer-reviewed	Yes	none	Corpus copy is arXiv v2; PMLR version authoritative
Wang2023NaviSTAR	S2	2023	2023 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)	A: published, peer-reviewed	Yes	10.1109/IROS55552.2023.10341395	Corpus copy is the arXiv author version; published version authoritative
Hirose2024SACSoN	S6	2024	IEEE Robotics and Automation Letters	A: published, peer-reviewed	Yes	10.1109/LRA.2023.3329626	Not reported
Holk2024PREDILECT	S4	2024	Proceedings of the 2024 ACM/IEEE International Conference on Human-Robot Interac	A: published, peer-reviewed	Yes	10.1145/3610977.3634970	Not reported
Karwowski2024RequirementsPlanningEvaluationReview	S0	2024	Sensors	A: published, peer-reviewed	Yes	10.3390/s24092794	Corpus copy is the published version
Li2024HAVLN	S6	2024	Advances in Neural Information Processing Systems 37 (NeurIPS 2024, Datasets and	A: published, peer-reviewed	Yes	10.52202/079017-3795	Corpus copy is the arXiv author version; published version authoritative
Marpally2026SocRATES	S6	2026	IEEE Robotics and Automation Letters	A: published, peer-reviewed	Yes	10.1109/LRA.2026.3737041	arXiv v1 (different title) is the corpus copy; RA-L version authoritative
Mirsky2024ConflictAvoidanceSurvey	S0	2024	ACM Transactions on Human-Robot Interaction	A: published, peer-reviewed	Yes	10.1145/3647983	Corpus copy is the arXiv author version; published version authoritative
Raj2024HybridSocialNav	S2	2024	2024 IEEE International Conference on Robotics and Automation (ICRA)	A: published, peer-reviewed	Yes	10.1109/ICRA57147.2024.10611710	Corpus copy is the arXiv author version; published version authoritative
Sathyamoorthy2024CoNVOI	S4	2024	2024 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)	A: published, peer-reviewed	Yes	10.1109/IROS58592.2024.10802716	Corpus copy is the arXiv author version; published version authoritative
Shcherbyna2025Arena40	S6	2025	2025 IEEE International Conference on Robotics and Automation (ICRA)	A: published, peer-reviewed	Yes	10.1109/ICRA55743.2025.11127635	Corpus copy is the arXiv author version; published version authoritative
Singamaneni2024SANTaxonomySurvey	S0	2024	The International Journal of Robotics Research	A: published, peer-reviewed	Yes	10.1177/02783649241230562	Corpus copy is the arXiv author version; published version authoritative
Sridhar2024NoMaD	S3	2024	2024 IEEE International Conference on Robotics and Automation (ICRA)	A: published, peer-reviewed	Yes	10.1109/ICRA57147.2024.10610665	Corpus copy is the arXiv author version; published version authoritative
Wang2024SALM	S5	2024	arXiv preprint (arXiv:2403.15648v3)	C: preprint only	No	none	Not reported

Cite key	Category	Year	Venue	Status class	Peer reviewed	DOI	arXiv relation
Yokoyama2024VLFM	S3	2024	2024 IEEE International Conference on Robotics and Automation (ICRA)	A: published, peer-reviewed	Yes	10.1109/ICRA57147.2024.10610712	Corpus copy is the arXiv author version; published version authoritative
Yu2024PANav	S5	2024	arXiv preprint (arXiv:2410.04302v1)	C: preprint only	No	none	Not reported
Zhou2024NavGPT	S3	2024	Proceedings of the AAAI Conference on Artificial Intelligence (AAAI-24)	A: published, peer-reviewed	Yes	10.1609/aaai.v38i7.28597	Corpus copy is the arXiv author version; published version authoritative
Alyassi2025LearningSocialNavReviewBenchmark	S0	2025	Frontiers in Robotics and AI	A: published, peer-reviewed	Yes	10.3389/frobt.2025.1658643	Corpus copy is the published version
Francis2025PrinciplesGuidelines	S0	2025	ACM Transactions on Human-Robot Interaction	A: published, peer-reviewed	Yes	10.1145/3700599	Corpus copy is the arXiv author version; published version authoritative
Kawabata2025SocialNavMoE	S5	2025	arXiv:2512.14757v1 [cs.CV], 15 Dec 2025	C: preprint only	No	none	arXiv only
Kong2025AutoSpatial	S4	2025	2025 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)	A: published, peer-reviewed	Yes	10.1109/IROS60139.2025.11245987	Corpus copy is the arXiv author version; published version authoritative
Liang2025MOSU	S5	2025	International Symposium on Experimental Robotics (ISER) 2025 - per citation in F	B: accepted/presented per citing paper; proceedings not indexed	Accepted (not yet indexed)	none	Not reported
Luo2025GSON	S4	2025	IEEE Robotics and Automation Letters	A: published, peer-reviewed	Yes	10.1109/LRA.2025.3595038	Corpus copy is the arXiv author version; published version authoritative
MartinezBaselga2025HeyRobot	S4	2025	2025 IEEE International Conference on Robotics and Automation (ICRA)	A: published, peer-reviewed	Yes	10.1109/ICRA55743.2025.11128826	Not reported
Munje2025SocialNavSUB	S6	2025	Proceedings of The 9th Conference on Robot Learning (CoRL 2025), PMLR 305	A: published, peer-reviewed	Yes	none	Corpus copy is the PMLR published version; arXiv 2509.08757v1 alternate
Narasimhan2025OLiViaNav	S4	2025	2025 IEEE International Conference on Robotics and Automation (ICRA)	A: published, peer-reviewed	Yes	10.1109/ICRA55743.2025.11128004	Corpus copy is the arXiv author version; published version authoritative
Payandeh2025SocialLLaVA	S4	2025	2025 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)	A: published, peer-reviewed	Yes	10.1109/IROS60139.2025.11247618	Corpus copy is the published IROS version; arXiv 2501.09024 carries a different title (alt
Ruo2025LLMSocialCues	S4	2025	2025 34th IEEE International Conference on Robot and Human Interactive Communica	A: published, peer-reviewed	Yes	10.1109/RO-MAN63969.2025.11217843	Not reported
Saito2025VLMCrowdMapping	S4	2025	2025 IEEE International Conference on Real-time Computing and Robotics (RCAR)	A: published, peer-reviewed	Yes	10.1109/RCAR65431.2025.11139707	Not reported
Song2025VLTGS	S3	2025	IEEE Robotics and Automation Letters	A: published, peer-reviewed	Yes	10.1109/LRA.2025.3559822	Corpus copy is the arXiv author version; published version authoritative
Song2025VLMSocialNav	S4	2025	IEEE Robotics and Automation Letters	A: published, peer-reviewed	Yes	10.1109/LRA.2024.3511409	Corpus copy is the arXiv author version; published version authoritative
Sotomi2025TransparentSocialNavVLM	S4	2025	2025 34th IEEE International Conference on Robot and Human Interactive Communica	A: published, peer-reviewed	Yes	10.1109/RO-MAN63969.2025.11217693	Not reported
Wang2025SAMALM	S5	2025	arXiv preprint (arXiv:2503.09758v1)	C: preprint only	No	none	Not reported
Weerakoon2025BehAV	S4	2025	2025 IEEE International Conference on Robotics and Automation (ICRA)	A: published, peer-reviewed	Yes	10.1109/ICRA55743.2025.11127890	Corpus copy is the arXiv author version; published version authoritative
Wen2025HSACLLM	S4	2025	2025 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)	A: published, peer-reviewed	Yes	10.1109/IROS60139.2025.11246738	Corpus copy is the arXiv author version; published version authoritative
Zhang2025FewShotPerception	S6	2025	arXiv preprint (arXiv:2512.16019v1)	C: preprint only	No	none	Not reported
Ahmadi2026HUMA	S5	2026	arXiv preprint (arXiv:2607.10991v1)	C: preprint only (CoRL 2026 submission per arXiv comment; not an acceptance)	No	none	Not reported

Cite key	Category	Year	Venue	Status class	Peer reviewed	DOI	arXiv relation
BejaranoSepulveda2026PAD	S4	2026	Companion Proceedings of the 21st ACM/IEEE International Conference on Human-Rob	A: published, peer-reviewed	Yes	10.1145/3776734.3794394	Corpus copy is the published version
Cai2026VLMDeployableSRNSurvey	S0	2026	arXiv preprint (arXiv:2606.28760v1 [cs.RO], 27 Jun 2026)	C: preprint only	No	none	arXiv only; v1 only
Chen2026LISN	S4	2026	2026 IEEE International Conference on Robotics and Automation (ICRA)	A: published, peer-reviewed	Yes	10.1109/ICRA57385.2026.11695757	Not reported
Chen2026SocialNavCVPR	S4	2026	Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognitio	A: published, peer-reviewed	Yes	none	Corpus copy is the CVF open-access version; arXiv 2511.21135v2 alternate. IEEE Xplore DOI
Dai2026DPedVLN	S6	2026	arXiv preprint (arXiv:2609.21504v1)	C: preprint only	No	none	Not reported
Dong2026HAVLN20	S6	2026	arXiv:2503.14229v5 [cs.AI], 18 Aug 2026 (document dated July 10, 2026)	B: accepted per authors/arXiv; proceedings not yet available	Accepted (not yet indexed)	none	arXiv only; published version not yet available
Elnoor2026ViLAM	S5	2026	arXiv:2503.09820v2 [cs.RO], 6 Mar 2026	C: preprint only	No	none	arXiv only; no venue on UMD project page
Fang2026GroupJoining	S5	2026	arXiv preprint (arXiv:2609.28467v1)	C: preprint only	No	none	Not reported
Fang2026ObstaclestoEtiquette	S4	2026	IEEE Robotics and Automation Letters	A: published, peer-reviewed	Yes	10.1109/LRA.2026.3662586	Corpus copy is the arXiv author version; published version authoritative
Gandotra2026RBTLR	S5	2026	Companion Proceedings of the 21st ACM/IEEE International Conference on Human-Rob	D: conference paper; acceptance verified, peer review not established	Not established	10.1145/3776734.3794437	Not reported
Hu2026SPARKVLN	S5	2026	arXiv preprint (arXiv:2607.16806v1)	C: preprint only	No	none	Not reported
Huang2026SONG	S6	2026	arXiv preprint (arXiv:2607.25219v1)	C: preprint only	No	none	Not reported
Huo2026NORMNav	S4	2026	2026 IEEE International Conference on Robotics and Automation (ICRA)	A: published, peer-reviewed	Yes	10.1109/ICRA57385.2026.11696526	Not reported
Kabir2026DRLSANSurvey	S0	2026	Applied Intelligence	A: published, peer-reviewed	Yes	10.1007/s10489-025-06982-5	Corpus copy is the arXiv author version; published version authoritative
Liao2026G2Nav	S5	2026	arXiv:2607.16956v1 [cs.RO], 18 Jul 2026	B: accepted per authors/arXiv; proceedings not yet available	Accepted (not yet indexed)	none	arXiv only; published version not yet available
Liu2026MUSON	S6	2026	arXiv:2512.22867v2 [cs.CV], 3 Jul 2026	C: preprint only	No	none	arXiv only; dataset on Hugging Face (MARSLab/MUSON)
Oush2026VLATurtleBot	S4	2026	2026 11th International Conference on Control and Robotics Engineering (ICCRE)	A: published, peer-reviewed	Yes	10.1109/ICCRE69951.2026.11593560	Not reported
Pan2026FoundationLMNavigationReview	S0	2026	ACM Computing Surveys	A: published, peer-reviewed	Yes	10.1145/3802539	Corpus copy is the published version
Park2026TACET	S5	2026	arXiv preprint (arXiv:2610.03828v1)	C: preprint only	No	none	Not reported
Payandeh2026Narrate2Nav	S4	2026	2026 IEEE International Conference on Robotics and Automation (ICRA)	A: published, peer-reviewed	Yes	10.1109/ICRA57385.2026.11696112	Not reported
Sethuraman2026ContextAwarePreferences	S5	2026	arXiv preprint (arXiv:2603.17510v2)	C: preprint only	No	none	Not reported
Wang2026LENav	S4	2026	IEEE Transactions on Automation Science and Engineering	A: published, peer-reviewed	Yes	10.1109/TASE.2026.3673004	Not reported
Wang2026MActionSocialNav	S4	2026	IEEE Robotics and Automation Letters	A: published, peer-reviewed	Yes	10.1109/LRA.2026.3706951	Corpus copy is the arXiv author version; published version authoritative
Wang2026SALSA	S5	2026	arXiv preprint (arXiv:2606.10495v2)	C: preprint only	No	none	Not reported
Xiao2026ESocialNav	S4	2026	ICASSP 2026 - IEEE International Conference on Acoustics, Speech and Signal Proc	A: published, peer-reviewed	Yes	10.1109/ICASSP55912.2026.11462076	Corpus copy is the arXiv author version; published version authoritative
Xiao2026PromptingVLMsJSAI	S5	2026	Proceedings of the Annual Conference of JSAI, JSAI2026 (40th Annual Conference o	D: conference paper; acceptance verified, peer review not established	Not established	10.11517/pjsai.JSAI2026.0_2E5GS10o01	Corpus copy is the JSAI proceedings PDF; arXiv 2601.14622 is a related, differently titled
Xu2026HCSG	S5	2026	arXiv:2605.13321v1 [cs.RO], 13 May 2026	C: preprint only	No	none	arXiv only
Zhang2026AHeadVLN	S4	2026	Proceedings of the Thirty-Fifth International Joint Conference on Artificial Int	A: published, peer-reviewed	Yes	10.24963/ijcai.2026/696	Not reported

Cite key	Category	Year	Venue	Status class	Peer reviewed	DOI	arXiv relation
Zhang2026GCL	S5	2026	arXiv:2603.11447v1 [cs.RO], 12 Mar 2026 (LNCS-style template)	C: preprint only	No	none	arXiv only; an IEICE technical report is a related (non-archival) version
Zhang2026SAATTNav	S5	2026	arXiv preprint (arXiv:2603.13698v2)	C: preprint only (arXiv comment: submitted to IROS 2026; acceptance not found)	No	none	Not reported
Zhang2026SOPDSocialNav	S5	2026	arXiv:2607.19850v1 [cs.RO], 22 Jul 2026	B: accepted per authors/arXiv; proceedings not yet available	Accepted (not yet indexed)	none	arXiv only; published version not yet available
Zhang2026WalkWithMe	S5	2026	arXiv preprint (arXiv:2604.26839v1)	C: preprint only	No	none	Not reported
Zhao2026EvoNav	S5	2026	arXiv preprint (arXiv:2605.11859v1)	C: preprint only	No	none	Not reported

Table S3. Adjacent foundation-model navigation works (not social-navigation evidence)

13 rows. Machine-readable version: `S3_adjacent_foundation_models.csv`.

Paper	Cite key	Year	Foundation-model mechanism	Navigation task	Concept inherited by social-navigation methods	Relevance to social navigation	Why NOT direct social-navigation evidence
SayCan	Ichter2022SayCan	2022	Product of LLM usefulness score and learned affordance value selects next skill; iterative planning in language.	method	Cited by corpus VLM-Social-Nav (Song 2025), PaLM-E, NavGPT, Code as Policies, Pan review.	Foundational 'LLM proposes, grounding module disposes' architecture; analogous to VLM social reasoning filtered by feasi	No social navigation, no human-aware behaviour.
LM Nav	Shah2022LMNav	2022	Modular zero-shot pipeline: LLM -> landmarks; VLM -> node grounding; VNM -> traversability and control; DP graph search.	method	Cited by corpus VLM-Social-Nav, VL-TGS, PAD, ViLAM, Code as Policies, VLMaps, GNM, Pan review.	Canonical 'compose pre-trained models' navigation architecture reused by VLM social navigation pipelines (VLM semantic l	No humans, no social reasoning; ignores verbs/ adverbs such as 'drive past the dog very slowly'.
RT 2	Zitkovich2023RT2	2023	Pretrained VLM fine-tuned jointly on web VL and robot data so that action tokens are generated like text (VLA).	model	Cited by corpus PAD, MUSON, Cai survey, SocialNav-SUB, Pan review.	Defines the VLA (D6) end of the integration spectrum; latency (1-3 Hz cloud) and data needs frame the feasibility de	Manipulation only; humans only as recognition targets; no social navigation behaviour.
PaLM E	Driess2023PaLME	2023	Continuous sensor embeddings interleaved with text tokens in a pretrained LLM, trained end-to-end; LLM outputs plans/answers.	model	Cited by corpus PAD, MUSON, Cai survey; NavGPT; Pan review.	Archetype of VLM as high-level planner and scene interpreter (R1/R4) with discrete skill outputs, and of positive transf	No humans as navigation agents, no social behaviour.
VLMaps	Huang2023VLMaps	2023	Spatially anchored open-vocabulary features enable language indexing of map locations and language-defined obstacle maps; LLM writes navigat	method	Cited by corpus CoNVOI, VL-TGS, ViLAM, Cai survey, NavGPT, G2-Nav (via search: 'Visual Language Maps').	Template for semantic/language maps that social systems extend with human/social layers (e.g., language-defined no-go or	Static environments only; no human modelling.
VoxPoser	Huang2023VoxPoser	2023	LLM-generated code builds dense affordance/ avoidance cost maps grounded by VLM perception, optimized by MPC with fast replanning.	method	Cited by corpus MUSON and Pan foundation-language-models navigation review.	Direct methodological precedent for VLM/ LLM-generated cost or constraint maps (D3) used by social navigation planners	Manipulation only; no humans or social norms.
Code as Policies	Liang2023CodeasPolicies	2023	LLM writes executable policy code composing perception and control APIs, with hierarchical function generation.	method	Cited by corpus Cai survey and VoxPoser (adjacent); not cited by the direct social-navigation papers in the reviewed corpus.	Precedent for LLMs producing parameterised constraints/costs/waypoints programmatically (e.g., translating 'a bit to the	No humans or social norms; navigation shown only qualitatively.
GNM	Shah2023GNM	2023	Single goal-conditioned visuomotor navigation policy trained on 60-70 h of heterogeneous multi-robot data with normalized waypoint action sp	model	Cited by corpus SocialNav (CVPR 2026, Chu); ViNT/NoMaD (adjacent) build on it.	Navigation foundation model lineage (GNM -> ViNT -> NoMaD) used as base policies/ baselines in VLM social navigation; tra	No language, no humans in evaluation, no social metrics.
ViNT	Shah2023ViNT	2023	Transformer NavFM trained on 80-160 h of cross-robot data; diffusion-proposed subgoals; soft-prompt adaptation to new goal modalities.	model	Cited by corpus VL-TGS, BehAV, Cai survey, SocialNav (CVPR 2026), ViLAM, Obstacles-to-Etiquette, G2-Nav.	Base/backbone for several VLM social-navigation systems (used for low-level execution) and source of the 'emergent pedes	No language, no social model; pedestrian behaviour only anecdotal.
NoMaD	Sridhar2024NoMaD	2024	Goal-masked Transformer context + conditional action diffusion yields one policy for undirected exploration and goal-conditioned navigation.	model	Cited by corpus SocialNav-MoE, VL-TGS, BehAV, SocialNav (CVPR 2026); Arena 4.0.	Common low-level navigator / baseline in VLM social-navigation work; demonstrates multimodal action modelling relevant t	No language, no human modelling, no social metrics.
VLFM	Yokoyama2024VLFM	2024	VLM image-text similarity projected into a top-down value map, fused with confidence weighting, ranks exploration frontiers.	method	Cited by corpus Cai survey.	Example of VLM as dense spatial scorer (D2) that avoids vision-to-text bottlenecks; directly analogous to VLM path/t	Object search only; no humans or social norms.
NavGPT	Zhou2024NavGPT	2024	LLM as zero-shot sequential decision maker over text-translated observations, with explicit reasoning traces and summarised history.	method	Cited by corpus HA-VLN (2024), HA-VLN 2.0, HCSG; also by Pan foundation-language-models navigation review (00).	Template for LLM-in-the-loop navigation with explainable reasoning (R4/D4) and for the 'vision-to-language bottleneck' l	No humans, no social behaviour, static scenes.
VL TGS	Song2025VLTGS	2025	Propose-and-select: generative (CVAE) geometric candidates + VLM visual-prompt selection of semantically appropriate trajectory	method	Fang et al. 2026 'From Obstacles to Etiquette' explicitly cites VL-TGS as closest propose-and-select work; GSON/OLiVia-s	Demonstrates VLM as path critic for conventions of human-centred spaces (crosswalks, pavements); direct template for lat	No humans in evaluated scenarios; objective is semantic traversability, not interaction with people

Table S4. Full final coding of all non-survey corpus works

85 rows. Machine-readable version: `S4_full_final_coding.csv`.

Cite key	Category	A_social_ representation	B_roles	D_interface	EV1_offline	EV2_closed_ loop_sim	EV3_physical_ robot	EV4_real_humans_in_ trials	EV5_human_evaluation	EV6_open_world_ deployment	Capability leaves Yes	Capability leaves Partial
Helbing1995SocialForce	S1	A1; A3; A4	NA (not a direct FM social-navigation method)	NA	No	No	none	not coded (non-direct work)	none	No		SAF1 SPA1 SPA2 SPA3 SPA4 HUM1 EXP1
Trautman2010UnfreezingIGP	S1	A2; A4	NA (not a direct FM social-navigation method)	NA	No	Yes	none	not coded (non-direct work)	none	No	SAF2 SPA4	SAF1 SPA1 HUM1 HUM2 EXP3
Kuderer2012FeatureBasedPrediction	S2	A2	NA (not a direct FM social-navigation method)	NA	Yes	Yes	demo	not coded (non-direct work)	none	No	SPA3 HUM1 EXP3	SAF1 SAF2 SPA1 HUM2 EXP2
Kretzschmar2016SociallyCompliantIRL	S2	A2; A4	NA (not a direct FM social-navigation method)	NA	No	Yes	demo	not coded (non-direct work)	none	No	SAF2 SPA3 HUM1 EXP3	SAF1 SPA1 SPA2 HUM2 NOR1 NOR4 EXP2
Okal2016BayesianIRL	S2	A1; A4	NA (not a direct FM social-navigation method)	NA	No	Yes	demo	not coded (non-direct work)	none	No	SPA1 SPA2 SPA4 NOR1 NOR3	SAF1 HUM1 HUM2 EXP1
Chen2017CADRL	S2	A1	NA (not a direct FM social-navigation method)	NA	No	Yes	none	not coded (non-direct work)	none	No	SAF1	SAF2 SPA3 SPA4 HUM1 EXP3
Chen2017SACADRL	S2	A1	NA (not a direct FM social-navigation method)	NA	No	Yes	systematic	not coded (non-direct work)	none	No	SAF1 SPA3 NOR1	SAF2 SPA1 SPA4 HUM1 NOR4 EXP1
Tai2018GAILDepth	S2	A6	NA (not a direct FM social-navigation method)	NA	No	Yes	demo	not coded (non-direct work)	none	No		SAF1 SAF2 SPA1 SPA2 SPA3
Chen2019SARLCrowdNav	S2	A1; A4	NA (not a direct FM social-navigation method)	NA	No	Yes	demo	not coded (non-direct work)	none	No	SAF1 SAF2 SPA4	SPA1 HUM1 EXP1
Tsoi2020SEAN	S6	A1	NA (not a direct FM social-navigation method)	NA	No	Yes	none	not coded (non-direct work)	none	No	SAF1	SAF2 SPA1
Kastner2021ArenaRosnav	S6	A0	NA (not a direct FM social-navigation method)	NA	No	Yes	none	not coded (non-direct work)	none	No		
Ichter2022SayCan	S3	A0	NA (not a direct FM social-navigation method)	NA	No	Yes	systematic	not coded (non-direct work)	text-raters	No		
Biswas2022SocNavBench	S6	A1; A2	NA (not a direct FM social-navigation method)	NA	No	Yes	none	not coded (non-direct work)	none	No	SAF1 SAF2	SPA1 SPA3 SPA4 NOR3 EXP2
Karnan2022SCAND	S6	A6	NA (not a direct FM social-navigation method)	NA	No	No	systematic	not coded (non-direct work)	robot-behaviour evaluation (type not differentiated)	No	EXP1	SAF1 SPA2 SPA3 SPA4 NOR1 NOR3
Kim2022GroupEstimation	S2	A3; A4	NA (not a direct FM social-navigation method)	NA	No	Yes	none	not coded (non-direct work)	none	No	SPA1 SPA2 SPA4	SAF1 HUM1 NOR1 EXP1
Shah2022LMNav	S3	A0	NA (not a direct FM social-navigation method)	NA	No	No	systematic	not coded (non-direct work)	none	No		
Zitkovich2023RT2	S3	A6	NA (not a direct FM social-navigation method)	NA	No	Yes	systematic	not coded (non-direct work)	none	No		
Driess2023PaLME	S3	A0	NA (not a direct FM social-navigation method)	NA	No	Yes	demo	not coded (non-direct work)	none	No		
Huang2023VLMaps	S3	A0	NA (not a direct FM social-navigation method)	NA	No	Yes	systematic	not coded (non-direct work)	none	No		
Huang2023VoxPoser	S3	A0	NA (not a direct FM social-navigation method)	NA	No	Yes	systematic	not coded (non-direct work)	none	No		
Liang2023CodeasPolicies	S3	A0	NA (not a direct FM social-navigation method)	NA	No	Yes	demo	not coded (non-direct work)	none	No		
Nguyen2023MuSoHu	S6	A6	NA (not a direct FM social-navigation method)	NA	Yes	No	none	not coded (non-direct work)	none	No		
Shah2023GNM	S3	A0	NA (not a direct FM social-navigation method)	NA	No	No	systematic	not coded (non-direct work)	none	No		
Shah2023ViNT	S3	A6	NA (not a direct FM social-navigation method)	NA	No	Yes	systematic	not coded (non-direct work)	none	No		
Wang2023NaviSTAR	S2	A1; A4; A6	NA (not a direct FM social-navigation method)	NA	No	Yes	systematic	not coded (non-direct work)	robot-behaviour evaluation (type not differentiated)	No	SAF1 SPA4 EXP1 EXP3	SAF2 SPA1 HUM1 HUM2 NOR1
Hirose2024SACSoN	S6	A6	NA (not a direct FM social-navigation method)	NA	No	No	systematic	not coded (non-direct work)	none	No	SAF1	SPA3 EXP1
Holk2024PREDILECT	S4	A1	R4	D7	No	Yes	none	simulated-only	none	No		SPA1 NOR3 EXP1
Li2024HAVLN	S6	A1; A5	NA (not a direct FM social-navigation method)	NA	No	Yes	demo	not coded (non-direct work)	text-raters	No	SAF1	SAF2 SPA1 SPA3 HUM1 HUM2 NOR2 NOR3
Marpally2026SocRATES	S6	A1	NA (not a direct FM social-navigation method)	NA	No	Yes	none	not coded (non-direct work)	none	No		NOR1 NOR3
Raj2024HybridSocialNav	S2	A1; A6	NA (not a direct FM social-navigation method)	NA	No	No	systematic	not coded (non-direct work)	robot-behaviour evaluation (type not differentiated)	No	SAF2 SPA3 EXP2	SAF1 SPA1 NOR1 NOR3 EXP1
Sathyamoorthy2024CoNVOI	S4	A1; A6	R1; R3	D4	No	No	systematic	present-type-unreported	none	No	SPA3 NOR3	SAF1 SPA2 NOR1 NOR2 EXP3
Shcherbyna2025Arena40	S6	A1; A4	NA (not a direct FM social-navigation method)	NA	No	No	none	not coded (non-direct work)	text-raters	No		
Sridhar2024NoMaD	S3	A0	NA (not a direct FM social-navigation method)	NA	No	No	systematic	not coded (non-direct work)	none	No		
Wang2024SALM	S5	A1; A4; A5	R3; R4	D6; D2	No	Yes	none	simulated-only	none	No	SAF1 SPA1 NOR2	SPA4 EXP1
Yokoyama2024VLFM	S3	A0	NA (not a direct FM social-navigation method)	NA	No	Yes	demo	not coded (non-direct work)	none	No		
Yu2024PANav	S5	A3	R1; R3	D2	Yes	No	demo	none	none	No		NOR1 NOR3
Zhou2024NavGPT	S3	A0	NA (not a direct FM social-navigation method)	NA	No	Yes	none	not coded (non-direct work)	none	No		
Kawabata2025SocialNavMoE	S5	A5	R1; R2; R3	D0	Yes	No	none	none	none	No		SPA3 SPA4 HUM1 NOR1 NOR3
Kong2025AutoSpatial	S4	A5	R1; R2; R3	D0	Yes	No	none	none	text-raters	No		SPA2 SPA3 SPA4 HUM1 HUM2 NOR1 NOR3

Cite key	Category	A_social_ representation	B_roles	D_interface	EV1_offline	EV2_closed_ loop_sim	EV3_physical_ robot	EV4_real_humans_in_ trials	EV5_human_evaluation	EV6_open_world_ deployment	Capability leaves Yes	Capability leaves Partial
Liang2025MOSU	S5	A6	R1; R3	D2; D3	Yes	No	demo	none	none	No		SPA2 SPA3 NOR1 NOR3
Luo2025GSON	S4	A1; A2; A3; A4	R2	D1	Yes	No	systematic	unscripted	none	No	SPA1 SPA2 SPA3	SAF1 SAF2 SPA4 HUM1 HUM2 NOR1 NOR3 EXP1 EXP3
MartinezBaselga2025HeyRobot	S4	A1; A2	R1; R4	D3	No	Yes	demo	staged	none	No	SAF1 SPA1 NOR2 NOR3	HUM1 EXP1 EXP3
Munje2025SocialNavSUB	S6	A1; A2; A4	NA (not a direct FM social-navigation method)	NA	No	No	none	not coded (non-direct work)	text-raters	No		SPA3 SPA4 HUM1 HUM2
Narasimhan2025OLiViaNav	S4	A5; A6	R1	D5; D7	Yes	No	systematic	staged	none	No	SPA1	SAF1 SPA2 SPA3 SPA4 HUM1 NOR1 NOR3 EXP3
Payandeh2025SocialLLaVA	S4	A5	R1; R2; R3	D0	Yes	No	demo	none	text-raters	No		SPA1 SPA2 SPA3 SPA4 HUM1 HUM2 NOR1 NOR3
Ruo2025LLMSocialCues	S4	A1; A3; A5	R2; R3; R4	D3; D4	Yes	Yes	none	simulated-only	none	No	NOR3	SAF1 SPA1 HUM2 NOR2 EXP1
Saito2025VLMCrowdMapping	S4	A5	R2	D3	No	No	demo	staged	none	No		SAF1 SPA4 NOR3
Song2025VLTGS	S3	A0	NA (not a direct FM social-navigation method)	NA	No	No	systematic	not coded (non-direct work)	none	No		
Song2025VLMSocialNav	S4	A1; A5	R1; R3	D3	No	No	systematic	staged	co-present participants	No	SAF1 SPA3 NOR1 EXP1	SAF2 SPA1 HUM1 HUM2 NOR2 NOR3 EXP2
Sotomi2025TransparentSocialNavVLM	S4	A5; A6	R1; R5	D1; D8	Yes	No	systematic	present-type-unreported	observers/video	No		SPA1 SPA2 NOR3 EXP1 EXP2
Wang2025SAMALM	S5	A1; A5	R3	D6; D2	No	Yes	none	simulated-only	none	No	SAF1 SPA1	SPA4 EXP1
Weerakoon2025BehAV	S4	A3; A5	R1; R3; R4	D3; D4	Yes	No	systematic	staged	none	No	SPA3 NOR2 NOR3	SAF1 HUM2 NOR1 EXP3
Wen2025HSACLLM	S4	A1; A5	R5	D8	No	Yes	demo	not-reported	none	No	SAF1 SAF2 SPA1	SPA3 HUM1 HUM2 NOR1 NOR2 NOR3 EXP1 EXP2
Zhang2025FewShotPerception	S6	A1; A5	NA (not a direct FM social-navigation method)	NA	No	No	systematic	not coded (non-direct work)	robot-behaviour evaluation (type not differentiated)	No		EXP1 EXP2
Ahmadi2026HUMA	S5	A5; A6	R2; R3	D5	No	Yes	demo	not-reported	none	No	SAF1	SPA4 HUM1 HUM2 NOR3
BejaranoSepulveda2026PAD	S4	A1; A2; A5	R1; R2; R5	D1; D8	No	No	systematic	staged	co-present participants	No	SAF2 EXP3	SAF1 SPA1 SPA3 HUM1 HUM2 NOR1 NOR2 NOR3 EXP2
Chen2026LISN	S4	A1; A3; A5	R1; R2; R4	D3; D4	No	Yes	none	simulated-only	none	No	SAF1 SPA1 NOR2 NOR3	SPA4 HUM1 NOR1 EXP1 EXP3
Chen2026SocialNavCVPR	S4	A6	R1; R3	D6; D7	No	Yes	systematic	not-reported	none	No	SPA3	SAF1 SPA4 NOR1 NOR3 EXP3
Dai2026DPedVLN	S6	A1	NA (not a direct FM social-navigation method)	NA	No	Yes	none	not coded (non-direct work)	none	No	SAF1 SPA1	NOR2
Dong2026HAVLN20	S6	A1; A3; A4; A5	NA (not a direct FM social-navigation method)	NA	No	Yes	systematic	not coded (non-direct work)	none	No	SAF1	SAF2 SPA1 SPA2 SPA3 SPA4 HUM1 HUM2 NOR1 NOR2 NOR3
Elnoor2026ViLAM	S5	A3; A6	R2	D7	No	No	systematic	present-type-unreported	none	No		SAF1 SPA1 SPA4 HUM1 HUM2 NOR1 NOR3 EXP3
Fang2026GroupJoining	S5	A1; A3; A4	R2; R4	D4	Yes	No	systematic	present-type-unreported	none	No	SPA1 SPA2 SPA3 NOR2	
Fang2026ObstaclestoEtiquette	S4	A1; A2; A3; A5	R2; R3	D2; D7	No	No	systematic	staged	none	No	SAF1 SPA1 SPA2 SPA3 HUM2 NOR1 NOR3	SAF2 HUM1 EXP2
Gandotra2026RBTLR	S5	A1; A5; A6	R1; R4; R5	D5; D4; D8	No	Yes	none	simulated-only	none	No		SAF1 SPA1 SPA3 NOR2 NOR3
Hu2026SPARKVLN	S5	A6	R1; R4	D5	No	Yes	none	simulated-only	none	No	SAF1 SPA1	SPA3 SPA4 NOR2
Huang2026SONG	S6	A1	NA (not a direct FM social-navigation method)	NA	No	Yes	systematic	not coded (non-direct work)	none	No	SAF1 SPA1	SPA4
Huo2026NORMNav	S4	A3; A5	R1; R4	D3	No	Yes	demo	present-type-unreported	none	No	SPA3 NOR2	SAF1 NOR1 NOR3 EXP3
Liao2026G2Nav	S5	A1; A3; A5	R1; R2	D3	Yes	No	systematic	present-type-unreported	none	No	SAF1 SAF2	SPA1 SPA2 SPA3 SPA4 HUM1 HUM2 NOR1 NOR3 EXP3
Liu2026MUSON	S6	A2; A5	NA (not a direct FM social-navigation method)	NA	Yes	No	none	not coded (non-direct work)	none	No		SAF1 SPA1 SPA3 SPA4 HUM1 NOR1 NOR3
Oush2026VLATurtleBot	S4	A1; A5	R1; R3	D6	No	Yes	none	simulated-only	none	No	SAF2	SAF1 SPA3
Park2026TACET	S5	A6	R1; R3	D3; D7	No	No	systematic	staged	observers/video	No	SAF1 SPA1 NOR1 NOR3	EXP1
Payandeh2026Narrate2Nav	S4	A5; A6	R1; R2	D7	Yes	No	systematic	present-type-unreported	none	No	SAF1	SAF2 SPA3 SPA4 HUM1 HUM2 NOR1 NOR3 EXP3
Sethuraman2026ContextAwarePreferences	S5	A1	R1; R4	D3	No	No	systematic	present-type-unreported	text-raters	No	NOR2 NOR3	SPA1 EXP1
Wang2026LENav	S4	A1; A5	R1; R2	D3	No	No	systematic	present-type-unreported	co-present participants	No	SAF1 SAF2 NOR3 EXP1	SPA4 EXP3
Wang2026MActionSocialNav	S4	A5	R1; R2; R3	D0	Yes	No	none	none	none	No		SAF1 SPA1 SPA3 SPA4 HUM1 NOR1 NOR3
Wang2026SALSA	S5	A6	R3	D6	Yes	No	demo	none	none	No	SAF2	SAF1 SPA4 EXP3
Xiao2026ESocialNav	S4	A5	R1; R2; R3	D0	Yes	No	none	none	none	No		SPA3 HUM1 NOR1 NOR3
Xiao2026PromptingVLMsJSAI	S5	A5	R1; R2; R3	D0	Yes	No	none	none	none	No		SPA3 HUM1 HUM2 NOR1 NOR3
Xu2026HCSG	S5	A1; A2; A5; A6	R2	D1	No	Yes	systematic	present-type-unreported	none	No	SAF1 NOR2	SAF2 SPA1 SPA2 SPA3 HUM1 HUM2 NOR1 NOR3
Zhang2026AHeadVLN	S4	A2; A5; A6	R1; R3; R4	D2; D6	No	Yes	none	simulated-only	none	No	SAF1	SPA3 SPA4 HUM1
Zhang2026GCL	S5	A5	R1; R2; R3	D0	Yes	No	none	none	none	No		SPA3 SPA4 HUM1 NOR1 NOR3
Zhang2026SAATTNav	S5	A1; A2; A3; A5	R2; R5	D1; D8	No	Yes	none	simulated-only	none	No	SAF1 SAF2 SPA1 HUM2	SPA2 EXP2
Zhang2026SOPDSocialNav	S5	A5	R1; R3	D0	Yes	No	demo	none	none	No		SPA1 SPA2 SPA3 NOR1 NOR3
Zhang2026WalkWithMe	S5	A6	R3; R4	D4; D6	No	No	systematic	present-type-unreported	none	No		SAF1 SPA3 SPA4 NOR3
Zhao2026EvoNav	S5	A1	R3	D7	No	Yes	none	simulated-only	none	No	SAF1 SPA1 SPA4	

Table S5, part A. Pre-FM comparison set: per-work selection and coding

40 rows. Machine-readable version: `S5_prefm_comparison_set.csv`.

Cite key	Title	Year	Venue	Paradigm	Reference surveys citing (of 7)	Selection	Coding source	Link	A (representation)	Evaluated capabilities (Yes)	Addressed-only capabilities (Partial)	EV1 offline (recorded data)	EV2 closed-loop simulation	EV3 physical robot	EV4 real people in trials	EV5 human evaluation	EV6 open-world deployment	Statistical testing	Evidence note	Metadata source
Thrun1999Minerva [1]	Probabilistic algorithms and the interactive museum tour-guide robot Minerva	2000	The International Journal of Robotics Research	Geometric and socially aware planning	3	cited by at least 3 of the 7 reference surveys	reference-survey descriptions	https://doi.org/10.1177/02783640022067922	A1	SAF1	SPA4 EXP1	No	No	systematic	unscripted	co-present	Yes	No	Two-week museum deployment; statistics on performance and collision avoidance; visitor impressions of the robot (Mavrogiannis et al. 2023, Sec. 4.2.4.3)	Crossref record
Sisbot2007SisbotHAMP [2]	A human aware mobile robot motion planner	2007	IEEE T-RO 23(5) 2007	Geometric and socially aware planning	3	cited by at least 3 of the 7 reference surveys	reference-survey descriptions	https://doi.org/10.1109/TRO.2007.904911	A1 A3	-	SAF1 SPA1 EXP1 EXP2	No	Yes	demo	staged	none	No	No	Safety, visibility and hidden-zone cost grids; robot-human interactions in a lab described as a demonstration (Mavrogiannis et al. 2023, Table 4)	bibliography record
vandenBerg2008RVO [3]	Reciprocal velocity obstacles for real-time multi-agent navigation	2008	2008 IEEE International Conference on Robotics and Automation	Geometric and socially aware planning	3	cited by at least 3 of the 7 reference surveys	reference-survey descriptions	https://doi.org/10.1109/robot.2008.4543489	A1	-	SAF1 SPA4	No	Yes	none	none	none	No	No	Simulated multi-agent scenarios; collision- and oscillation-free motion shown qualitatively	Crossref record
Pandey2010Framework [4]	A framework towards a socially aware mobile robot motion in human-centered dynamic environment	2010	2010 IEEE/RSJ International Conference on Intelligent Robots and Systems	Geometric and socially aware planning	3	cited by at least 3 of the 7 reference surveys	reference-survey descriptions	https://doi.org/10.1109/iros.2010.5649688	A1 A3	-	SAF1 SPA1 SPA2 SPA3 NOR1	No	ND	not determinable	not determinable	not determinable	No	not determinable	Rule-based conventions (overtake on the right, avoid passing through groups, keep right in hallways) per Kruse et al. 2013; evaluation not determinable	Crossref record
Svenstrup2010Trajectory [5]	Trajectory planning for robots in dynamic human environments	2010	2010 IEEE/RSJ International Conference on Intelligent Robots and Systems	Geometric and socially aware planning	3	cited by at least 3 of the 7 reference surveys	full text	https://doi.org/10.1109/iros.2010.5651531	A1 A2 A3	-	SAF1 SPA1 EXP3	No	Yes	none	none	none	No	No	RRT/MPC through a person-dependent potential field; demonstrated in a simulated pedestrian street	Crossref record
Mavrogiannis2018Social Momentum [6]	Social momentum: A framework for legible navigation in dynamic multi-agent environments	2018	HRI 2018	Geometric and socially aware planning	3	cited by at least 3 of the 7 reference surveys	full text	https://rpal.cs.cornell.edu/docs/MavEtal_HRI_2018.pdf	A1 A4	EXP2	SAF1 SPA3 HUM1	No	Yes	none	none	observers/video	No	Yes	Simulation; online video study of legibility (N = 180) with t-tests	bibliography record
Mavrogiannis2019Mavrogiannis HRI2019 [7]	Effects of distinct robot navigation strategies on human behavior in a crowded environment	2019	HRI 2019	Geometric and socially aware planning	3	cited by at least 3 of the 7 reference surveys	full text	https://rpal.cs.cornell.edu/docs/MavEtal_HRI_2019.pdf	A1 A4	EXP1	SAF1 SPA4 EXP2 EXP3	No	No	systematic	staged	co-present	No	Yes	Lab study, N = 105, three participants at a time; trajectories and questionnaire; statistical tests	bibliography record
Pacchierotti2006Passing [8]	Evaluation of passing distance for social robots	2006	ROMAN 2006 - The 15th IEEE International Symposium on Robot and Human Interactive Communication	Proxemics and social costmaps	3	cited by at least 3 of the 7 reference surveys	full text	https://doi.org/10.1109/roman.2006.314436	A1 A3	SPA1 EXP1	SAF1 SPA3	No	No	systematic	staged	co-present	No	No	Corridor passing with 10 participants; 5-point comfort ratings per lateral distance	Crossref record
Chung2010Spatial [9]	A mobile robot that understands pedestrian spatial behaviors	2010	2010 IEEE/RSJ International Conference on Intelligent Robots and Systems	Proxemics and social costmaps	4	cited by at least 3 of the 7 reference surveys	reference-survey descriptions	https://doi.org/10.1109/iros.2010.5649718	A1 A2 A3	-	SPA1 HUM1 NOR3	No	ND	not determinable	not determinable	not determinable	No	not determinable	Map of place-specific "spatial effects" learned from observed pedestrians (Kruse et al. 2013); evaluation not determinable	Crossref record

Cite key	Title	Year	Venue	Paradigm	Reference surveys citing (of 7)	Selection	Coding source	Link	A (representation)	Evaluated capabilities (Yes)	Addressed-only capabilities (Partial)	EV1 offline (recorded data)	EV2 closed-loop simulation	EV3 physical robot	EV4 real people in trials	EV5 human evaluation	EV6 open-world deployment	Statistical testing	Evidence note	Metadata source
Charalampous2016Social Maps [10]	Robot navigation in large-scale social maps: An action recognition approach	2016	Expert Systems with Applications	Proxemics and social costmaps	3	cited by at least 3 of the 7 reference surveys	reference-survey descriptions	https://doi.org/10.1016/j.eswa.2016.09.026	A1 A3 A5	-	SPA1 HUM2 NOR3	No	ND	demo	not determinable	not determinable	No	not determinable	Deep action recognition assigns activity labels that select rule-based proxemic zones (Moller et al. 2021); custom robot base (Singamaneni et al. 2024)	Crossref record
Helbing1995SocialForce [11]	Social force model for pedestrian dynamics	1995	Physical Review E	Social-force models	-	corpus work (retained)	full text	https://doi.org/10.1103/PhysRevE.51.4282	A1 A3 A4	-	SAF1 SPA1 SPA2 SPA3 SPA4 HUM1 EXP1	No	No	none	not coded (corpus work)	none	No	not coded (corpus work)		corpus record
Ferrer2013Companion [12]	Robot companion: A social-force based approach with human awareness-navigation in crowded environments	2013	2013 IEEE/RSJ International Conference on Intelligent Robots and Systems	Social-force models	3	cited by at least 3 of the 7 reference surveys	reference-survey descriptions	https://doi.org/10.1109/iros.2013.6696576	A1 A2 A4	-	SAF1 SPA1 EXP1	No	ND	demo	not determinable	not determinable	No	not determinable	Extended SFM with person-person, object-person and robot-person interactions (Charalampous et al. 2017); robot Tibi (Singamaneni et al. 2024)	Crossref record
Shiomi2014Acceptable [13]	Towards a socially acceptable collision avoidance for a mobile robot navigating among pedestrians using a pedestrian model	2014	International Journal of Social Robotics	Social-force models	4	cited by at least 3 of the 7 reference surveys	full text	https://doi.org/10.1007/s12369-014-0238-y	A1 A2	SAF2 EXP1	SAF1 SPA4 EXP3	No	No	systematic	unscripted	co-present	No	No	Lab comparison rated by participants; two-hour shopping-mall field trials counting unsafe pedestrian reactions	Crossref record
Kim2022GroupEstimation [14]	Group Estimation for Social Robot Navigation in Crowded Environments	2022	2022 22nd International Conference on Control, Automation and Systems (ICCAS)	Group and F-formation navigation	-	corpus work (retained)	full text	https://doi.org/10.23919/ICCAS55662.2022.10003761	A3 A4	SPA1 SPA2 SPA4	SAF1 HUM1 NOR1 EXP1	No	Yes	none	not coded (corpus work)	none	No	not coded (corpus work)		corpus record
Truong2017Approach [15]	To approach humans?: A unified framework for approaching pose prediction and socially aware robot navigation	2018	IEEE Transactions on Cognitive and Developmental Systems	Group and F-formation navigation	2	added to reach 3 works in its paradigm	reference-survey descriptions	https://doi.org/10.1109/tcds.2017.2751963	A1 A3 A4	-	SPA1 SPA2	No	ND	demo	not determinable	not determinable	No	not determinable	Approaching poses for individuals and groups using dynamic social zones; custom robot (Singamaneni et al. 2024)	Crossref record
Wang2022GMPC [16]	Group-based motion prediction for navigation in crowded environments	2021	CoRL 2021	Group and F-formation navigation	1	added to reach 3 works in its paradigm	full text	https://arxiv.org/abs/2107.11637	A2 A3 A4	SAF1 SAF2 SPA2 SPA4	HUM1	No	Yes	none	none	none	No	Yes	Simulated pedestrian scenes from ETH/UCY; group intrusions and minimum distance; Mann-Whitney tests; group-prediction ablation	bibliography record
Katyal2022GroupAware [17]	Learning a group-aware policy for robot navigation	2022	2022 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)	Group and F-formation navigation	1	added to reach 3 works in its paradigm	full text	https://doi.org/10.1109/iros47612.2022.9981183	A1 A4	SAF1 SPA1 SPA2 SPA4	-	No	Yes	none	none	none	No	Yes	CrowdNav simulation; collisions, individual discomfort, group intersections; t-tests	Crossref record
Trautman2010UnfreezingIGP [18]	Unfreezing the Robot: Navigation in Dense, Interacting Crowds	2010	2010 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)	Trajectory prediction and prediction-coupled planning	-	corpus work (retained)	full text	https://doi.org/10.1109/IROS.2010.5654369	A2 A4	SAF2 SPA4	SAF1 SPA1 HUM1 HUM2 EXP3	No	Yes	none	not coded (corpus work)	none	No	not coded (corpus work)		corpus record
Pellegrini2009ETH [19]	You'll never walk alone: Modeling social behavior for multi-target tracking	2009	ICCV 2009	Trajectory prediction and prediction-coupled planning	4	cited by at least 3 of the 7 reference surveys	reference-survey descriptions	https://doi.org/10.1109/ICCV.2009.5459260	A1 A2 A4	HUM1	-	Yes	No	none	none	none	No	No	Social motion model for tracking; prediction evaluated on recorded crowd video (ETH)	bibliography record
al2016SocialLSTM [20]	Social LSTM: Human trajectory prediction in crowded spaces	2016	CVPR 2016	Trajectory prediction and prediction-coupled planning	4	cited by at least 3 of the 7 reference surveys	full text	https://openaccess.thecvf.com/content_cvpr_2016/papers/Alahi_Social_LSTM_Human_CVPR_2016_paper.pdf	A2 A4	HUM1	-	Yes	No	none	none	none	No	No	ADE/FDE on ETH and UCY	bibliography record

Cite key	Title	Year	Venue	Paradigm	Reference surveys citing (of 7)	Selection	Coding source	Link	A (representation)	Evaluated capabilities (Yes)	Addressed-only capabilities (Partial)	EV1 offline (recorded data)	EV2 closed-loop simulation	EV3 physical robot	EV4 real people in trials	EV5 human evaluation	EV6 open-world deployment	Statistical testing	Evidence note	Metadata source
Vemula2018SocialAttention [21]	Social attention: Modeling attention in human crowds	2018	2018 IEEE International Conference on Robotics and Automation (ICRA)	Trajectory prediction and prediction-coupled planning	3	cited by at least 3 of the 7 reference surveys	full text	https://doi.org/10.1109/icra.2018.8460504	A2 A4	HUM1	-	Yes	No	none	none	none	No	No	ADE/FDE on ETH and UCY	Crossref record
Kuderer2012FeatureBased Prediction [22]	Feature-Based Prediction of Trajectories for Socially Compliant Navigation	2012	Robotics: Science and Systems VIII (RSS 2012)	Inverse reinforcement learning	-	corpus work (retained)	full text	https://doi.org/10.15607/RSS.2012.VIII.025	A2	SPA3 HUM1 EXP3	SAF1 SAF2 SPA1 HUM2 EXP2	Yes	Yes	demo	not coded (corpus work)	none	No	not coded (corpus work)		corpus record
Kretzschmar2016Socially CompliantIRL [23]	Socially Compliant Mobile Robot Navigation via Inverse Reinforcement Learning	2016	The International Journal of Robotics Research	Inverse reinforcement learning	-	corpus work (retained)	full text	https://doi.org/10.1177/0278364915619772	A2 A4	SAF2 SPA3 HUM1 EXP3	SAF1 SPA1 SPA2 HUM2 NOR1 NOR4 EXP2	No	Yes	demo	not coded (corpus work)	none	No	not coded (corpus work)		corpus record
Okal2016BayesianIRL [24]	Learning Socially Normative Robot Navigation Behaviors with Bayesian Inverse Reinforcement Learning	2016	2016 IEEE International Conference on Robotics and Automation (ICRA)	Inverse reinforcement learning	-	corpus work (retained)	full text	https://doi.org/10.1109/ICRA.2016.7487452	A1 A4	SPA1 SPA2 SPA4 NOR1 NOR3	SAF1 HUM1 HUM2 EXP1	No	Yes	demo	not coded (corpus work)	none	No	not coded (corpus work)		corpus record
Henry2010Crowded [25]	Learning to navigate through crowded environments	2010	2010 IEEE International Conference on Robotics and Automation	Inverse reinforcement learning	4	cited by at least 3 of the 7 reference surveys	full text	https://doi.org/10.1109/robot.2010.5509772	A1	-	SPA3 SPA4 EXP3	No	Yes	none	none	none	No	No	Crowd simulator; path similarity to simulated human paths	Crossref record
Luber2012Learning [26]	Socially-aware robot navigation: A learning approach	2012	2012 IEEE/RSJ International Conference on Intelligent Robots and Systems	Inverse reinforcement learning	4	cited by at least 3 of the 7 reference surveys	reference-survey descriptions	https://doi.org/10.1109/iros.2012.6385716	A1 A2	-	SAF1 SPA1 EXP3	No	Yes	demo	not determinable	none	No	not determinable	Motion prototypes learned from overhead video; better than a proxemics baseline in simulation (Moller et al. 2021); listed as demonstration (Mavrogiannis et al. 2023)	Crossref record
Vasquez2014IRLFeatures [27]	Inverse reinforcement learning algorithms and features for robot navigation in crowds	2014	2014 IEEE/RSJ International Conference on Intelligent Robots and Systems	Inverse reinforcement learning	4	cited by at least 3 of the 7 reference surveys	reference-survey descriptions	https://doi.org/10.1109/iros.2014.6942731	A1	-	SPA4 EXP3	Yes	Yes	none	none	none	No	No	Experimental comparison of IRL algorithms and crowd features on recorded crowd data (Alyassi et al. 2025; Mavrogiannis et al. 2023)	Crossref record
Kim2016SociallyAdaptive [28]	Socially adaptive path planning in human environments using inverse reinforcement learning	2016	International Journal of Social Robotics	Inverse reinforcement learning	4	cited by at least 3 of the 7 reference surveys	full text	https://doi.org/10.1007/s12369-015-0310-2	A1 A3	-	SAF1 SPA1 SPA4 EXP3	No	No	systematic	present-type-unreported	none	No	No	Robotic wheelchair; scenario-based and field runs compared with human trajectories	Crossref record
Tai2018GAILDepth [29]	Socially Compliant Navigation through Raw Depth Inputs with Generative Adversarial Imitation Learning	2018	2018 IEEE International Conference on Robotics and Automation (ICRA)	Imitation learning	-	corpus work (retained)	full text	https://doi.org/10.1109/ICRA.2018.8460968	A6	-	SAF1 SAF2 SPA1 SPA2 SPA3	No	Yes	demo	not coded (corpus work)	none	No	not coded (corpus work)		corpus record
Hamandi2019DeepMoTIon [30]	DeepMoTIon: Learning to navigate like humans	2019	2019 28th IEEE International Conference on Robot and Human Interactive Communication (RO-MAN)	Imitation learning	1	added to reach 3 works in its paradigm	full text	https://doi.org/10.1109/ro-man46459.2019.8956408	A6	SAF1	SPA3 EXP3	No	Yes	none	none	none	No	No	Trained on ETH pedestrians; simulated evaluation with collision counts and similarity to human paths	Crossref record
Chen2017CADRL [31]	Decentralized Non-communicating Multiagent Collision Avoidance with Deep Reinforcement Learning	2017	2017 IEEE International Conference on Robotics and Automation (ICRA)	Deep reinforcement learning	-	corpus work (retained)	full text	https://doi.org/10.1109/ICRA.2017.7989037	A1	SAF1	SAF2 SPA3 SPA4 HUM1 EXP3	No	Yes	none	not coded (corpus work)	none	No	not coded (corpus work)		corpus record

Cite key	Title	Year	Venue	Paradigm	Reference surveys citing (of 7)	Selection	Coding source	Link	A (representation)	Evaluated capabilities (Yes)	Addressed-only capabilities (Partial)	EV1 offline (recorded data)	EV2 closed-loop simulation	EV3 physical robot	EV4 real people in trials	EV5 human evaluation	EV6 open-world deployment	Statistical testing	Evidence note	Metadata source
Chen2017SACADRL [32]	Socially Aware Motion Planning with Deep Reinforcement Learning	2017	2017 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)	Deep reinforcement learning	-	corpus work (retained)	full text	https://doi.org/10.1109/IROS.2017.8202312	A1	SAF1 SPA3 NOR1	SAF2 SPA1 SPA4 HUM1 NOR4 EXP1	No	Yes	systematic	not coded (corpus work)	none	No	not coded (corpus work)		corpus record
Everett2018GA3CCADRL [33]	Motion planning among dynamic decision-making agents with deep reinforcement learning	2018	IROS 2018	Deep reinforcement learning	3	cited by at least 3 of the 7 reference surveys	full text	https://arxiv.org/abs/1805.01956	A1	SAF1 SPA4	SPA3	No	Yes	demo	staged	none	No	No	Simulation with up to 10 agents; hardware demonstration among pedestrians	bibliography record
Liu2020DRLCrowded [34]	Robot navigation in crowded environments using deep reinforcement learning	2020	2020 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)	Deep reinforcement learning	3	cited by at least 3 of the 7 reference surveys	reference-survey descriptions	https://doi.org/10.1109/iros45743.2020.9341540	A1 A6	-	SAF1 SPA4	No	Yes	not determinable	not determinable	not determinable	No	not determinable	Imitation plus DRL with separate static and dynamic channels (abstract via publisher record); robot evidence not determinable	Crossref record
Chen2019SARLCrowdNav [35]	Crowd-Robot Interaction: Crowd-aware Robot Navigation with Attention-based Deep Reinforcement Learning	2019	2019 International Conference on Robotics and Automation (ICRA)	Attention, graph and transformer models	-	corpus work (retained)	full text	https://doi.org/10.1109/ICRA.2019.8794134	A1 A4	SAF1 SAF2 SPA4	SPA1 HUM1 EXP1	No	Yes	demo	not coded (corpus work)	none	No	not coded (corpus work)		corpus record
Wang2023NaviSTAR [36]	NaviSTAR: Socially Aware Robot Navigation with Hybrid Spatio-Temporal Graph Transformer and Preference Learning	2023	2023 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)	Attention, graph and transformer models	-	corpus work (retained)	full text	https://doi.org/10.1109/IROS55552.2023.10341395	A1 A4 A6	SAF1 SPA4 EXP1 EXP3	SAF2 SPA1 HUM1 HUM2 NOR1	No	Yes	systematic	not coded (corpus work)	co-present	No	not coded (corpus work)	human evaluation type not differentiated	corpus record
Chen2020RGL [37]	Relational graph learning for crowd navigation	2020	IROS 2020	Attention, graph and transformer models	4	cited by at least 3 of the 7 reference surveys	full text	https://arxiv.org/abs/1909.13165	A1 A4	SAF1 SPA4	HUM1	No	Yes	demo	present-type-unreported	none	No	No	CrowdNav simulation; Pioneer real-world demonstrations	bibliography record
Raj2024HybridSocialNav [38]	Rethinking Social Robot Navigation: Leveraging the Best of Two Worlds	2024	2024 IEEE International Conference on Robotics and Automation (ICRA)	Hybrid model-based and learned	-	corpus work (retained)	full text	https://doi.org/10.1109/ICRA57147.2024.10611710	A1 A6	SAF2 SPA3 EXP2	SAF1 SPA1 NOR1 NOR3 EXP1	No	No	systematic	not coded (corpus work)	co-present	No	not coded (corpus work)	human evaluation type not differentiated	corpus record
Sathyamoorthy2020Frozone [39]	Frozone: Freezing-free pedestrian-friendly navigation in human crowds	2020	IEEE Robotics and Automation Letters	Hybrid model-based and learned	3	cited by at least 3 of the 7 reference surveys	full text	https://doi.org/10.1109/lra.2020.2996593	A1 A2 A3	SAF1 SPA4	SPA1 SPA3	No	Yes	demo	present-type-unreported	none	No	No	Gazebo crowds (success, freezing rate, pedestrian-friendliness); Jackal and Turtlebot runs in indoor crowds	Crossref record
Brito2021Subgoal [40]	Where to go next: Learning a subgoal recommendation policy for navigation in dynamic environments	2021	IEEE Robotics and Automation Letters	Hybrid model-based and learned	2	added to reach 3 works in its paradigm	full text	https://doi.org/10.1109/lra.2021.3068662	A1 A2	SAF1 SPA4	-	No	Yes	none	none	none	No	Yes	Simulation with 6-10 agents; Mann-Whitney tests	Crossref record

Table S5, part B. Pre-FM comparison set by paradigm

10 rows. Machine-readable version: `S5b_prefm_by_paradigm.csv`.

Paradigm	Works (full text)	Works (cite keys)	Human representation (A: works)	Evaluated capabilities (leaf: works)	Systematic real robot	Human evaluation of behaviour
Geometric and socially aware planning	7 (3)	Thrun1999Minerva [1], Sisbot2007SisbotHAMP [2], vandenBerg2008RVO [3], Pandey2010Framework [4], Svenstrup2010Trajectory [5], Mavrogiannis2018Social Momentum [6], Mavrogiannis2019MavrogiannisHRI 2019 [7]	A1:7 A2:1 A3:3 A4:2	SAF1:1 EXP1:1 EXP2:1	2	3 (1 ND)
Proxemics and social costmaps	3 (1)	Pacchierotti2006Passing [8], Chung2010Spatial [9], Charalampous2016SocialMaps [10]	A1:3 A2:1 A3:3 A5:1	SPA1:1 EXP1:1	1	1 (2 ND)
Social-force models	3 (2)	Helbing1995SocialForce [11], Ferrer2013Companion [12], Shiomi2014Acceptable [13]	A1:3 A2:2 A3:1 A4:2	SAF2:1 EXP1:1	1	1 (1 ND)
Group and F-formation navigation	4 (3)	Truong2017Approach [15], Wang2022GMPC [16], Kim 2022GroupEstimation [14], Katyal2022GroupAware [17]	A1:2 A2:1 A3:3 A4:4	SAF1:2 SAF2:1 SPA1:2 SPA2:3 SPA4:3	0	0 (1 ND)
Trajectory prediction and prediction-coupled planning	4 (3)	Pellegrini2009ETH [19], Trautman2010Unfreezing IGP [18], al2016SocialLSTM [20], Vemula2018Social Attention [21]	A1:1 A2:4 A4:4	SAF2:1 SPA4:1 HUM1:3	0	0
Inverse reinforcement learning	7 (5)	Henry2010Crowded [25], Kuderer2012FeatureBased Prediction [22], Luber2012Learning [26], Vasquez2014 IRLFeatures [27], Kretzschmar2016SociallyCompliant IRL [23], Okal2016BayesianIRL [24], Kim2016Socially Adaptive [28]	A1:5 A2:3 A3:1 A4:2	SAF2:1 SPA1:1 SPA2:1 SPA3:2 SPA4:1 HUM1:2 NOR1:1 NOR3:1 EXP3:2	1	0
Imitation learning	2 (2)	Tai2018GAILDepth [29], Hamandi2019DeepMoTIon [30]	A6:2	SAF1:1	0	0
Deep reinforcement learning	4 (3)	Chen2017CADRL [31], Chen2017SACADRL [32], Everett 2018GA3CCADRL [33], Liu2020DRLCrowded [34]	A1:4 A6:1	SAF1:3 SPA3:1 SPA4:1 NOR1:1	1	0 (1 ND)
Attention, graph and transformer models	3 (3)	Chen2019SARLCrowdNav [35], Chen2020RGL [37], Wang 2023NaviSTAR [36]	A1:3 A4:3 A6:1	SAF1:3 SAF2:1 SPA4:3 EXP1:1 EXP3:1	1	1
Hybrid model-based and learned	3 (3)	Sathyamoorthy2020Frozone [39], Brito2021Subgoal [40], Raj2024HybridSocialNav [38]	A1:3 A2:2 A3:1 A6:1	SAF1:2 SAF2:1 SPA3:1 SPA4:2 EXP2:1	1	1

Table S6. Robustness of headline counts

6 rows. Machine-readable version: `S6_robustness.csv`.

Indicator	All direct papers	Peer-reviewed (class A)	Public before Cai et al. cut-off
Linguistic description of people (A5)	30/45 (66.7%)	19/24 (79.2%)	26/38 (68.4%)
Social-instruction following evaluated (NOR2)	8/45 (17.8%)	4/24 (16.7%)	7/38 (18.4%)
Crowd navigation evaluated (SPA4)	1/45 (2.2%)	0/24 (0.0%)	1/38 (2.6%)
FM output never reaches motion (D0/D8 only)	9/45 (20.0%)	5/24 (20.8%)	8/38 (21.1%)
Systematic real-robot evaluation	18/45 (40.0%)	11/24 (45.8%)	15/38 (39.5%)
Human evaluation of robot behaviour	5/45 (11.1%)	4/24 (16.7%)	4/38 (10.5%)

Table S7. Social-capability coverage

20 rows. Machine-readable version: `S7_capability_coverage.csv`.

Code	Capability	Direct: evaluated (Yes)	Direct: addressed (Yes or Partial)	Pre-FM comparison set: evaluated (n = 40)	Datasets / benchmarks: evaluated (n = 15)	Example direct papers (Yes)
SAF family		20/45	32/45	16/40	7/15	
SAF1	Collision avoidance	17/45	32/45	12/40	7/15	Wang2024SALM, MartinezBaselga2025HeyRobot, Song2025VLMSocialNav, Wang2025SAMALM
SAF2	Near-collision / risk margin	7/45	12/45	6/40	1/15	Wen2025HSACLLM, BejaranoSepulveda2026PAD, Liao2026G2Nav, Oush2026VLATurtleBot
SPA family		18/45	44/45	16/40	2/15	
SPA1	Personal space	13/45	26/45	4/40	2/15	Wang2024SALM, Luo2025GSON, MartinezBaselga2025HeyRobot, Narasimhan2025OLiViaNav
SPA2	Group / conversation space	3/45	13/45	4/40	0/15	Luo2025GSON, Fang2026GroupJoining, Fang2026ObstaclestoEtiquette
SPA3	Encounter conventions	8/45	28/45	4/40	0/15	Sathyamoorthy2024CoNVOI, Luo2025GSON, Song2025VLMSocialNav, Weerakoon2025BehAV
SPA4	Crowd navigation	1/45	22/45	11/40	0/15	Zhao2026EvoNav
HUM family		2/45	24/45	5/40	0/15	
HUM1	Motion understanding / prediction	0/45	21/45	5/40	0/15	-
HUM2	Intention / activity inference	2/45	16/45	0/40	0/15	Fang2026ObstaclestoEtiquette, Zhang2026SAATTNav
NOR family		14/45	39/45	2/40	0/15	
NOR1	Etiquette / politeness	3/45	26/45	2/40	0/15	Song2025VLMSocialNav, Fang2026ObstaclestoEtiquette, Park2026TACET
NOR2	Explicit social-instruction following	8/45	15/45	0/40	0/15	Wang2024SALM, MartinezBaselga2025HeyRobot, Weerakoon2025BehAV, Chen2026LISN
NOR3	Context-dependent conduct	9/45	36/45	1/40	0/15	Sathyamoorthy2024CoNVOI, MartinezBaselga2025HeyRobot, Ruo2025LLMSocialCues, Weerakoon2025BehAV
NOR4	Cultural / norm adaptation	0/45	0/45	0/40	0/15	-
EXP family		3/45	25/45	8/40	1/15	
EXP1	Perceived safety / comfort	2/45	13/45	4/40	1/15	Song2025VLMSocialNav, Wang2026LENav
EXP2	Legibility / predictability	0/45	6/45	2/40	0/15	-
EXP3	Naturalness	1/45	14/45	3/40	0/15	BejaranoSepulveda2026PAD

Table S8. Code dictionary

50 rows. Machine-readable version: `S8_code_dictionary.csv`.

Dimension	Code	Label	Definition or values
A Social representation	A0	Not represented	Humans absent, or present only as instruction givers
A Social representation	A1	Kinematic agent	Tracked position, velocity or disc
A Social representation	A2	Predicted motion	Trajectory forecasts or distributions
A Social representation	A3	Social-space field	Proxemic potentials, personal or group spaces, social cost layers
A Social representation	A4	Relational or interaction structure	Pairwise interactions, graphs, attention over agents, groups as relations
A Social representation	A5	Symbolic or linguistic description	Attributes, activities, intents or rules expressed in language or symbols
A Social representation	A6	Implicit or latent	Raw images interpreted by a learned model; humans only as embeddings
B FM role	R1	Scene and context interpretation	Non-human context: place type, traversability, objects, situation
B FM role	R2	Human-state and relation inference	Motion, intent, activity, identity, group membership, crowding
B FM role	R3	Social-norm reasoning and behaviour selection	The FM itself infers what is socially appropriate and selects the behaviour
B FM role	R4	Instruction and preference grounding	Norms, preferences or tasks given in language are translated into plans, constraints or parameters
B FM role	R5	Human-facing communication	Dialogue or explanations delivered to people
C Social capability	SAF1	Collision avoidance	Evaluated: dedicated metric or systematic scenario-specific criterion; addressed: targeted without such a measure
C Social capability	SAF2	Near-collision or risk margin	Evaluated: dedicated metric or systematic scenario-specific criterion; addressed: targeted without such a measure
C Social capability	SPA1	Personal space	Evaluated: dedicated metric or systematic scenario-specific criterion; addressed: targeted without such a measure
C Social capability	SPA2	Group and conversation space	Evaluated: dedicated metric or systematic scenario-specific criterion; addressed: targeted without such a measure
C Social capability	SPA3	Encounter conventions (passing side, yielding, queueing)	Evaluated: dedicated metric or systematic scenario-specific criterion; addressed: targeted without such a measure
C Social capability	SPA4	Crowd navigation	Evaluated: dedicated metric or systematic scenario-specific criterion; addressed: targeted without such a measure
C Social capability	HUM1	Motion understanding or prediction	Evaluated: dedicated metric or systematic scenario-specific criterion; addressed: targeted without such a measure
C Social capability	HUM2	Intention or activity inference	Evaluated: dedicated metric or systematic scenario-specific criterion; addressed: targeted without such a measure
C Social capability	NOR1	Etiquette and politeness	Evaluated: dedicated metric or systematic scenario-specific criterion; addressed: targeted without such a measure
C Social capability	NOR2	Explicit social-instruction following	Evaluated: dedicated metric or systematic scenario-specific criterion; addressed: targeted without such a measure
C Social capability	NOR3	Context-dependent conduct	Evaluated: dedicated metric or systematic scenario-specific criterion; addressed: targeted without such a measure
C Social capability	NOR4	Cultural or norm adaptation	Evaluated: dedicated metric or systematic scenario-specific criterion; addressed: targeted without such a measure
C Social capability	EXP1	Perceived safety and comfort	Evaluated: dedicated metric or systematic scenario-specific criterion; addressed: targeted without such a measure
C Social capability	EXP2	Legibility and predictability	Evaluated: dedicated metric or systematic scenario-specific criterion; addressed: targeted without such a measure
C Social capability	EXP3	Naturalness	Evaluated: dedicated metric or systematic scenario-specific criterion; addressed: targeted without such a measure
D Navigation interface	D0	Non-executed decision	FM output compared with annotations; not executed
D Navigation interface	D1	Semantic or social state	Groups, intents, activities passed to a planner
D Navigation interface	D2	Candidate selection	FM chooses among planner-generated options
D Navigation interface	D3	Objective, constraint or parameter shaping	FM sets costs, constraints or controller parameters
D Navigation interface	D4	Goal, waypoint or reference trajectory	FM specifies where to go
D Navigation interface	D5	Runtime latent conditioning	FM embeddings or tokens condition a learned policy
D Navigation interface	D6	Direct action	FM or an FM-based policy (e.g. a VLA) outputs velocities or trajectories without an intervening planner selecting the motion; an FM-independent safety guard is coded separately
D Navigation interface	D7	Training-time supervision	FM supplies labels, rewards or distillation targets; no FM at runtime
D Navigation interface	D8	Human-facing channel	Dialogue or explanations sent to people
E Evidence profile	EV1	Offline evaluation	Yes / No
E Evidence profile	EV2	Closed-loop simulation	Yes / No

Dimension	Code	Label	Definition or values
E Evidence profile	EV3	Physical robot	none / demonstration / systematic
E Evidence profile	EV4	Real people in robot trials	none / simulated only / staged / unscripted / present, type unreported / not reported
E Evidence profile	EV5	Human evaluation	none / text raters / observers or video raters / co-present participants
E Evidence profile	EV6	Open-world or long-duration deployment	Yes / No
Cross-cutting attribute	coupling	Temporal coupling	per decision / asynchronous slow-fast / event-triggered / offline / training only
Cross-cutting attribute	guard	FM-independent safety guard	Safety that does not rely on the FM being correct (CBF filter, reflex layer, constrained optimiser, feasible-candidate restriction)
Cross-cutting attribute	uncertainty	Uncertainty treatment	explicit / partial / none
Cross-cutting attribute	hallucination	Hallucination mitigation	structural / prompt or training level / none
Cross-cutting attribute	location	Inference location	on board / local server / remote server / cloud API / hybrid
Cross-cutting attribute	memory	Memory and continual adaptation	Persistent state across episodes
Cross-cutting attribute	privacy	Privacy constraint	Privacy treated as an objective or constraint
Cross-cutting attribute	explanation	Explanation output	Text delivered to people

Table S9. Navigation interfaces and evidence

12 rows. Machine-readable version: `S9_interface_evidence.csv`.

Consumer	Interface	Label	Papers	Systematic robot evaluation	FM-independent safety guard	Cloud / hybrid / remote inference
Planner-facing	D1	Semantic or social state	5/45	4/5	3/5	2/5
Planner-facing	D2	Candidate selection	6/45	1/6	2/6	6/6
Planner-facing	D3	Objective shaping	12/45	6/12	10/12	10/12
Planner-facing	D4	Goal or waypoints	7/45	4/7	6/7	4/7
Policy-facing	D5	Latent conditioning	4/45	1/4	1/4	1/4
Policy-facing	D6	Direct action	7/45	2/7	1/7	5/7
Training-facing	D7	Training-time supervision	8/45	6/8	4/8	3/8
Human-facing	D8	Human-facing channel	5/45	2/5	3/5	2/5
Not executed	D0	Non-executed decision	8/45	0/8	0/8	0/8
Year 2024		all direct papers of the year	4/45	1/4		human evaluation of behaviour: 0/4
Year 2025		all direct papers of the year	14/45	5/14		human evaluation of behaviour: 2/14
Year 2026		all direct papers of the year	27/45	12/27		human evaluation of behaviour: 3/27

Table S10. Metric families

14 rows. Machine-readable version: `S10_metric_families.csv`.

Metric family	Metric names found (as printed)	Objective / subjective	Level	Applicability	What it measures	What it fails to measure	Direct papers using family	Representative papers
Navigation success / efficiency	Success; Overall success rate; Path length; Path length ratio; Navigation Error (NE); Success Rate (SR); Scenario generation success; Success rate…	Objective	Robot	Sim / real	Task completion and efficiency	Safety margins, social quality; >= 8 incompatible "success" definitions	23/45	Kastner2021ArenaRosnav, Biswas2022SocNav Bench, Li2024HAVLN, Marpally2026Soc RATES, Wang2024SALM
Collision / human-collision rate	Collisions; Pedestrian collisions; Total pedestrian collisions; Time-to-collision; Total Collision Rate (TCR); Collision Rate (CR); Collision rate…	Objective	Robot	Sim / real	Physical contact (SAF1)	Near misses and discomfort; definitions vary (counterfactual in MUSON; manual interventions in VLM-Social-Nav)	13/45	Tsoi2020SEAN, Kastner2021ArenaRosnav, Biswas2022SocNavBench, Li2024HAVLN, MartinezBaselga2025HeyRobot
Near-collision / risk margin	Min. distance; Minimum Distance to Human (MDH); Stopping distance (avg.); Safety Violation Time (SVT); Risk rate; Near-collision reduction…	Objective	Robot	Sim / real	Proximity safety (SAF2)	Whether proximity is socially acceptable	7/45	MartinezBaselga2025HeyRobot, Wen2025 HSACLLM, BejaranoSepulveda2026PAD, Liao 2026G2Nav, Wang2026LENav
Personal-space violation / compliance	Social score; Time of Disturbing Individuals; Comfort Distance; Personal space violation duration (PSV); Social/discomfort score; Comfort Space…	Objective	Robot	Sim / real	Proxemic compliance (SPA1)	Perceived comfort; thresholds differ (0.25 m vs 1.2 m vs Hall bands)	11/45	Wang2024SALM, Luo2025GSON, Narasimhan 2025OLiViaNav, Wang2025SAMALM, Wen2025 HSACLLM
Group-space disruption	Accurate / Miss / Extra / Error grouping rate; Time of Disturbing Groups; Social-zone Interruption Time (SIT); Maximum Social-zone Interruption Ratio…	Objective	Robot	Sim / real	Group / F-formation respect (SPA2)	Group members' perception	2/45	Luo2025GSON, Fang2026ObstaclestoEtiquette
Rule / instruction / region compliance	Assistant success rate; Behavior Following Accuracy (BFA); Average Subject Score; Average Region Score; Distance Compliance Rate (DCR); Time…	Objective	Robot	Sim / real	Instruction and convention compliance (NOR2, SPA3)	Whether rules are socially appropriate; task-specific	5/45	MartinezBaselga2025HeyRobot, Weerakoon2025 BehAV, Chen2026LISN, Chen2026SocialNav CVPR, Huo2026NORMNav
Similarity to human trajectories	Average Hausdorff distance; Frechet Distance w.r.t. Human Teleoperation; Mean squared error (MSE); Hausdorff loss; Frechet Dist. w.r.t. Human…	Objective	Model / robot	Offline / real	Human-likeness proxy (EXP3 Partial)	Perceived naturalness; demonstrator bias	8/45	Karnan2022SCAND, Sathyamoorthy2024Co NVOI, Narasimhan2025OLiViaNav, Weerakoon 2025BehAV, Chen2026SocialNavCVPR
Smoothness / comfort proxies	Average acceleration; Average jerk; Curvature; Jerk; Roughness; Path Smoothness; Acceleration / jerk	Objective	Robot	Sim / real	Motion quality	Human comfort (proxy only)	3/45	Biswas2022SocNavBench, Luo2025GSON, Chen 2026LISN, Wang2026LENav
VLM action / decision accuracy	LLM extraction accuracy; Reference Path Error; % Unacceptable Paths; Accurate / Miss / Extra / Error grouping rate; VLM crowd label; Explanation…	Objective	Model	Offline	Decision correctness vs annotation	Whether the decision is executable or safe in closed loop	10/45	Holk2024PREDILECT, Sathyamoorthy2024Co NVOI, Luo2025GSON, Saito2025VLMCrowd Mapping, Sotomi2025TransparentSocialNavVLM
Text similarity of reasoning	Cosine Similarity; BERTScore-P / -R / -F1; SBERT-cos; SMS (Sentence Mover's Similarity); Relative Position similarity; Predicted Intention…	Objective (proxy)	Model	Offline	Agreement with reference text	Correctness and grounding of the reasoning; robot behaviour	7/45	Sathyamoorthy2024CoNVOI, Kawabata2025 SocialNavMoE, Payandeh2025SocialLLaVA, Liu 2026MUSON, Xiao2026ESocialNav
FM-as-judge / expert-system scores	LLM extraction accuracy; CODA Benchmark score; Expert-system cross-validation score (P&P, R, A, E); Human evaluator ranking (P&P, R, A, E); Human…	Subjective (model)	Model	Offline	Quality as judged by an FM or human judges of text	Validity of FM judges; robot behaviour	3/45	Holk2024PREDILECT, Kong2025AutoSpatial, Payandeh2025SocialLLaVA
Human questionnaire ratings	Perceived safety rating; Usability questionnaire; User Study Score; Survey agreement and preference score PS; Perceived clarity; Perceived…	Subjective (human)	Robot (human-centred)	Real (human study)	Perceived safety, comfort, naturalness, clarity, trust (EXP)	Small samples; rarely tested statistically (3/45 papers report any statistics)	6/45	Karnan2022SCAND, Shcherbyna2025Arena40, Song2025VLMSocialNav, Sotomi2025 TransparentSocialNavVLM, BejaranoSepulveda 2026PAD

Metric family	Metric names found (as printed)	Objective / subjective	Level	Applicability	What it measures	What it fails to measure	Direct papers using family	Representative papers
Runtime / latency	Generation time / cost; Inference Time; Generation component latency; Parameter count; FPS; Inference time; LMM inference time; Runtime; Query…	Objective	System	Sim / real	Deployability	End-to-end staleness unless measured with moving humans (SPARK-VLN)	21/45	Marpally2026SocRATES, Sathyamoorthy2024CoNVOI, Shcherbyna2025Arena40, Kawabata2025SocialNavMoE, Liang2025MOSU
Task-specific social measures	Privacy risk; Social conflicts detected / sudden stops; Stopping distance (avg.); Dialogue turns (avg.); Acoustic noise	Objective	Robot	Real / sim	Privacy exposure, noise, negotiation	Comparability across tasks and papers	4/45	Yu2024PANav, Sotomi2025TransparentSocialNavVLM, BejaranoSepulveda2026PAD, Park2026TACET

Table S11. Architecture of the direct papers (per paper)

45 rows. Machine-readable version: `S11_direct_architecture.csv`.

Year	Paper	Cite key	Status	FM type - models	Inputs	Human representation (A)	FM role (B)	Interface (D)	Planner / controller	Temporal coupling	FM-independent safety guard	Other attributes	Main contribution
2025	GSON	Luo2025GSON	published	LVLM/LMM - gpt-4o-2024-08-06; compared Gemini 2.0 Flash; Gemini 2.5 Pro; Claude 3	RGB, LiDAR, Language, Tracks/ states	A1; A2; A3; A4	R2	D1	A* + BIT* re-planning + NMPC-CBF	per-decision	Yes	cloud	LMM group inference defines social spaces
2025	Transparent Social Nav VLM	Sotomi2025TransparentSocialNavVLM	published	VLM LLM - BLIP; GPT-3.5 Turbo	RGB	A5; A6	R1; R5	D1; D8	ROS2 re-planning (mechanism unspecified)	per-decision	No	explanation cloud	VLM explanations of social re-planning for trust
2026	PAD	BejaranoSepulveda2026PAD	published	VLM LLM - Not reported	RGB, LiDAR, Language	A1; A2; A5	R1; R2; R5	D1; D8	Global planner + SFM/DWA	per-decision	Yes	explanation	VLM/LLM awareness map chooses dialogue vs re-planning
2026	HCSG	Xu2026HCSG	preprint	VLM CLIP-style - Qwen3-VL-2B-Instruct; CLIP text encoder; YOLO-Pose	RGB-D, Language, Tracks/states	A1; A2; A5; A6	R2	D1	Learned graph-transformer waypoint selection	per-decision	No	-	VLM human-activity descriptions in a topological VLN map
2026	SAATT Nav	Zhang2026SAATTNav	preprint	LLM - LLM	Tracks/states	A1; A2; A3; A5	R2; R5	D1; D8	Rollout-based selection in 2D planner	per-decision	Yes	explanation	LLM pedestrian-intent hypotheses + rider explanations
2024	SALM	Wang2024SALM	preprint	LLM/VLM - GPT-4o	Language, Tracks/ states	A1; A4; A5	R3; R4	D6; D2	Fusion with DRL policy (LLM feedback scorer)	per-decision	No	cloud	LLM and DRL navigation commands fused by an LLM feedback model
2024	PANav	Yu2024PANav	preprint	VLM - GPT-4 Turbo	RGB	A3	R1; R3	D2	A* candidate paths	per-decision	Yes	cloud	VLM selects the privacy-minimising path
2025	MOSU	Liang2025MOSU	accepted (not yet indexed)	VLM - Gemini	RGB, LiDAR	A6	R1; R3	D2; D3	CVAE candidates + motion planner	per-decision	No	cloud	VLM ranks outdoor trajectories by social cues
2025	SAMALM	Wang2025SAMALM	preprint	LLM/VLM - GPT-4o; other zero-shot LLMs as baselines	Language, Tracks/ states	A1; A5	R3	D6; D2	None (direct velocities)	per-decision	No	cloud	Multi-agent LLM actors with local/ global critics
2026	Obstacles to Etiquette	Fang2026ObstaclestoEtiquette	published	VLM - Qwen-2.5 7B fully fine-tuned; GPT-4o	RGB, LiDAR, Language, Tracks/ states	A1; A2; A3; A5	R2; R3	D2; D7	Modified ORCA for pedestrian avoidance along selected path	asynchronous	Yes	async cloud	Fine-tuned VLM selects the socially compliant candidate path
2026	AHead VLN	Zhang2026AHeadVLN	published	LLM/VLM - gpt-4o-2024-05-13; gpt-3.5-turbo	RGB-D, Language	A2; A5; A6	R1; R3; R4	D2; D6	Waypoint candidates; LLM-written discrete action scripts	per-decision	No	cloud	Forecast human-risk events compiled for hierarchical LLM path choice
2025	Hey Robot	MartinezBaselga2025HeyRobot	published	LLM VLM - GPT-4o-mini	RGB, Language, Tracks/states	A1; A2	R1; R4	D3	MPC with collision constraints	asynchronous	Yes	async cloud	LLM writes and tunes MPC costs from user requests
2025	LLM Social Cues	Ruo2025LLMSocialCues	published	LLM - llama3.1:8b; command-r7b	LiDAR, Language	A1; A3; A5	R2; R3; R4	D3; D4	CBF quadratic-program controller	per-instruction	Yes	-	Local LLM sets navigation mode, speed bound and social-zone distance from user cues
2025	VLM Crowd Mapping	Saito2025VLMCrowdMapping	published	VLM - gpt-4o-mini	RGB, LiDAR, Language	A5	R2	D3	Dijkstra on fused cost map	per-decision	Yes	cloud	VLM crowd labels become a GPR crowd-density map
2025	VLM Social Nav	Song2025VLMSocialNav	published	VLM - GPT-4V	RGB, LiDAR, Language, Goal	A1; A5	R1; R3	D3	DWA	per-decision	Yes	cloud	VLM output converted into a social cost term
2025	BehAV	Weerakoon2025BehAV	published	LLM VLM CLIP-style - GPT-4; CLIPSeg; GPT-4o; GPT4v, Gemini compared	RGB, LiDAR, Language	A3; A5	R1; R3; R4	D3; D4	MPC on behavioural cost map	per-decision	Yes	cloud	Language rules grounded into behaviour cost maps
2026	LISN	Chen2026LISN	published	VLM open-vocabulary perception - GPT-4o; Grounded-SAM 2; RoboPoint	RGB, LiDAR, Language	A1; A3; A5	R1; R2; R4	D3; D4	SFM controller + social costmap	asynchronous	Yes	async cloud	Slow VLM modulates a fast social-force controller; LISN-Bench
2026	NORM Nav	Huo2026NORMNav	published	VLM/LLM open-vocabulary segmentation - Qwen-VL-Max; Grounded-SAM 2	RGB-D, LiDAR, Language	A3; A5	R1; R4	D3	A* + TEB on multi-layer costmaps	per-instruction	Yes	cloud	Language norms parsed into costmap constraints
2026	G2 Nav	Liao2026G2Nav	accepted (not yet indexed)	VLM - Qwen3.5-2B; GPT-5.4-Nano; RAM++; GroundingDINO; SAM	RGB, LiDAR, Tracks/ states, Goal	A1; A3; A5	R1; R2	D3	Gradient trajectory optimisation + pure pursuit	asynchronous	Yes	async cloud	Grounded VLM costmaps with LiDAR reflex safety layer
2026	TACET	Park2026TACET	preprint	VLM - Qwen2.5-VL-7B; Gemini 2.5 Flash	RGB	A6	R1; R3	D3; D7	Social costmap + quiet-locomotion RL policy	asynchronous	Yes	async	VLM behaviour tokens for context-appropriate gait and noise
2026	Context Aware Preferences	Sethuraman2026ContextAwarePreferences	preprint	VLM LLM - Gemini 2.0 Flash; Mistral-Large-2.1 / GPT-4o	RGB, Language	A1	R1; R4	D3	Preference-conditioned MORL policy	asynchronous	No	async memory cloud	VLM/LLM translate context and rules into MORL weights
2026	LE Nav	Wang2026LENav	published	MLLM - GPT-4o; Qwen-VL-Max; Kimi-latest	RGB, LiDAR, Language	A1; A5	R1; R2	D3	TEB/DWA with generated hyperparameters	asynchronous	Yes	async cloud	MLLM scene text conditions CVAE planner tuning
2024	CoNVOI	Sathyamoorthy2024CoNVOI	published	VLM CLIP-style - CLIP; GPT-4V; Gemini	RGB, LiDAR, Language	A1; A6	R1; R3	D4	Modified DWA	per-decision	Yes	cloud	VLM reference paths encoding context-dependent behaviours
2026	Group Joining	Fang2026GroupJoining	preprint	vision-language model (contrastive) - SigLIP2	RGB, Language	A1; A3; A4	R2; R4	D4	Local planner to predicted goal	asynchronous	Yes	async	Language-guided, F-formation-aware group joining
2026	RBTLR	Gandotra2026RBTLR	acceptance verified; review not established	VLM (contrastive) VLM LLM - CLIP ViT-B/32; Qwen VLM; LLM	RGB, Language	A1; A5; A6	R1; R4; R5	D5; D4; D8	RRT-Connect candidates + learned selector	per-decision	Yes	explanation	CLIP/Qwen social context conditions path selection; LLM goal assignment
2026	Walk With Me	Zhang2026WalkWithMe	preprint	VLM VLA - high-level VLM; low-level VLA	RGB, Language	A6	R3; R4	D4; D6	Low-level VLA on remote server	hierarchical	No	async cloud	VLM-VLA hierarchy for human-accompanying outdoor navigation
2025	OLiVia Nav	Narasimhan2025OLiViaNav	published	VLM CLIP-style - GPT-4o; SC-CLIP: ViT-L/14 image encoder + transformer text encoder	RGB, LiDAR, Language, Goal	A5; A6	R1	D5; D7	Learned trajectory planner + selector, PID	per-decision	No	memory cloud	Distilled VLM context embeddings condition a learned planner
2026	HUMA	Ahmadi2026HUMA	preprint	VLM - Qwen3-VL-2B	RGB-D	A5; A6	R2; R3	D5	RL policy (Falcon)	event-triggered	No	async	Proximity-triggered VLM reasoning conditions an RL policy
2026	SPARK VLN	Hu2026SPARKVLN	preprint	VLM - VILA-based VLM reasoner	RGB, Language	A6	R1; R4	D5	Flow-matching planner	asynchronous	No	async	Token-wise VLM latent streaming against stale observations
2026	SocialNav CVPR	Chen2026SocialNavCVPR	published	VLA VLM NavFM - Qwen2.5-VL 3B; Diffusion Transformer flow-matching action expert; Qwen	RGB, Tracks/states, Goal	A6	R1; R3	D6; D7	Flow-matching action expert + locomotion policy	per-decision	No	cloud	VLA-style social-navigation FM with SocNav benchmark
2026	VLA TurtleBot	Oush2026VLATurtleBot	published	LLM - LLM	RGB, LiDAR, Goal	A1; A5	R1; R3	D6	Motion controller behind a LiDAR safety layer	event-triggered	Yes	async	LLM picks proceed, yield, avoid or stop when a person is detected
2026	SALSA	Wang2026SALSA	preprint	VLA - OmniVLA	RGB, Goal	A6	R3	D6	None (VLA actions)	per-decision	No	-	Social-safety post-training of a navigation VLA

Year	Paper	Cite key	Status	FM type - models	Inputs	Human representation (A)	FM role (B)	Interface (D)	Planner / controller	Temporal coupling	FM-independent safety guard	Other attributes	Main contribution
2025	SocialNav MoE	Kawabata2025SocialNavMoE	preprint	VLM - SocialNav-MoE: SigLIP + Phi-2-2.7B with sparse MoE layers; variants: Q	RGB, Language	A5	R1; R2; R3	D0	None	offline	No	-	Small MoE VLM with RL fine-tuning for social actions
2025	AutoSpatial	Kong2025AutoSpatial	published	VLM - LLaVA-1.6 OneVision-7B; GPT-4o, Gemini 2.0 Flash, Claude 3.5 Sonnet	RGB, Language	A5	R1; R2; R3	D0	None	offline	No	-	Structured spatial grounding for social VLM reasoning
2025	Social LLaVA	Payandeh2025SocialLLaVA	published	VLM - LLaVA-v1.5-7B + LoRA; GPT-4V; Gemini 1.5 Pro	RGB, Language	A5	R1; R2; R3	D0	None (hard-coded demo mapping)	offline	No	-	SNEI dataset; perception-to-action social VLM reasoning
2026	MAction SocialNav	Wang2026MActionSocialNav	published	VLM - NVILA-Lite-2B; GPT-4o, Claude	RGB, Language	A5	R1; R2; R3	D0	None	offline	No	-	Small VLM ranking multiple acceptable actions
2026	E SocialNav	Xiao2026ESocialNav	published	VLM - SigLIP + Phi-2-2.7B; variants: CLIP, DINO vision towers; TinyLlama-1.1	RGB, Language	A5	R1; R2; R3	D0	None	offline	No	-	Efficient small VLM via SFT + DPO
2026	Prompting VLMs JSAI	Xiao2026PromptingVLMsJSAI	acceptance verified; review not established	VLM - TinyLLaVA framework with SigLIP + Phi, StableLM, TinyLlama language mo	RGB, Language	A5	R1; R2; R3	D0	None	offline	No	-	Prompt-design study for small VLMs
2026	GCL	Zhang2026GCL	preprint	VLM - Qwen2.5-VL-3B, Qwen2.5-VL-7B, Qwen3-VL-2B, Qwen3-VL-4B, Qwen3-VL-8B	RGB, Language	A5	R1; R2; R3	D0	None	offline	No	-	Competitive co-distillation of small VLMs
2026	SOPD SocialNav	Zhang2026SOPDSocialNav	accepted (not yet indexed)	VLM - Teacher: Qwen3-VL-8B; Student: Qwen2.5-VL-3B	RGB, Language	A5	R1; R3	D0	None (qualitative robot demo)	offline	No	-	Selective on-policy distillation of a social VLM
2024	PREDILECT	Holk2024PREDILECT	published	LLM - GPT-4	Language	A1	R4	D7	RL policy (SAC/PPO) + SFM repulsion	training-only	Yes	-	LLM turns free-text preference explanations into reward features
2026	ViLAM	Elnoor2026ViLAM	preprint	VLM - GPT-4o; VANP as base	RGB, Goal	A3; A6	R2	D7	DWA-style planner with distilled attention cost	training-only	Yes	-	Distils VLM crowd reasoning into attention maps
2026	Narrate2Nav	Payandeh2026Narrate2Nav	published	VLM vision-language encoder vision foundation encoder - Qwen2-VL-72 B-Instruct; CLIP text encoder; DINOv2	RGB, Goal	A5; A6	R1; R2	D7	Student policy + pure pursuit	training-only	No	-	VLM narrations distilled into an RGB-only student
2026	EvoNav	Zhao2026EvoNav	preprint	LLM - gpt-oss-120B		A1	R3	D7	DRL policy (CrowdNav++)	training-only	No	-	LLM evolves social reward code
2025	HSAC LLM	Wen2025HSACLLM	published	LLM - GPT-4	LiDAR, Language, Tracks/states, Goal	A1; A5	R5	D8	DRL policy (HSAC)	per-decision	No	explanation cloud	LLM speech for robot-pedestrian passage negotiation

Table S12. Capabilities and evidence of the direct papers (per paper)

45 rows. Machine-readable version: `S12_direct_evidence.csv`.

Paper	Evaluated capabilities (Yes leaves)	Addressed-only families (Partial)	EV1 offline	EV2 closed-loop sim	EV3 physical robot	EV4 real humans in trials	EV5 human evaluation	EV6 open-world	Trials (as reported)	Statistics	Key social / safety / human metrics	Runtime reported	Uncertainty	Hallucination mitigation	Code / data	Key limitation
GSON	SPA1 SPA2 SPA3	SAF HUM NOR EXP	Yes	No	systematic	unscripted	none	No	3 runs per scenario	No	Time of Disturbing Individuals; Time of Disturbing Groups; Comfort Distance	GPT-4o ~4 s; Qwen2.5-VL-32B ~2 s (offline test); 1.5-2 s in dense-crowd tests…	No	Partial	code No; data No	No human evaluation; 3 runs per scenario
Transparent Social Nav VLM	-	SPA NOR EXP	Yes	No	systematic	present-type-unreported	observers/video	No	average of 4 runs per condition	No	Social conflicts detected / sudden stops; Survey agreement and preference score PS	~20 s mean explanation latency (6-51 s)	No	No	code No; data NA	4 runs; ~20 s explanation latency
PAD	SAF2 EXP3	SPA HUM NOR	No	No	systematic	staged	co-present participants	No	1 trial per condition per participant (3…	No	Stopping distance (avg.); Perceived clarity; Perceived naturalness; Preferred strategy	Not reported (VLM updates at lower frequency due to latency)	No	No	code No; data NA	n = 8, no statistics; FMs unnamed
HCSG	SAF1 NOR2	SPA HUM	No	Yes	systematic	present-type-unreported	none	No	1,839 unseen val episodes; 10 real trials…	No	Total Collision Rate (TCR); Collision Rate (CR)	Not reported	No	No	code Partial; data NA	No safety layer; no human study
SAATT Nav	SAF1 SAF2 SPA1 HUM2	EXP	No	Yes	none	simulated-only	none	No	360 trials	Yes	Collisions / min distance; Personal-space intrusion	Not reported	Partial	Yes	code No; data NA	Custom 2D simulation; LLM unnamed
SALM	SAF1 SPA1 NOR2	EXP	No	Yes	none	simulated-only	none	No	50 test cases per task	No	Social score	Not reported	No	Partial	code No; data NA	Simulation only; cloud LLM in control loop
PANav	-	NOR	Yes	No	demo	none	none	No	3 areas x 3 episodes	No	Privacy risk	Not reported	Partial	Yes	code No; data NA	No humans in quantitative evaluation
MOSU	-	SPA NOR	Yes	No	demo	none	none	No	Not reported	No	-	2.30 s per trajectory-generation cycle	No	Partial	code No; data NA	No social metric; robot runs unquantified
SAMALM	SAF1 SPA1	EXP	No	Yes	none	simulated-only	none	No	50 test cases per setting	No	Social/discomfort score	Not reported	Partial	Yes	code No; data NA	Simulation only
Obstacles to Etiquette	SAF1 SPA1 SPA2 SPA3 HUM2 NOR1 NOR3	EXP	No	No	systematic	staged	none	No	3 trials per method per scenario	No	Personal Space Violation duration (PSV); Time Facing Pedestrians (TFP); Social-zone Interruption Time (SIT); Maximum…	Not reported numerically	Partial	Yes	code No; data No	3 trials per method; staged actors
AHead VLN	SAF1	SPA HUM	No	Yes	none	simulated-only	none	No	216 evaluation trajectories	No	TCR; CR	Not reported numerically (parallel scripting claimed to reduce latency)	No	Yes	code No; data NA	Simulation only; collisions in 44% of episodes; cloud LLMs
Hey Robot	SAF1 SPA1 NOR2 NOR3	HUM EXP	No	Yes	demo	staged	none	No	10 simulations per query (6 queries)	No	Collision rate; Min. distance	~3 s for parameter updates; ~7 s with new cost generation	No	Yes	code Promised; data NA	Keep-distance request not fully met; lab demo
LLM Social Cues	NOR3	SAF SPA HUM EXP	Yes	Yes	none	simulated-only	none	No	Not reported (example runs per use case)	No	-	Not reported	No	Yes	code No; data NA	Simulation only; no baselines; positions assumed known
VLM Crowd Mapping	-	SAF SPA NOR	No	No	demo	staged	none	No	2	No	-	Not reported	Partial	No	code No; data NA	Two qualitative trials
VLM Social Nav	SAF1 SPA3 NOR1 EXP1	HUM	No	No	systematic	staged	co-present participants	No	21 runs per method per scenario	No	Collision Rate; User Study Score	~3 s average VLM inference; 2-3 s average response after prompt formatting…	No	Partial	code No; data NA	Small user study; no statistics
BehAV	SPA3 NOR2 NOR3	SAF HUM EXP	Yes	No	systematic	staged	none	No	at least 10 trials per scenario	No	Behavior Following Accuracy (BFA)	GPT-4o landmark inference ~6 s via internet (p.5); 3.59 s in Table II; overall…	Partial	No	code Yes; data NA	Most rules concern terrain/ objects
LISN	SAF1 SPA1 NOR2 NOR3	HUM EXP	No	Yes	none	simulated-only	none	No	5-9 repetitions per scenario	No	Collision Rate; Path Smoothness; Average Subject Score; Average Region Score	VLM slow loop ~7.1 s; fast loop 5.27 ms	No	Partial	code No; data No	Simulation only; ~7 s VLM latency
NORM Nav	SPA3 NOR2	SAF EXP	No	Yes	demo	present-type-unreported	none	No	Not reported (success in 10% increments)	No	Behavioral Following Accuracy (BFA)	Not reported	No	Yes	code No; data NA	Pedestrian behaviour only qualitative
G2 Nav	SAF1 SAF2	SPA HUM NOR EXP	Yes	No	systematic	present-type-unreported	none	No	6 trials per method (real)	No	Safety Violation Time (SVT)	avg query 1.5 s (local) / 4 s (online)	Partial	Yes	code Promised; data NA	6 trials per method; no human study
TACET	SAF1 SPA1 NOR1 NOR3	EXP	No	No	systematic	staged	observers/video	No	10 trials per condition	No	Personal-space compliance (PSC); Appropriateness ratings	2.6 s median reasoner query, 95th percentile 3.5 s (p3)	No	Partial	code No; data No	Video raters only (N = 15)
Context Aware Preferences	NOR2 NOR3	SPA EXP	No	No	systematic	present-type-unreported	text-raters	No	Not reported	No	Rule/explanation ratings	context 9.59 s, rule update 5.94 s, translation 3.34 s (asynchronous)	No	Partial	code Yes; data NA	Humans rated text, not robot behaviour
LE Nav	SAF1 SAF2 NOR3 EXP1	SPA	No	No	systematic	present-type-unreported	co-present participants	No	>100 real-world trials	No	Risk rate; Acceleration / jerk; Questionnaire scores	MLLM ~0.5 Hz; packet loss handled by masking	Partial	Partial	code Yes; data No	No statistics reported for user study
CoNVOI	SPA3 NOR3	SAF EXP	No	No	systematic	present-type-unreported	none	No	Not reported	No	-	Inference time 3.28-6.36 s across configurations (Table I); t_query assumed 5 s	No	Partial	code Yes; data NA	Evaluation mostly terrain/ crosswalk context
Group Joining	SPA1 SPA2 SPA3 NOR2	-	Yes	No	systematic	present-type-unreported	none	No	5 per real scenario	No	Personal-space violation duration (PSV)	0.936 s grounding	No	NA	code No; data No	Small trial counts; no member ratings
RBTLR	-	SAF SPA NOR	No	Yes	none	simulated-only	none	No	Not reported	No	-	0.33 s per selection cycle	No	No	code No; data NA	Early-stage; demonstration only; no social metrics
Walk With Me	-	SAF SPA NOR	No	No	systematic	present-type-unreported	none	No	20 real trials	No	-	Not reported	No	No	code No; data NA	60% success; no latency reported

Paper	Evaluated capabilities (Yes leaves)	Addressed-only families (Partial)	EV1 offline	EV2 closed-loop sim	EV3 physical robot	EV4 real humans in trials	EV5 human evaluation	EV6 open-world	Trials (as reported)	Statistics	Key social / safety / human metrics	Runtime reported	Uncertainty	Hallucination mitigation	Code / data	Key limitation
OLiVia Nav	SPA1	SAF HUM NOR EXP	Yes	No	systematic	staged	none	No	3 trials per scenario	Yes	Personal space violation duration (PSV)	average 5 Hz planning (OLi Via-Nav); VLM-Social-Nav with GPT4o average 0.6 Hz…	No	No	code No; data No	No safety layer; 4x3 trials
HUMA	SAF1	SPA HUM NOR	No	Yes	demo	not-reported	none	No	Not reported	No	Human collision (H-Coll)	2,425 ms per VLM call	No	No	code No; data NA	High human-collision rate; robot demo only
SPARK VLN	SAF1 SPA1	NOR	No	Yes	none	simulated-only	none	No	Not reported in text read	No	PSC; Col; AvgD	0.185 s average (max 1.030 s) per inference with streaming vs 0.788 s…	No	No	code No; data No	Simulation only; ~29% human-collision rate
SocialNav CVPR	SPA3	SAF NOR EXP	No	Yes	systematic	not-reported	none	No	180 benchmark cases; 20 real trials per…	No	Distance Compliance Rate (DCR); Time Compliance Rate (TCR)	Operates at over 5 Hz (p.8)	No	No	code No; data No	No safety layer; success-rate-only real trials
VLA TurtleBot	SAF2	SPA	No	Yes	none	simulated-only	none	No	Not reported	No	Minimum distance to obstacles; Safety-override frequency	Not reported numerically (LLM delay discussed)	No	Yes	code No; data NA	Simulation only; single simulated person; LLM not named
SALSA	SAF2	SPA EXP	Yes	No	demo	none	none	No	Not reported	No	Near-collision reduction	Not reported	No	No	code No; data NA	Quantitative results offline only
SocialNav MoE	-	SPA HUM NOR	Yes	No	none	none	none	No	NA	No	-	1.709 FPS (vs GPT-4o 0.212, Claude 0.087); hardware not reported	No	No	code Promised; data NA	Offline text similarity only
AutoSpatial	-	SPA HUM NOR	Yes	No	none	none	text-raters	No	NA	No	-	Not reported	No	Partial	code No; data No	Offline; LLM/human scoring of text
Social LLaVA	-	SPA HUM NOR	Yes	No	demo	none	text-raters	No	NA	No	-	Not reported	No	Partial	code No; data Yes	Offline VQA; single robot demo
MAction SocialNav	-	SAF SPA HUM NOR	Yes	No	none	none	none	No	NA	Yes	-	1.524 FPS (vs GPT-4o CoT 0.314 FPS, Claude CoT 0.452 FPS)	Partial	No	code No; data No	Offline; 79 test samples
E SocialNav	-	SPA HUM NOR	Yes	No	none	none	none	No	NA	No	-	2.354 FPS (inference hardware not reported)	No	Partial	code Yes; data No	Offline; 60 test samples
Prompting VLMs JSAI	-	SPA HUM NOR	Yes	No	none	none	none	No	NA	No	-	Not reported	No	No	code No; data NA	Offline; peer review not established
GCL	-	SPA HUM NOR	Yes	No	none	none	none	No	NA	No	-	Not reported	No	No	code No; data NA	Offline; no robot experiments
SOPD SocialNav	-	SPA NOR	Yes	No	demo	none	none	No	NA	No	-	0.984 FPS model; 2.74 s end-to-end (0.365 FPS)	No	No	code No; data NA	Offline; robot only qualitative
PREDILECT	-	SPA NOR EXP	No	Yes	none	simulated-only	none	No	400 preferences per condition	No	-	Not reported	No	No	code No; data NA	Simulation only; LLM limited to text parsing
ViLAM	-	SAF SPA HUM NOR EXP	No	No	systematic	present-type-unreported	none	No	10 trials per method per scenario	No	-	~20 Hz inference	No	No	code No; data No	FM used offline only; no human evaluation
Narrate2Nav	SAF1	SPA HUM NOR EXP	Yes	No	systematic	present-type-unreported	none	No	10 real trials per scenario per method	No	Collision	Not reported (real-time claimed, no rate given)	No	No	code No; data No	Crowd scenario not among real-world tests
EvoNav	SAF1 SPA1 SPA4	-	No	Yes	none	simulated-only	none	No	Not reported	No	Collision rate; Intrusion time ratio (ITR) / social distance	NA (offline reward design)	No	Partial	code No; data NA	ORCA-only humans; no real robot
HSAC LLM	SAF1 SAF2 SPA1	HUM NOR EXP	No	Yes	demo	not-reported	none	No	500 tests per scene (2D sim)	No	Collision; Human Collisions; Minimum Distance to Human (MDH); Comfort Space Compliance (CSC)	Not reported	No	No	code No; data NA	Simulated training pedestrians respond to requests; demo only

Table S13. Datasets, simulators and benchmarks (per resource)

16 rows. Machine-readable version: `S13_datasets_benchmarks.csv`.

Resource	Cite key	Year	Type	Real / synthetic	Size	Sensors	Dynamic humans	Human trajectories	Group / intention labels	Language annotations	Preference labels	Multi-action labels	Closed loop	VLM-ready	FM role in resource	Known limitation
SEAN	Tsoi2020SEAN	2020	simulator	synthetic	2 scenes (lab, city); 2 robots	simulated LiDAR, RGB	Yes	Yes (simulated)	group: No; intent: No	No	No	No	Yes	Medium (photorealistic Unity…	None	Few scenes, robots and metrics (stated by the authors)
Arena Rosnav	Kastner2021ArenaRosnav	2021	training/ benchmark platform (2D ROS simulator)	synthetic	2 maps (office, empty); 5/10/20 dynamic obstacles x 3 velocities; 100 runs per planner per scenario	2D LiDAR (simulated)	No (generic random-moving…	No	group: No; intent: No	No	No	No	Yes	No	None	Not social navigation (no humans); text/ table inconsistencies
SocNavBench	Biswas2022SocNavBench	2022	benchmark + simulator	hybrid (real trajectories, synthetic rendering)	33 curated episodes; random sampling	RGB-D (rendered), top view	Yes (non-reactive)	Yes (real, replayed)	group: No (groups present…; intent: No	No	No	No	Yes (robot closed-loop…	Medium (photorealistic RGB-D…	None	Non-reactive pedestrians invalid for negotiation scenarios (stated by the authors)
SCAND	Karnan2022SCAND	2022	dataset	real	8.7 h, 138 trajectories, 25 miles	3D LiDAR, RGB, stereo, monocular, IMU, odometry, joystick	Yes (natural)	No (no human annotations)	group: Partial (tag…; intent: No	Partial (textual interaction tags)	No	No (but authors show >1 valid…	No (offline demonstrations)	Medium-High (egocentric RGB…	None	Single city/campus; teleoperated style of few demonstrators; no human labels
MuSoHu	Nguyen2023MuSoHu	2023	dataset	real	~100 km, 20 h, 300 trials, 13 human demonstrators	3D LiDAR, stereo/depth, 360 camera, IMU, microphone array…	Yes (natural)			No			No	Medium (egocentric RGB; SNEI…	None	Human (not robot) embodiment; single region
SACSoN	Hirose2024SACSoN	2024	dataset	real	>75 h, 58.7 km, >4,000 human interactions	RGB (fisheye/ 360), odometry	Yes (natural)						No	Medium-High	None	Office settings only
HA VLN	Li2024HAVLN	2024	simulator + dataset + benchmark	synthetic humans in real-scan environments (Matterport3D)	21,567 instructions; 435 motion models; 90 buildings; 374 humans	RGB-D egocentric renders	Yes	Yes (SMPL motion sequences…	group: No; intent: No	Yes (instructions with human activity…	No	No	Yes	High (language instructions +…	FM generates instructions/scenes	Humans do not react to agent; no social nuances or group dynamics; indoor only (authors)
Arena 4.0	Shcherbyna2025Arena40	2025	simulator/ benchmark platform	synthetic	procedurally generated (e.g. 5000 worlds); >100 obstacle and >15 human models	simulator-dependent (not specified)	Yes	Simulated (HuNav Sim)	group: Partial…; intent: No	Partial (text-to-world prompts…	No	No	Yes (simulator)	Medium (language interface…	LLM-generated worlds (unnamed)	Users requested more realistic simulators (Isaac, Habitat); evaluation pipeline found…
SocialNav SUB	Munje2025SocialNavSUB	2025	benchmark (VQA) + human-labelled dataset	real	60 scenarios; 4,968 questions; >=5 human answers each	robot front RGB + odometry; derived BEV	Yes	Yes (estimated 3D…	group: No; intent: Yes (social…	Yes (multiple-choice VQA)	Yes (distribution of human…	Partial (human answer…	No	High (designed for VLMs)	VLMs evaluated (benchmark of VLM scene understanding)	Single campus; limited models/ scenarios (authors)
SNEI (via Social-LLaVA)	Payandeh2025SocialLLaVA	2025	dataset (VQA) - SNEI	real	[a] ~2K scenarios; 1,961 images (originating paper); >40K QA; public release 1,508 rows	front-facing robot RGB image (from SCAND)	Yes (images)	No	group: Partial (agent type…; intent: Yes (robot…	Yes (categorical labels + free-form…	No	No (single final action)	No	Yes	Human-annotated VQA for VLM fine-tuning	All annotations reviewed by one person; split sizes not reported; dataset size reported…
FewShot Perception	Zhang2025FewShotPerception	2025	human-perception dataset extension	real	404 labelled episodes from 69 participants	robot and human tracking					Yes (human ratings)				LLM predicts human ratings (judge)	
SocRATES	Marpally2026SocRATES	2026	benchmark/ evaluation tool	synthetic						Yes (scenario text)			Yes (simulation)	NA (FM is the generator)	LLM generates test scenarios	Scenario validity checked manually; LLM behaviour-tree errors.
DPed VLN	Dai2026DPedVLN	2026	benchmark	synthetic	33,093 episodes; 66,186 instruction-trajectory pairs		Yes			Yes			Yes		VLM generates instructions	
HA VLN 2.0	Dong2026HAVLN20	2026	simulator + dataset + benchmark + leaderboard	synthetic SMPL humans in scanned environments	16,844 instructions; 910 human models; 90 scans	RGB-D egocentric	Yes	Yes (SMPL motion with…	group: Yes (2-, 3-…; intent: No	Yes (socially grounded instructions)	No	No	Yes (DE and CE)	High (benchmarks Na Vid/NaVILA)	LLM generates instructions	Simulated humans scripted (reactivity not described); collisions defined by proximity…
SONG	Huang2026SONG	2026	simulator + benchmark	synthetic (photorealistic)	1,000 3DGS scenes; 500 human avatars; 500 evaluation episodes		Yes						Yes	High (photorealistic RGB)	LLM generates pedestrian intents	
MUSON	Liu2026MUSON	2026	dataset/ benchmark	real	10,110 samples	RGB (egocentric, from MuSoHu human-worn sensors)	Yes (images)	No (language prediction only)	group: No; intent: Partial (language…	Yes (5-step CoT)	No	No (single action per sample)	No	High	GPT annotation (initial)	Single-frame; discrete 6 actions; human-collected ego views (not robot); long-tailed…

Fig. S1. FM roles per direct paper (Dimension B), grouped by primary navigation interface



Papers per role (non-exclusive, n = 45)

R1 Scene / context interpretation
29/45 (64%)

R2 Human-state / relation inference
21/45 (47%)

R3 Social-norm reasoning / behaviour selection
25/45 (56%)

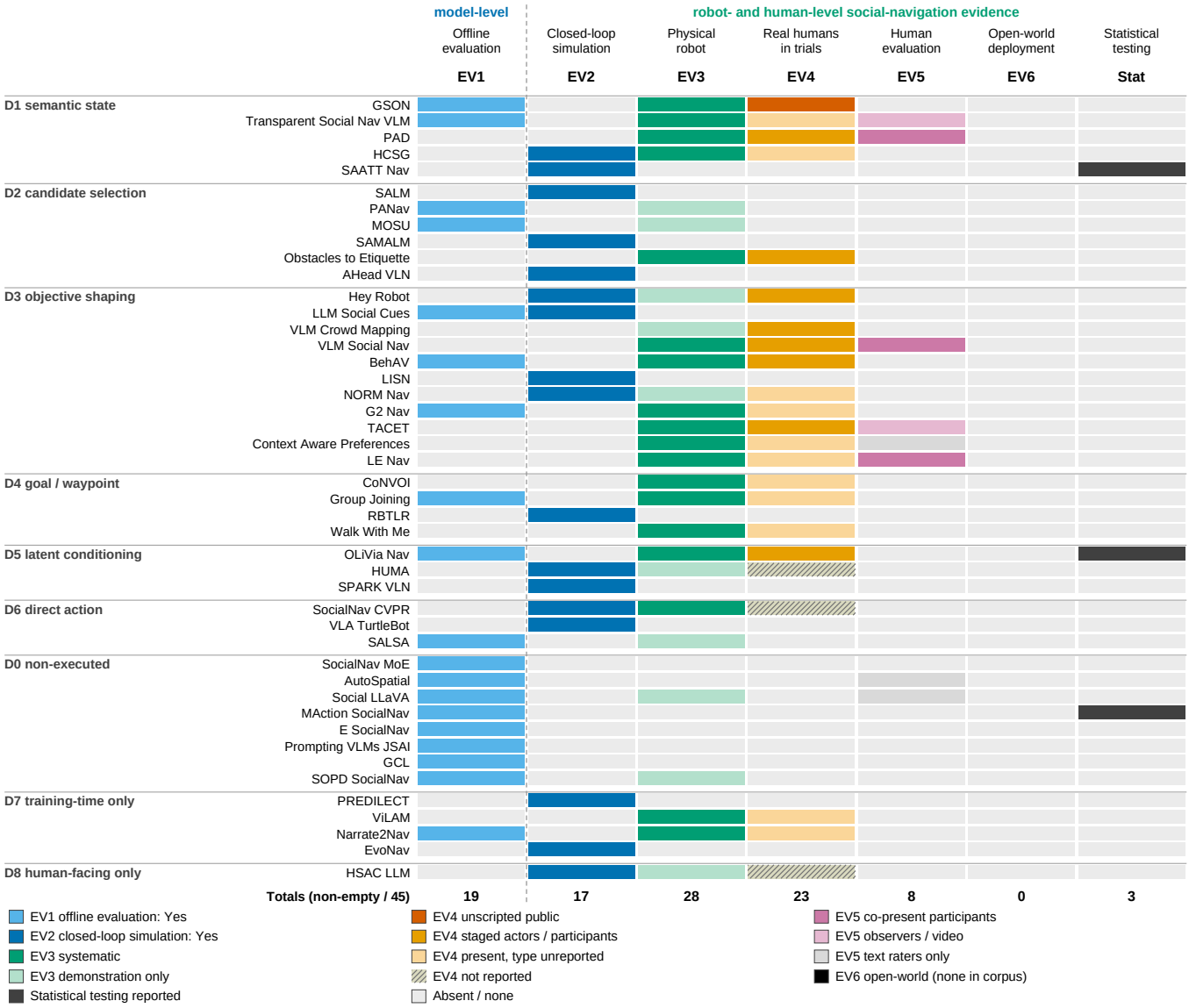
R4 Instruction / preference grounding
13/45 (29%)

R5 Human-facing communication
5/45 (11%)

Roles per paper (number of papers):
1 role: 10; 2 roles: 22; 3 roles: 13

Rows grouped by primary navigation interface (Axis D; categorical, not a ranking).

Fig. S2. Evidence profile per direct paper (Dimension E), grouped by primary navigation interface



EV4 records real people during robot trials (simulated humans only, or none, shown as absent). Rows grouped by primary navigation interface.

Fig. S3. Capability coding per direct paper (Dimension C)

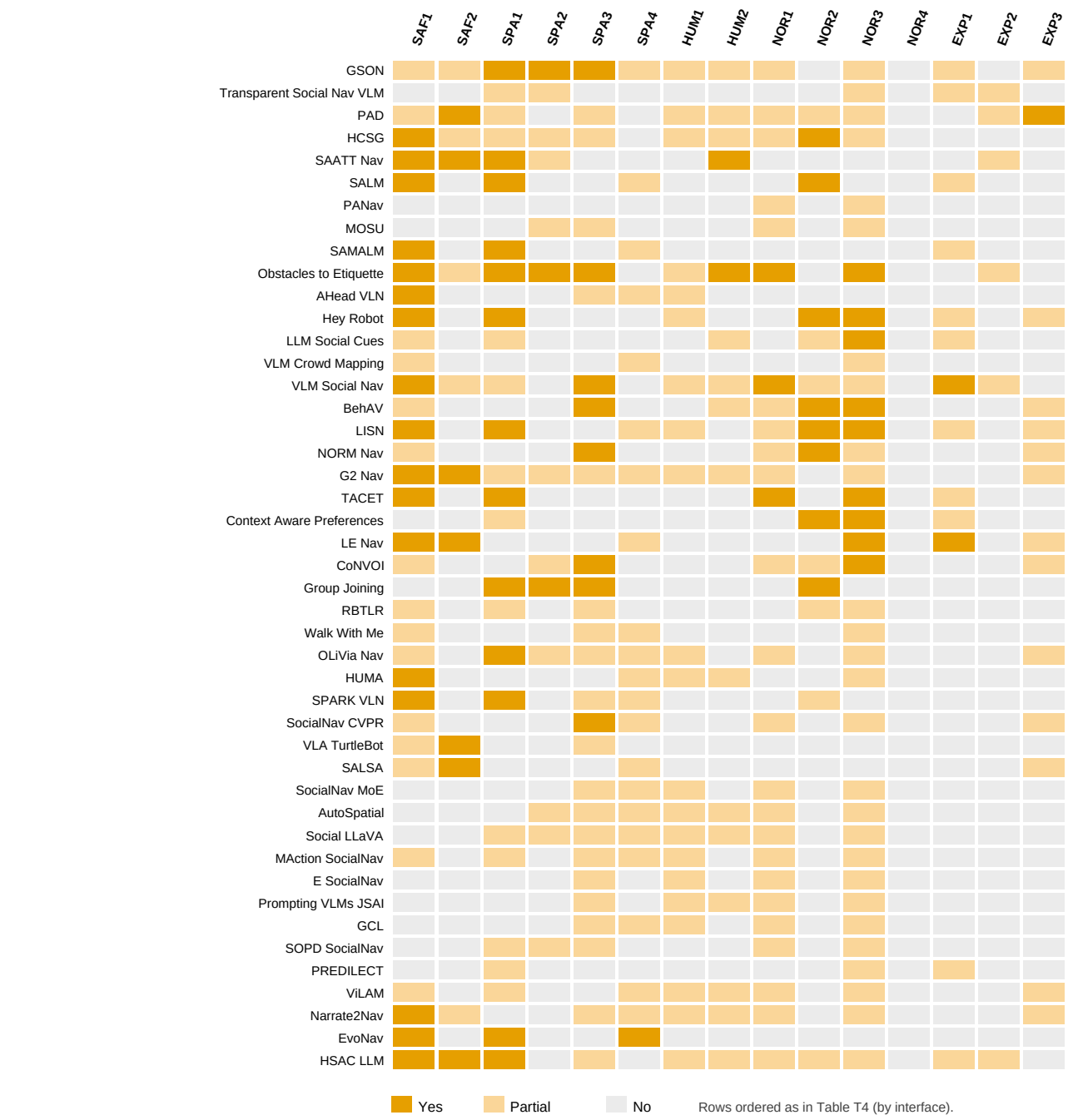
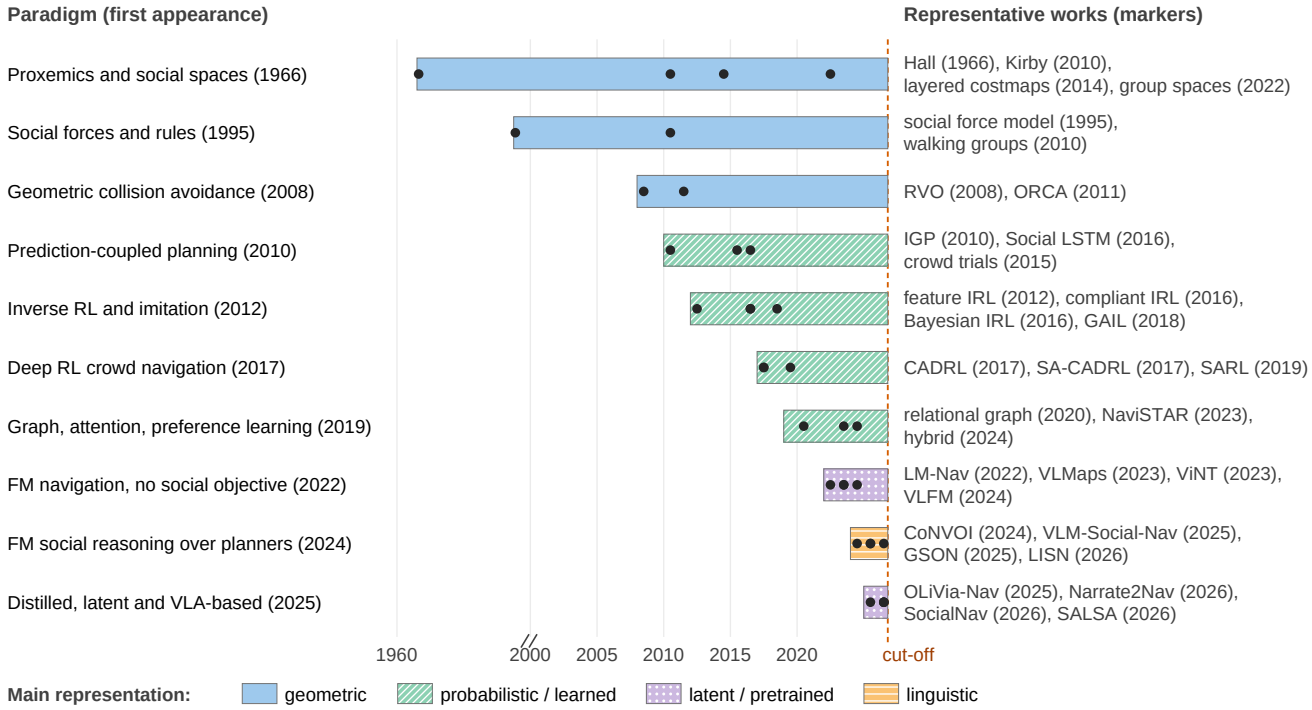


Fig. S4. Dated timeline of the paradigms of social representation. Each band runs from the approximate first appearance of a paradigm to the literature cut-off; markers show representative works at their publication year. Overlapping bands indicate coexistence, not replacement or causal progression



Bands run from first appearance to the literature cut-off; overlapping bands indicate coexistence, not replacement.
The axis is compressed before 2000.

References cited in Table S5 (pre-FM comparison set)

Numbering is independent of the main manuscript; every work of the pre-FM comparison set is listed.

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